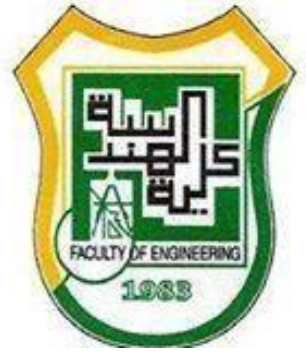




Republic of Yemen
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University of Sana'a
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Design & implantation of 6 Axis Robotic Arm Based on Vision & Closed Loop Trajectory Following

تصميم وتنفيذ ذراع روبوتية سداسية المحاور تعتمد على الرؤية الحاسوبية
وتتبع المسار بالتحكم ذي الحلقة المغلقة

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A graduation project document submitted to the department of Mechatronics in partial fulfillment to the requirements for Bachelor Degree in Engineering

Abstract

This thesis presents the design, implementation, and experimental validation of a design and implantation of autonomous system for path identification and robot execution with 3D vision and ROS2 . The system integrates a six-degree-of-freedom robotic arm based on the PAROL6 platform, MKS SERVO42C closed-loop Field-Oriented Control motor drivers with MT6816 absolute encoders, a Microsoft Kinect v2 RGB-D sensor, a ROS 2-based vision pipeline, and a MoveIt 2 motion planning framework. The software environment is containerized using Docker and version-controlled to enable reproducible deployment.

The main contribution of this work is a four-mode welding path identification pipeline that converts RGB-D data into kinematically feasible 6-DOF trajectories. The four modes—YOLO-based instance segmentation, HSV color masking, red-line annotation, and GUI-based path input—improve robustness across different weld geometries and environmental conditions. A three-stage Kinect calibration process reduced maximum depth error from 17.8 mm to below 2.4 mm, achieving 3D seam localization accuracy of 3.1 mm RMS.

The embedded control system is based on a multi-rate firmware architecture implemented on an STM32 microcontroller, including encoder acquisition, a 500 Hz PID position controller with velocity feedforward, and a 40 kHz Direct Digital Synthesis pulse generation layer. The system achieved a 16.7-fold reduction in joint tracking error compared to open-loop control and an end-effector repeatability of 0.70 mm. Finite element analysis confirmed structural safety factors above 1.5.

Experimental validation showed a Cartesian path planning success rate of 97.3%, processing latency of 340 ms, and path execution deviation of 1.3 mm, with a total system cost below 1500\$.

Keywords: Robotic welding, RGB-D vision, ROS 2, MoveIt 2, Field-Oriented Control, point cloud processing, finite element analysis.

Authorization

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Dedication

With immense pride and gratitude, we dedicate our graduation project to those who have been by our side throughout this remarkable journey. Your unwavering belief in our potential, your guidance through challenges, and your encouragement to reach higher have been our driving forces. To our mentors, whose wisdom and expertise have shaped our project and enriched our understanding, we extend our deepest appreciation. Your patience and dedication to our growth have been truly inspiring. To our families, friends, and loved ones, whose unwavering support and encouragement have lifted us through every hurdle, we dedicate our achievement. Your belief in us has been a constant source of strength. To the journey itself, with its highs and lows, its lessons and triumphs, we offer our gratitude. It has been a transformative experience that has shaped us into resilient and forward-thinking individuals. As we stand at this milestone, we carry forward the knowledge, skills, and values imparted by this endeavor. This dedication is a tribute to the collective efforts that have brought us here and a promise to continue pushing boundaries in pursuit of excellence.

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We would like to express our deepest gratitude and appreciation to all those who have contributed to the successful completion of our project, which served as our engineering graduation project at the Mechatronics Department of Sanaa University.

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Last but not least, we would like to express our heartfelt appreciation to our families and friends for their unwavering support, encouragement, and understanding throughout this challenging endeavor. Their belief in us has been a constant source of motivation and inspiration.

To all those mentioned above and to anyone else who has contributed in one way or another, although not explicitly mentioned, we extend our sincere thanks and gratitude. Your collective efforts and contributions have been invaluable in making This project a success.

Thank you all

Supervisor Certification

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Design & implantation of 6 Axis Robotic Arm Based on Vision & Closed Loop Trajectory
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Chapter 1

Introduction

1.1 Overview

In modern manufacturing, the demand for high precision, minimal errors, and consistent quality has pushed industries toward advanced automation solutions. Among the most versatile and capable of these solutions is the 6-axis robotic arm, whose six degrees of freedom provide a level of dexterity and flexibility comparable to, and in many cases surpassing, that of the human arm. This configuration enables the robot to execute complex movements, reach challenging positions, and maintain precise orientation control, making it suitable for a wide variety of applications including assembly, material handling, painting, and most notably, welding.^[1] Robotic welding has become a central pillar of modern manufacturing, driven by the need for consistency, safety, and productivity. Traditional welding operations expose human workers to intense heat, arc radiation, and toxic fumes. Automation addresses these hazards while simultaneously improving repeatability. According to Ain Shams Engineering Journal (2024), the integration of advanced vision sensor seam tracking into robotic-arm arc welding (RAAW) systems directly improves weld quality and process monitoring reliability.^[2] Wang et al. (2021) further demonstrate that digital twin platforms for human robot interactive welding enable real-time welder behavior analysis and safer collaborative operation.^[3] While industrial robotic arms have existed for decades, their capabilities have significantly evolved with the integration of machine vision and artificial intelligence (AI). Traditional robotic arms follow pre-programmed paths, which limits their adaptability in dynamic environments where object positions, orientations, and shapes vary. In contrast, a robot equipped with a 3D vision system can perceive its surroundings in detail, detect and classify objects of interest, and adjust its actions in real time. This is particularly critical in welding applications, where precise edge detection and accurate path following determine the strength and quality of the weld. A persistent challenge in robotic welding is the automatic identification of the welding path, especially when workpiece placement varies or when joint geometry is complex. The system described in this thesis directly addresses this challenge through a multi-modal vision pipeline offering four input methods: AI segmentation, manual red-line annotation, GUI-drawn paths, and manual color masking. These feed a shared processing pipeline that ultimately generates a trajectory for the robot using ROS and MoveIt.^[4] This project aims to design and implement a 3D-printed 6-axis robotic arm as a proof of concept, integrating AI-powered 3D vision, motion and trajectory

planning, and closed-loop motor control with Field-Oriented Control (FOC). The arm is adapted from the PAROL6

1.2 Problem Statement

Welding is one of the most demanding manufacturing processes in terms of precision and consistency. Manual welding is inherently limited by human factors including operator fatigue, skill variability, and inability to maintain sub-millimeter precision over extended production runs. Industrial welding robots address many of these limitations but typically require expensive proprietary hardware, specialized programming expertise, and rigid pre-programmed trajectories that cannot adapt to variability in workpiece position or geometry.

The core challenge addressed by this project is the development of an autonomous welding path identification and execution system that is simultaneously: (1) affordable and accessible, using open-source hardware and software; (2) intelligent, capable of perceiving and analyzing the welding environment using computer vision and AI; and (3) precise, with closed-loop motor control ensuring accurate trajectory execution. Prior work has demonstrated vision-guided welding robots using expensive laser scanners, industrial cameras, or proprietary platforms. This project aims to achieve comparable capabilities using consumer-grade hardware (Kinect v2, 3D-printed PAROL6 arm, STM32 microcontroller) and open-source frameworks (ROS, MoveIt, PyQt5, Git, Docker).

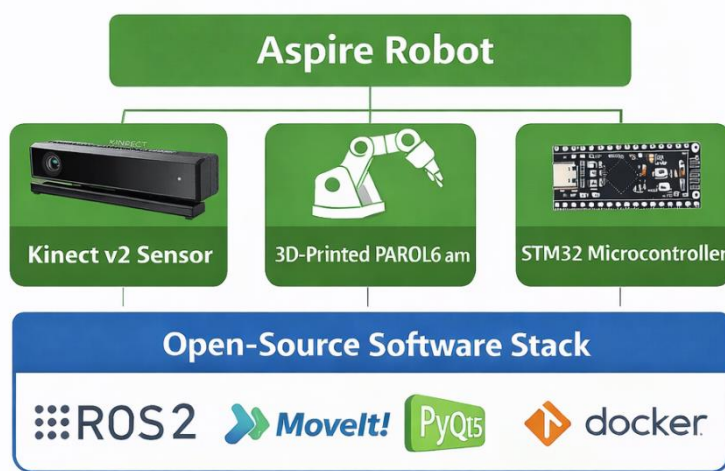


Figure 1. 1: Hardware and software architecture of the Aspire Robot, illustrating the perception, planning, control, and software framework components.

A secondary challenge is the reproducibility and maintainability of the software system. Complex robotics projects involve multiple interdependent software components running on different machines or operating system configurations. Without proper version control and containerization, replicating the development environment on a new machine -- or reverting to a known-good state after a failed change -- becomes a significant burden. This project addresses these software engineering challenges using Git for version control, GitHub for collaboration and code hosting, and Docker for environment containerization.

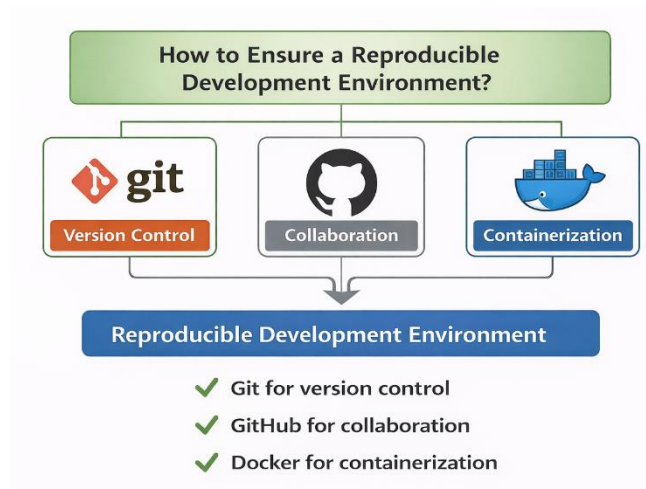


Figure 1. 2: Tools and workflow used to maintain version control, collaborative development, and environment reproducibility in the Aspire Robot project.

The problem of welding path identification itself presents multiple sub-challenges. Weld joint geometry varies widely across applications: butt joints, fillet joints, lap joints, T-joints, and corner joints all present different visual signatures. Workpiece surfaces may be reflective, oxidized, or contaminated. The welding environment is harsh, with arc light, spatter, and smoke interfering with vision systems during active welding. The path identification system must be robust to these variations while remaining computationally efficient enough for real-time operation on a standard workstation. Four distinct input modes are provided to handle different operational scenarios, from fully automated AI segmentation to manual operator guidance.

1.3 Project Objectives

The primary objective of this project is to develop and demonstrate a complete, integrated, vision-guided autonomous welding robot system. The specific objectives are:

- Design and fabricate a 3D-printed 6-axis robotic arm with integrated internal gear mechanisms, then processed with the printing, assembling, and installing of all the required parts of the robotic arm. This first stage would be divided into multiple smaller stages.
- Implement a closed-loop DC motor or stepper motors control system with position, speed, and acceleration feedback for smooth, accurate movement.
- Perform robotic analysis including forward/inverse kinematics, Jacobian analysis, path and trajectory planning.
- Implement a triple Raspberry Pi camera 3D vision system with stereo and double-stereo algorithms for depth perception and camera calibration.
- Develop AI algorithms to detect objects of interest, identify welding seams, and generate welding paths.
- Integrate all subsystems using ROS2 on Raspberry Pi 5 with optional computational offloading to a PC.
- Test and evaluate system accuracy, smoothness, and performance in example welding tasks.

1.4 Project Scope and Limitations

The scope of this project encompasses the full design and implementation cycle of the vision-guided robotic welding system, from mechanical assembly and electronic integration through software development and system-level testing. The project is intended as a proof-of-concept demonstrating the feasibility of integrating low-cost consumer hardware with open-source robotics software for autonomous welding path identification and execution.

Hardware scope: The project uses the PAROL6 3D-printed robotic arm platform, adapted with MKS SERVO42C motors and MT6816 encoders. The vision system is limited to the Microsoft Kinect v2 RGB-D sensor. The embedded controller is the WeAct STM32F411CE (Black Pill).

These hardware choices provide a well-defined and reproducible hardware baseline; industrial-grade alternatives (laser scanners, industrial 6-axis arms, dedicated embedded controllers) are explicitly out of scope.

Software scope: The motion planning stack is based on ROS Noetic (not ROS2) and MoveIt 1.x. The deep learning component uses a pre-trained segmentation model (not trained from scratch during this project). The GUI is implemented in PyQt5 with ROS integration. Git/GitHub/Docker are used for development infrastructure but are not the primary research contribution.



Figure 1. 3: Software scope for the Project

Limitations and exclusions: The system is designed for controlled laboratory conditions; it does not address real-time robustness to welding arc light, spatter, or smoke during active welding. The Kinect v2 is not appropriate for outdoor use or direct proximity to the welding arc. The project does not address weld quality metrics (penetration depth, bead geometry, porosity) or perform quantitative weld testing. The system operates in a static environment; dynamic obstacle avoidance during path execution is not implemented. The STM32-based motor control uses PWM step commands rather than direct torque control, which limits the precision of force feedback.

1.5 Project Methodology

The project follows a structured engineering development methodology: (1) requirements analysis and system architecture design; (2) subsystem design and implementation in parallel (mechanical assembly, embedded firmware, ROS stack, vision pipeline, GUI); (3) subsystem integration and interface verification; (4) system-level testing and validation. Each subsystem is implemented in a dedicated Git branch and merged to the development branch after unit testing. Docker ensures that all team members work in an identical software environment throughout development.

The vision pipeline development followed an iterative approach: initially implementing the simplest mode (color mask detection), then adding manual annotation modes, and finally integrating the AI segmentation mode. This bottom-up approach allowed early end-to-end system testing with simpler inputs before the full AI-based pipeline was operational. MoveIt configuration was developed in parallel using a Gazebo simulation, allowing trajectory planning to be validated without requiring the physical robot to be fully assembled.

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The motion control architecture was validated through a series of tests of increasing complexity: first, individual joint step-response testing to tune PID parameters on each MKS SERVO42C; second, multi-joint coordinated motion in joint space using MoveIt's joint trajectory planning; and finally, Cartesian path following using `computeCartesianPath()` with the welding speed constraint. Each stage's performance metrics (tracking error, path deviation, repeatability) were measured and documented.

1.6 Report Organization

This report is organized into two chapters. Chapter 1 (Introduction) provides an overview of the project motivation and context, a precise problem statement, the specific objectives, the project scope and limitations, and the methodology. Chapter 2 (Background and Literature Review) provides the technical background necessary to understand the design decisions made in this project, followed by a review of the peer-reviewed and technical literature most relevant to each subsystem.

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Chapter 2 is organized as follows: Section 2.1 (Background) covers fourteen technical areas in depth: robotic welding history and technology, 6-DOF kinematics and the PAROL6 platform, ROS architecture, MoveIt motion planning, trajectory planning algorithms including splines and NURBS, depth sensing technologies and their evolution, RGB-D camera calibration theory, 3D reconstruction and point cloud processing, image processing for path identification, deep learning for image segmentation, closed-loop motor control and FOC theory, STM32 embedded systems, Git and GitHub version control, and Docker containerization. Section 2.2 (Literature

Review) surveys the peer-reviewed literature organized by subsystem, culminating in a gap analysis positioning this project relative to prior work. The References section lists all 50 cited sources in IEEE format.

The report structure follows the guidelines established by the university for graduation theses in the engineering college. Each chapter begins with a brief overview paragraph that contextualizes the chapter content within the broader project. Figures, tables, and equations are numbered sequentially within each chapter (Figure 2.1, Table 2.1, Equation 2.1). All technical claims are supported by citations to peer-reviewed literature or authoritative technical references. The reference list uses the IEEE citation format, numbered sequentially in order of first appearance in the text, for compatibility with the Zotero reference management tool used during the literature review process.

Chapter 2

Background and Literature Review

This chapter provides the theoretical and technical foundation required to understand every design decision in this project. It is written to be accessible to readers with a general engineering background who may not be specialists in robotics, computer vision, or embedded systems. Each section introduces the relevant concepts from first principles, traces the evolution of the relevant technology, and explains precisely how it is applied in this project. The background is followed by a structured literature review that examines peer-reviewed work in each technical area.

2.1 Background

The depth of coverage reflects the needs of the target audience: a graduation thesis examination committee with general engineering expertise but limited specialization in robotics, computer vision, or embedded systems. Concepts are introduced assuming only a background in fundamental physics, mathematics, and programming, then progressively built up to the level of detail needed to understand the design decisions. Where possible, concrete numerical examples and performance specifications are provided to make abstract concepts tangible. The technical depth increases from section 2.1.1 (welding motivation, accessible to any engineer) to section 2.1.14 (Docker containerization, relevant to software-engineering readers), reflecting the project's interdisciplinary nature.

The following sections cover the fourteen core technical domains that underpin this project. The presentation is deliberately graduated: beginning with the broader motivation for robotic welding, progressing through the mechanical and kinematic design of the robot arm, the software infrastructure (ROS, MoveIt), the mathematical tools for trajectory planning, the vision system (depth sensing technologies, RGB-D calibration, 3D reconstruction), the image processing and AI pipeline for path detection, the motor control theory and hardware, the embedded microcontroller, and finally the software engineering tools (Git/GitHub and Docker) that ensure reproducibility and maintainability.

2.1.1 Robotic Welding: Background and Motivation

Robotic welding has been practiced since the 1960s, beginning with spot welding in the automotive industry. The first generation of welding robots used the teach-and-playback paradigm: a human operator physically guided the robot through the desired welding path, recording the joint angles at each point, and the robot then reproduced this motion repeatedly. This approach produces consistent results when workpieces are precisely fixtured in identical positions, but fails when any variability is introduced -- even small deviations in workpiece placement, thermal distortion during welding, or changes in joint geometry cause the torch to deviate from the intended seam. ^{[5][6]}



Early seam tracking systems used through-the-arc sensing (TAST), which exploits the relationship between welding arc current and voltage and the torch-to-seam distance. As the torch deviates laterally from the seam center, the arc geometry changes, producing measurable changes in the electrical parameters. TAST is simple and requires no external sensor, but it can only detect lateral deviations (not height variations), requires an active welding arc, and produces noisy signals that require filtering. Despite these limitations, TAST remains in use in some industrial systems due to its simplicity and zero additional hardware cost.

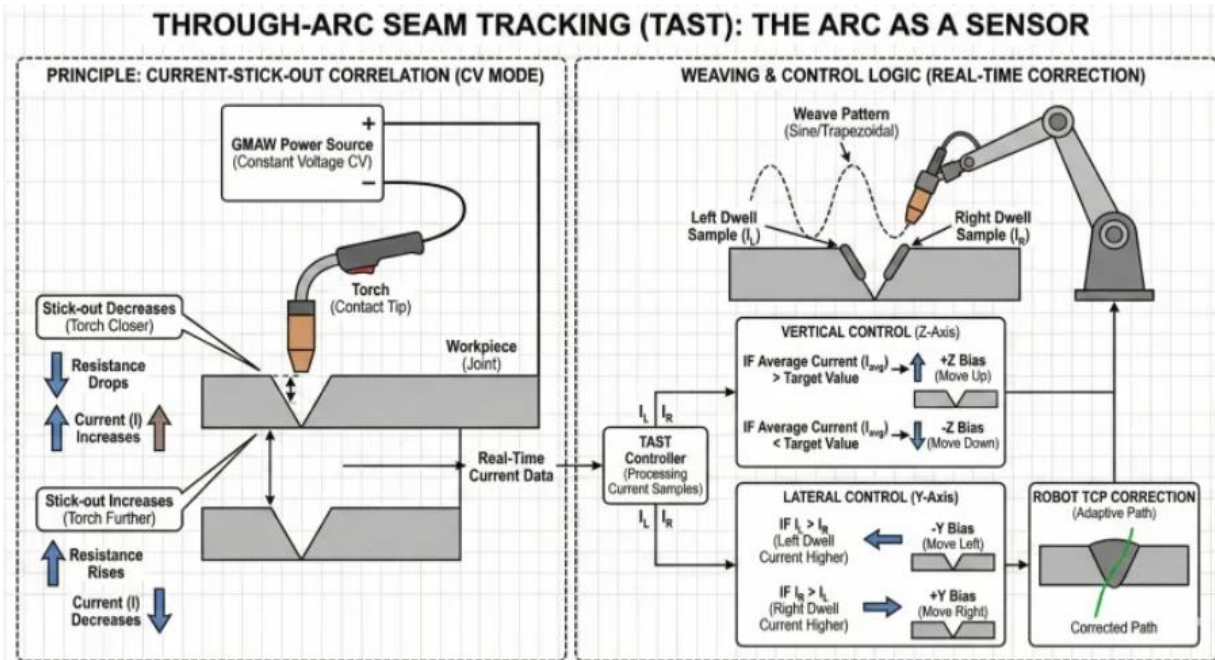


Figure 2. 2: Early seam tracking systems used through-the-arc sensing (TAST)

Laser stripe sensors, introduced in the 1980s and widely adopted in the 1990s, provided a major step forward. A laser line is projected across the seam, and a camera images the resulting stripe. The 3D geometry of the seam cross-section is computed from the distortion of the stripe using laser triangulation. Laser stripe sensors provide full 3D seam profiles (groove width, depth, root opening, bevel angle) and are the current standard in industrial robotic welding. However, they require precise sensor-torch integration (the sensor must be rigidly mounted to the torch at a precise offset), are sensitive to arc light and spatter, and are expensive.

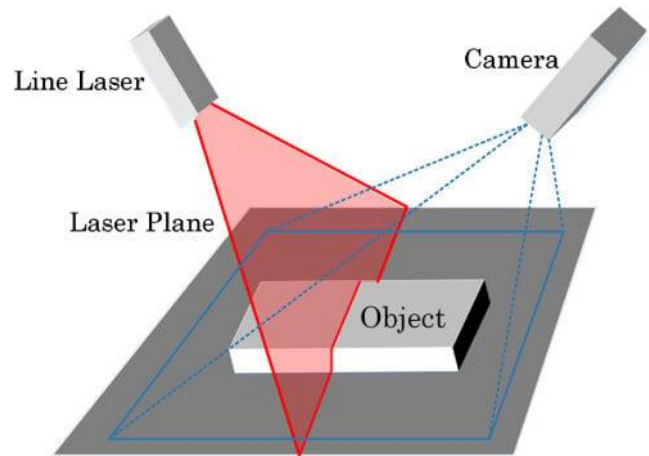


Figure 2. 1: Laser stripe sensors provide full 3D seam profiles

RGB-D cameras represent the most recent generation of seam sensing technology, combining color and depth information in a single sensor. They enable full 3D reconstruction of the weld zone without contact, at substantially lower cost than laser stripe sensors. Wang (2024) provides a comprehensive review of active visual sensing methods for robotic welding, categorizing approaches into arc sensing, structured light, and 3D vision methods, and identifying the key open challenges, As shown in the fig 1.6: sensor-torch integration, real-time performance, robustness to arc light and weld spatter, and the gap between laboratory demonstrations and industrial deployment. [4]

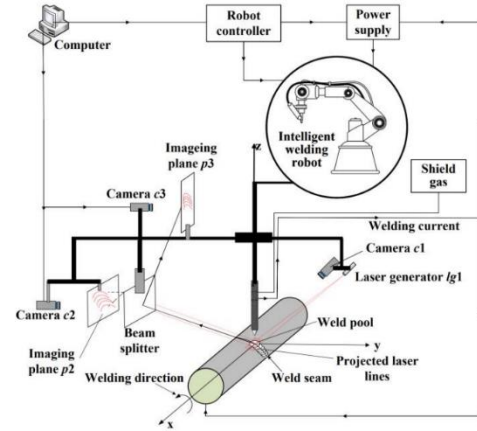


Figure 2. 3: The schematic diagram of the ray-intersection-based weld pool

Ain Shams Engineering Journal (2024) integrates an advanced vision sensor into a robotic-arm arc welding system for real-time seam tracking with EWMA-based (Exponentially Weighted Moving Average) fault detection, achieving sub-millimeter tracking deviations and fast reaction times of 47-55 ms , as shown in fig 1.7 . Wang et al. (2021) demonstrate a digital twin platform for human-robot interactive welding, where a synchronized digital model of the robot and weld pool enables real-time analysis of welder behavior and safer human-robot collaboration. [2], [3]

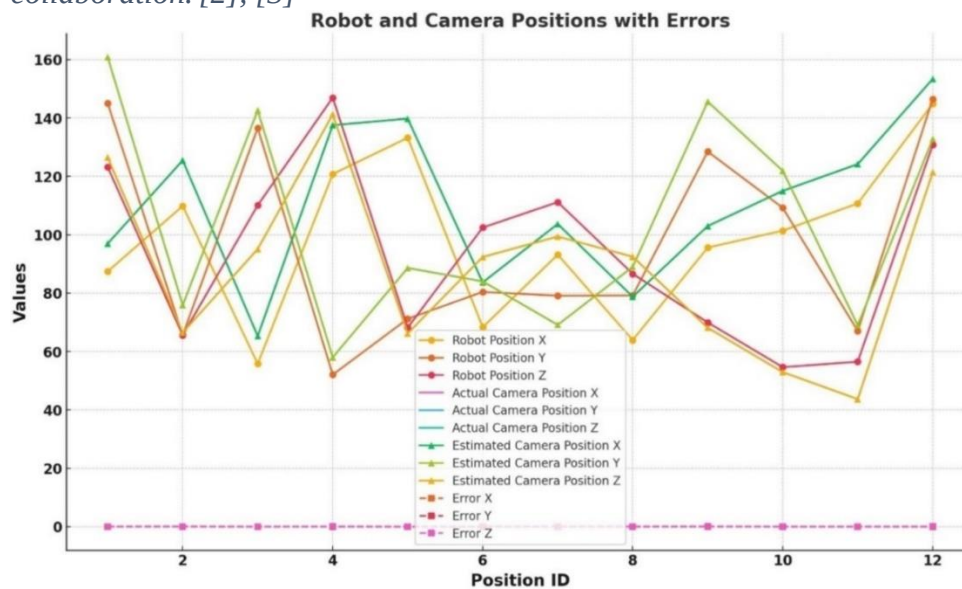


Figure 2. 4: Tracking Variances of Robot Motions and Camera Pose Accuracy with the Respective Estimation Errors

To understand why 6 DOF is the canonical configuration for industrial manipulators, consider the degrees of freedom of a rigid body in 3D space. A rigid body has 6 degrees of freedom: three translational (x, y, z position) and three rotational (roll, pitch, yaw orientation), as shown in fig 1.9. To position and orient an end-effector arbitrarily -- to reach any pose within its workspace -- a robot needs at least 6 independent joint variables. With fewer than 6 joints, certain orientations or positions are geometrically unreachable. With exactly 6 revolute joints (the most common type), the robot can reach any pose within its workspace, subject to joint limits and singularity avoidance.

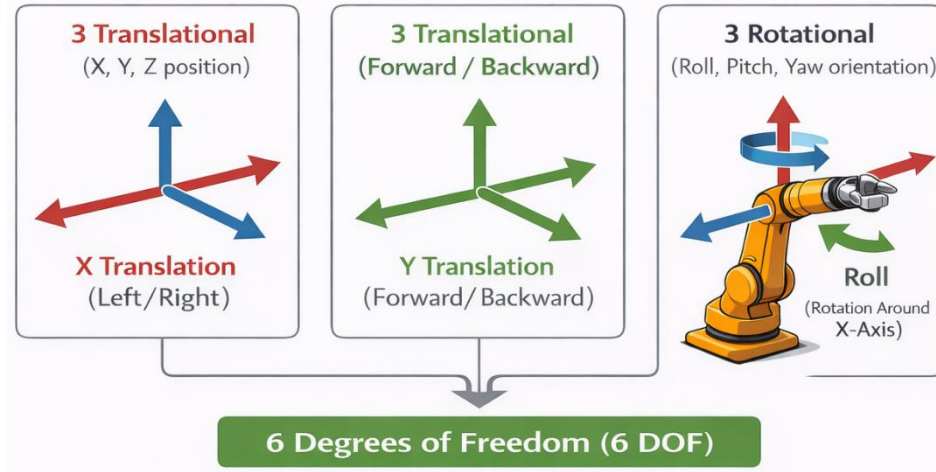


Figure 2. 6: Figure 1. 9: A rigid body in three-dimensional space has six degrees of freedom

The mathematical description of a 6-DOF arm relies on the Denavit-Hartenberg (DH) convention, which represents each joint transformation as a 4x4 homogeneous transformation matrix encoding rotation and translation along the joint axis. The forward kinematics (FK) function computes the end-effector pose from a given set of joint angles $q = [q_1, q_2, \dots, q_6]$ by multiplying the six DH transformation matrices in sequence: $T_{0_6} = T_{0_1} * T_{1_2} * \dots * T_{5_6}$. The result is a 4x4 matrix whose upper-left 3x3 submatrix is the rotation matrix R and whose last column (top 3 elements) is the position vector p .^[7]

The DH convention assigns four parameters to each joint: d (link offset along the previous joint axis), θ (joint angle about the previous joint axis -- the variable for revolute joints), a (link length -- offset along the joint axis), and α (twist angle between consecutive joint axes). These four parameters completely describe any serial-chain kinematic structure, regardless of its geometry. The DH table for the PAROL6 arm is derived from its CAD model and encoded



Figure 2. 8: HELENE six-axis open-source robotic manipulator

The gearbox in each PAROL6 joint introduces a phenomenon known as reflected inertia. From the perspective of the motor, the load inertia is multiplied by the square of the gear ratio: $J_{\text{reflected}} = \frac{J_{\text{load}}}{N^2}$ where N is the gear reduction ratio. For the PAROL6 arm with a gear ratio of approximately 50:1 at joints 1-3 and 20:1 at joints 4-6, the load inertia reflected to the motor shaft is 2500x or 400x smaller than the actual joint load inertia. This dramatically reduces the motor torque required to accelerate the joint and makes the system less sensitive to external load variations -- a desirable property for closed-loop position control. The tradeoff is increased backdrivability resistance (the gearbox prevents the joint from being easily moved by external forces), which reduces the risk of damage from unintended contact forces.

A critical consideration in robotic arm design is the distinction between workspace and singularities. The workspace is the set of all end-effector poses reachable by the arm given its joint limits. Singularities are configurations where the Jacobian matrix (relating joint velocities to end-effector velocities) becomes rank-deficient, meaning the robot loses one or more degrees of freedom at the end-effector level. At singularities, small Cartesian velocities require infinite joint velocities, causing erratic motion. MoveIt includes singularity avoidance in its planning pipeline, and the URDF model includes joint limits that keep the arm away from the most problematic singular configurations.

Background and Literature Review

Repeatability and accuracy are two distinct and frequently confused metrics for robotic arms. Accuracy is the ability to reach a commanded absolute position in space; repeatability is the ability to return to the same position when commanded repeatedly. Most robotic arms are specified by repeatability (typically 0.02-0.5 mm for industrial arms, 0.1-1 mm for research-grade open-source arms), not accuracy, because repeatability is much easier to achieve and directly relevant to most production tasks. The PAROL6 specification of 0.2 mm repeatability means that after calibration, the arm returns to a commanded position within a 0.2 mm sphere - sufficient for the welding path application, which requires 1-3 mm path-following accuracy.

The concept of compliance -- the ability of a manipulator to accommodate external forces without damage to the robot or workpiece -- is increasingly important in contact-rich applications such as welding and assembly. A fully rigid position-controlled robot will fight against any external force, which can cause workpiece damage or joint overload if the planned path deviates from the actual surface. Force/torque sensors at the end-effector, combined with admittance or impedance controllers implemented in MoveIt via the MoveIt Servo interface, allow the robot to detect contact forces and adjust its motion accordingly. While force control is not implemented in this project (the MKS SERVO42C provides torque feedback through the closed-loop current sensing, but force at the end-effector is not measured), the PAROL6 mechanical design includes compliance margins in the joint limits that allow small deviations from commanded positions without hardware damage.

Workspace analysis of the PAROL6 arm is critical for verifying that all welding paths within the target workpiece zone are reachable. The reachable workspace of a 6-DOF arm (the set of end-effector positions achievable from at least one joint configuration) is a complex 3D solid whose shape depends on the DH parameters and joint limits. The dexterous workspace (the subset of positions reachable from all possible orientations) is substantially smaller. For welding, the robot must reach each path waypoint with a specific end-effector orientation (torch perpendicular to the surface), which further constrains the effective workspace to the orientable subset of the reachable workspace. The PAROL6 workspace analysis, performed in Gazebo using Monte Carlo sampling of random joint configurations, confirms that the arm can reach all

Beyond topics, ROS provides two other communication patterns. Services implement synchronous request-reply interactions (like a function call): a client node sends a request message, the server processes it and returns a response. This is used for discrete operations like triggering a path detection computation. Actions extend services with feedback and preemption: a client sends a goal, the server begins execution and sends periodic feedback messages, and the client can cancel the goal at any time. MoveIt uses the action interface for trajectory execution, allowing the operator to monitor progress and abort if needed.

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The ROS coordinate system, managed by the tf2 library, is foundational to multi-sensor robotic systems. Every component -- each robot link, the camera frame, the end-effector, the workspace origin -- is represented as a named coordinate frame, and tf2 maintains the tree of transformations between all frames in real time. Any node can query the transformation between any two frames at any timestamp, enabling seamless fusion of information from multiple sensors and the robot model. In this project, tf2 manages the transformation chain from the Kinect v2 camera frame to the robot base frame, enabling the path generator to convert camera-frame 3D waypoints to robot-base-frame waypoints for MoveIt. This extrinsic calibration (the camera-to-robot transformation) is determined once using a calibration target, stored as a static transform in a launch file, and published by tf2 at system startup.

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The URDF (Unified Robot Description Format) is an XML format used to describe a robot's kinematic and dynamic model. It specifies each link (with visual mesh, collision mesh, mass, and inertia) and each joint (type, parent/child links, axis, limits). MoveIt reads the URDF at startup to build the robot model used for kinematics, collision checking, and visualization. In this project, the PAROL6 URDF is stored in the GitHub repository (Abdulkareem771/parol6_urdf), versioned with Git, and loaded by the MoveIt configuration package at startup.

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ROS also provides a rich toolset for development, debugging, and visualization. RViz (ROS Visualization) is a 3D visualization tool that can display the robot model (from URDF), the robot state (joint positions published on the /joint_states topic), sensor data (point clouds,

images, depth maps), planning scenes (collision objects), and motion plans (animated trajectories). During development, RViz was used to verify IK solutions, visualize planned trajectories before execution, and overlay the detected welding path on the point cloud for debugging. Rosbag allows recording and playing back all topic data, enabling off-robot debugging: the Kinect data stream can be recorded in the lab, and the vision pipeline can be debugged on a development machine without the camera or robot connected.

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ROS launch files are XML configuration files that describe a complete ROS system startup: which nodes to launch, their parameters, namespace assignments, and topic remappings. A single launch file for this project starts the ROS master (if not already running), the Kinect2 bridge driver node, the robot state publisher, the MoveIt move_group node (with the PAROL6 SRDF and URDF loaded from the GitHub repository), the hardware interface node, the vision pipeline node, and the PyQt5 GUI node. Launch file arguments allow runtime configuration (e.g., selecting between UART and Ethernet communication, enabling or disabling simulation mode). This single-command startup (`roslaunch parol6_welding full_system.launch`) is a significant practical advantage over manually starting each component, and combined with Docker, it reduces the system startup procedure to two commands: `docker-compose up` and `roslaunch`.

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The ROS parameter server provides a centralized key-value store for configuration parameters accessible to all nodes at runtime. Parameters used in this project include: the UART/Ethernet communication mode selection; the Kinect v2 camera intrinsic and extrinsic calibration parameters (loaded from YAML calibration files); the MoveIt planning pipeline configuration (planner selection, planning time limit, collision checking configuration); the welding speed (mm/s) target for Cartesian path execution; and the path detection mode (used by the GUI to configure which vision pipeline branch is active). Parameters can be set in the launch file (static configuration) or modified at runtime using the `rosparam` command-line tool or the ROS parameter API in the GUI, enabling operator adjustment without restarting the system.

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group containing the end-effector); generate the self-collision matrix (a precomputed table of joint configuration pairs that are always collision-free or always in collision, used to skip unnecessary collision checks during planning); define end-effector and virtual joint configurations; and set the robot's default joint positions (the "home" and "ready" configurations displayed in the GUI). The resulting MoveIt configuration package is stored in the GitHub repository alongside the URDF and serves as the single source of truth for the robot's kinematic model.

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The MoveIt Servo subsystem provides real-time, incremental Cartesian and joint velocity commands via a joystick or keyboard, enabling manual control of the end-effector position. While the primary operation mode of this project is autonomous trajectory execution from the detected path, the MoveIt Servo interface is used during system setup and calibration: the operator can jog the robot end-effector to the calibration target position, perform the camera-to-robot extrinsic calibration, and position the robot in the starting configuration for a welding run. The Servo node subscribes to incremental velocity commands and translates them to joint velocity commands using the Jacobian pseudoinverse, subject to joint limit and singularity avoidance constraints.

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MoveIt supports multiple motion planners through plugin interfaces. The most commonly used planners from the OMPL (Open Motion Planning Library) are: RRT (Rapidly-exploring Random Tree) -- a sampling-based planner that builds a tree of random configurations expanding toward the goal; RRTConnect -- a bidirectional variant of RRT that grows two trees simultaneously from the start and goal, significantly faster than single-tree RRT; PRM (Probabilistic Roadmap Method) -- pre-builds a graph of collision-free configurations, efficient for repeated queries in the same environment; CHOMP (Covariant Hamiltonian Optimization for Motion Planning) -- optimizes a trajectory using gradient descent for smoothness and collision avoidance; and Pilz Industrial Motion Planner -- generates deterministic Cartesian trajectories (linear, circular, point-to-point) suitable for manufacturing tasks.

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Background and Literature Review

The Pilz Industrial Motion Planner is particularly relevant to this project's welding application. Unlike sampling-based planners (RRT, PRM) which find collision-free paths through the configuration space but do not guarantee a specific end-effector trajectory shape, the Pilz LIN (linear) command moves the end-effector in a straight line in Cartesian space at a specified Cartesian velocity. This is essential for welding, where the torch must move along the seam at a specific, controlled speed (typically 3-8 mm/s for MIG welding) to achieve proper heat input and bead formation. The Pilz planner also supports synchronized deceleration and acceleration at trajectory waypoints, ensuring smooth transitions between sequential path segments.

For the welding path following task, MoveIt's `computeCartesianPath()` function is the key interface. This function accepts a sequence of end-effector Cartesian waypoints, an interpolation step size (e.g., 1 mm), and returns a joint trajectory. Internally, it calls the IK solver at each interpolated waypoint, checking for joint limit violations and collisions. The function returns the fraction of the path successfully planned; in practice, values above 0.95 (95% of waypoints successfully reached) are acceptable, with slight adjustments to waypoint spacing or robot configuration resolving the remaining cases.

Time parameterization is the process of assigning timestamps to each trajectory waypoint such that velocity and acceleration limits are respected. MoveIt includes two algorithms: IPTP (Iterative Parabolic Time Parameterization) and TOTG (Time-Optimal Trajectory Generation). TOTG computes the globally time-minimum trajectory subject to per-joint velocity and acceleration bounds, using bang-coast-bang velocity profiles (each joint segment accelerates at the maximum rate, coasts at maximum velocity if needed, then decelerates). TOTG produces significantly shorter-duration trajectories than IPTP for most paths. For the welding application, where the end-effector must move at a specified Cartesian speed, a Cartesian speed constraint is applied to ensure the end-effector does not exceed the desired welding speed.

MoveIt's collision checking uses the Flexible Collision Library (FCL) to check whether any configuration in the planned trajectory intersects with any collision object in the planning scene. Collision objects can be geometric primitives (box, sphere, cylinder) or arbitrary meshes. In this

welding project, the workpiece (the object being welded) is added to the planning scene as a collision object so that MoveIt plans trajectories that approach the weld seam from above without colliding with the rest of the workpiece. The collision model uses a safety margin (padding) around each object to provide a buffer zone that prevents near-collisions that could occur due to trajectory tracking errors.

2.1.5 Trajectory Planning: Splines, Interpolation, and Path Smoothing

Trajectory planning is the process of computing a time-parameterized sequence of robot configurations that moves the robot from an initial state to a goal state while satisfying physical constraints. It is important to distinguish between path and trajectory: a path specifies only the geometric sequence of configurations, while a trajectory adds a time dimension -- the exact joint position, velocity, and acceleration at every instant. This distinction is critical for physical execution because violating velocity or acceleration limits causes tracking errors or hardware damage.

Polynomial interpolation is the most fundamental method for generating smooth trajectories between discrete waypoints. A cubic polynomial $p(t) = a_0 + a_1*t + a_2*t^2 + a_3*t^3$ between two waypoints has four coefficients (a_0, a_1, a_2, a_3) that are determined by four boundary conditions: the position and velocity at the start and end of the segment. This ensures continuous position and velocity, but the acceleration is piecewise linear and has a discontinuity at waypoints. A quintic polynomial (fifth-degree) adds two more boundary conditions (the acceleration at start and end), producing continuous acceleration -- this is used when smooth jerk behavior is required.

Cubic spline interpolation is the standard method for producing smooth trajectories through N waypoints simultaneously. A cubic spline is a piecewise cubic polynomial that is continuous in position, velocity, and acceleration at every internal waypoint. The natural spline additionally enforces that the second derivative is zero at the endpoints, minimizing the total curvature of the path. The `ros_control` joint trajectory controller receives the planned trajectory waypoints

For the welding path, an additional spline fitting step is applied to the raw detected path pixels before they are sent to MoveIt. The path detection pipeline produces a dense set of 2D pixel coordinates (potentially noisy), which must be converted to a smooth, evenly spaced sequence of 3D waypoints. The process is: (1) the 2D path pixels are ordered along the path direction; (2) they are back-projected to 3D using the depth image; (3) outlier 3D points are removed using a median filter; (4) a cubic spline is fit to the remaining 3D points (parametrized by arc length); (5) the spline is sampled at uniform arc length intervals (e.g., one waypoint every 5 mm); (6) the resulting evenly spaced 3D waypoints are passed to MoveIt's `computeCartesianPath()` function.

NURBS (Non-Uniform Rational B-Splines) provide a more general curve representation used in CAD systems and high-accuracy path planning. A NURBS curve is defined by a set of control points, a knot vector, and a set of rational basis functions (B-spline basis functions weighted by control point weights). By adjusting the weights, NURBS can represent any conic section (lines, circles, ellipses, parabolas, hyperbolas) exactly, and arbitrary smooth curves approximately. In welding path planning, NURBS are used to fit smooth trajectories to complex weld seam geometries, particularly for saddle-shaped seams or circular welds where simple cubic splines produce insufficient smoothness.

B-spline fitting (a subset of NURBS with uniform weights of 1) is widely used for welding path smoothing on point cloud data. The key advantage of B-splines over interpolating splines (which must pass through every data point) is that B-splines are approximating -- they minimize a weighted sum of distance to the data points and a smoothness penalty, naturally filtering out measurement noise. The smoothness penalty is the integral of the squared second derivative over the curve length (the bending energy), and the trade-off between data fidelity and smoothness is controlled by a single regularization parameter λ . Geng et al. (2022) use B-spline fitting for welding path smoothing on complex structural part point clouds.

Background and Literature Review

The concept of jerk (the rate of change of acceleration) is important for mechanical robustness and weld quality. High jerk causes vibrations in the robot structure that are manifested as end-effector oscillations, degrading weld bead quality by producing periodic variations in torch height and position. Minimum-jerk trajectory planning (using fifth-degree polynomials) produces trajectories where acceleration changes smoothly, reducing structural vibrations. The `ros_control` trajectory interpolation uses cubic Hermite splines (continuous acceleration), which provides a compromise between the simplicity of cubic interpolation and the smoothness of fifth-degree polynomials. At the low end-effector speeds used for welding (3-8 mm/s), jerk-induced vibrations are typically below 0.1 mm amplitude and do not significantly affect weld quality.

Path smoothing has direct impact on weld quality. A path with many sharp direction changes (high curvature waypoints) causes the robot to decelerate sharply before each waypoint, producing non-uniform torch speed and hence non-uniform heat input. Excessive heat input in slow sections causes burn-through (melting through the workpiece), while insufficient heat input in fast sections produces cold laps (incomplete fusion). The B-spline smoothing step in the path detection pipeline reduces path curvature by fitting a smooth global curve to the detected path pixels, eliminating the high-frequency noise that would otherwise appear as curvature variations in the trajectory. The trade-off between smoothing (which may deviate from the true seam geometry) and fidelity (which preserves geometry but may introduce curvature noise) is controlled by the B-spline regularization parameter, tunable in the ROS parameter server.

The concept of arc-length parameterization is critical for uniform sampling of spline paths. A naively-parameterized spline (parameterized by the knot index t) produces non-uniformly spaced waypoints when sampled at uniform t intervals, because different segments of the curve may have different lengths. Arc-length re-parameterization computes the mapping between the knot parameter t and the actual distance along the curve, then samples at uniformly spaced arc lengths. This is essential for welding, where the torch must move at a uniform speed along the

Time-of-Flight (ToF) depth sensing is the technology used in the Microsoft Kinect v2. Rather than projecting a pattern, a ToF sensor illuminates the entire scene with modulated infrared light (typically using an LED or VCSEL array) and measures the phase shift of the returning light at each pixel to compute the round-trip time and hence the depth. The Kinect v2 uses Amplitude-Modulated Continuous Wave (AMCW) ToF, which illuminates with a sinusoidal amplitude-modulated infrared light and demodulates the received signal using four phase samples per modulation period (0 degrees, 90 degrees, 180 degrees, 270 degrees). This allows computation of both the depth (from phase) and the amplitude (related to surface reflectivity and angle).

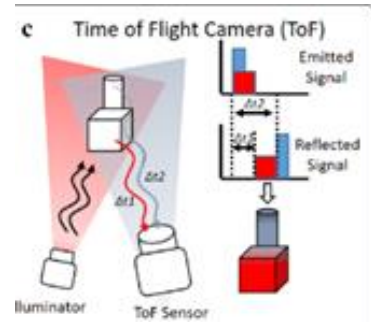


Figure 2.9: Time-of-Flight (ToF) depth sensing

AMCW ToF produces a depth image at the full sensor resolution (512 x 424 pixels for the Kinect v2) at up to 30 frames per second. The depth range is limited by the unambiguous range, which is $c/(2 * f_{mod})$ where c is the speed of light and f_{mod} is the modulation frequency. The Kinect v2 uses three modulation frequencies (approximately 16 MHz, 80 MHz, and 120 MHz) simultaneously, using the Chinese Remainder Theorem to extend the unambiguous range to approximately 8 meters while maintaining the depth resolution of the highest modulation frequency (approximately 1 mm depth precision at 2 meters).

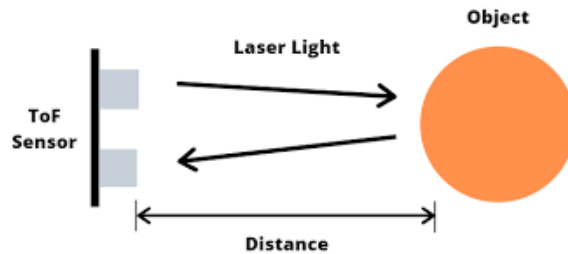


Figure 2.10: AMCW ToF

The ToF measurement principle introduces characteristic systematic errors that require compensation. The wiggling error (cyclic error) is a sinusoidal bias in the measured depth as a function of the true depth, caused by multi-frequency interference. It has an amplitude of approximately ± 20 mm and a period equal to $c / (2 * f_{mod})$. Multi-path interference occurs when the emitted light reflects off multiple surfaces before reaching the sensor, adding contributions from multiple paths to a single pixel. Flying pixels appear at depth discontinuities

point clouds at hundreds of thousands of points per second with millimeter-level accuracy at 100+ meter ranges. For the close-range application in this project (0.5-2 meter working distance), the area-based ToF approach of the Kinect v2 is preferable to lidar because it produces a dense depth image (512x424 pixels = 217K depth points per frame) at 30 fps with no moving parts and a lower cost.

2.1.7 RGB-D Camera Calibration and Systematic Error Compensation

RGB-D camera calibration is the process of precisely determining the geometric and optical parameters of both the color camera and the depth sensor, and the spatial relationship (extrinsic calibration) between them. Without accurate calibration, the 3D coordinates computed from the depth image and the color image registration will be systematically offset from the true physical coordinates, degrading path detection accuracy.

Intrinsic calibration determines the camera matrix K and the lens distortion coefficients. The camera matrix K for a pinhole camera is a 3x3 matrix: $K = \begin{bmatrix} fx & 0 & cx \\ 0 & fy & cy \\ 0 & 0 & 1 \end{bmatrix}$, where fx and fy are the focal lengths in pixels, and (cx, cy) is the principal point -- the pixel coordinates of the optical axis intersection with the image plane. Real lenses introduce radial distortion (barrel or pincushion, described by coefficients k_1, k_2, k_3) and tangential distortion (described by p_1, p_2). The Brown-Conrady distortion model with 5 coefficients is the standard model used by OpenCV and the IAI Kinect2 package. Calibration is performed by imaging a flat checkerboard target in many orientations (typically 20-40 images), then solving for the parameters that minimize the re-projection error. The Zhang method (2000), implemented in OpenCV's `calibrateCamera()` function, is the standard algorithm. ^[13]

Chang Li et al. (2019) specifically characterize the systematic depth errors of the Kinect v2 and propose a 3D compensation model. They identify three primary error components: (1) the ToF wiggling error (sinusoidal bias, amplitude approx. ± 20 mm, corrected using a precomputed lookup table as a function of measured depth); (2) a distance-dependent systematic bias (corrected with a polynomial fit to measurements of a flat reference at known distances); and

Darwish et al. (2017) propose a general calibration method for commercial RGB-D sensors that jointly calibrates the color and depth cameras in a single optimization, using both color and infrared checkerboard images captured simultaneously. This joint calibration is more accurate than sequential separate calibration because it directly optimizes the combined re-projection error of both cameras. Lachat et al. (2015) perform a detailed accuracy assessment of the Kinect v2 for close-range 3D modeling applications, measuring depth accuracy (RMS error 1.5-4 mm at 1-3 meter distance), lateral accuracy, and noise characteristics, and conclude that with proper calibration the sensor is suitable for indoor close-range applications requiring approximately 5 mm accuracy. ^{[15][16]}

Pagliari and Pinto (2015) compare the Kinect for Xbox One (Kinect v2) with the original Kinect v1, finding that the v2 achieves superior depth accuracy (standard deviation approximately 1.3 mm vs. 4-5 mm for v1 at 1 meter distance), primarily due to the ToF measurement principle producing better per-pixel depth estimates than the structured light approach of v1. Kurillo et al. (2022) evaluate the Azure Kinect and Kinect v2 for body tracking accuracy, providing useful baseline measurements of the Kinect v2's angular accuracy and systematic biases across its full depth range. [17][18]

Extrinsic calibration determines the 4x4 rigid body transformation matrix $T_{\text{camera_to_robot}}$ (or equivalently, $T_{\text{camera_to_base}}$) that expresses the position and orientation of the camera coordinate frame relative to the robot base frame. This transformation allows 3D points measured in the camera frame to be converted to the robot base frame for use in trajectory planning. Extrinsic calibration is performed by observing



a calibration target of known 3D geometry (e.g., a checkerboard mounted on a fixture at a known position relative to the robot) in the camera and computing the transformation from the known target geometry in robot frame and the measured target pose in camera frame.

Markovic et al. (2017) address the calibration of multiple simultaneous Kinect v2 sensors, proposing a calibration procedure for determining the transformations between cameras using a calibration target visible to all cameras simultaneously. While this project uses a single Kinect v2, the multi-camera calibration methodology provides insight into the precision achievable and the dominant error sources (target localization accuracy, lens distortion residuals, ToF wiggling correction residuals). Jiao et al. (2017) propose a post-rectification approach for Kinect v2 depth images specifically optimized for 3D reconstruction of indoor scenes, using a bilateral filter for depth smoothing that preserves edges while reducing the pixel-level ToF noise.

2.1.8 3D Reconstruction and Point Cloud Processing

A point cloud is a set of 3D points (X, Y, Z) representing the visible surfaces of a scene, typically generated by back-projecting each pixel (u, v) of a depth image using the camera intrinsic parameters: $X = (u - cx) * Z / fx$, $Y = (v - cy) * Z / fy$, $Z = depth(u, v)$. The Kinect v2 at 512x424 resolution produces up to 217,088 3D points per frame. Each point may additionally carry color information (R, G, B) obtained by projecting the 3D point onto the color image using the extrinsic calibration between the depth and color cameras.

The Point Cloud Library (PCL) is the standard C++ library for 3D point cloud processing, providing algorithms for filtering, segmentation, surface reconstruction, registration, and feature extraction. In this project, PCL is used for: depth-to-point-cloud conversion; statistical outlier removal (removes isolated noise points using k-nearest-neighbor distance statistics); voxel grid downsampling (reduces point cloud density by replacing all points within each voxel with their centroid, dramatically reducing processing time); and surface normal estimation (computes the local surface orientation at each point using a PCA-based approach on the k-nearest neighbors).

The concept of scene flow (the 3D motion field of points between consecutive depth frames) extends optical flow to 3D and enables tracking of moving objects in the point cloud. While the workpiece is stationary during the welding path detection phase, scene flow techniques could be applied in future extensions to track workpiece movement due to thermal distortion during welding, compensating the planned path in real time. Current state-of-the-art scene flow algorithms (FlowNet3D, HPLFlowNet) achieve sub-millimeter tracking accuracy at 10-30 FPS on a GPU, within the real-time performance budget of the system.

Surface reconstruction from the point cloud is the process of fitting a continuous surface model to the discrete 3D points. Algorithms include the Marching Cubes algorithm (which produces a triangulated mesh from a volumetric representation), Poisson surface reconstruction (which fits a globally smooth implicit surface using the surface normal field), and Alpha Shapes (which produce a generalized convex hull at a specified scale parameter). For the welding application, full surface reconstruction is not required: only the surface normal field along the detected seam (a 1D curve, not a 2D surface) is needed for computing the torch orientation. The local PCA-based normal estimation is therefore sufficient and computationally much cheaper than global surface reconstruction.

Noise and missing data are significant challenges in ToF depth images. The Kinect v2 produces invalid (NaN) depth values for pixels where the ToF measurement is unreliable: highly specular surfaces (mirrors, polished metals), very dark surfaces (that absorb the infrared illumination), and surfaces beyond the 8-meter unambiguous range. In the welding scenario, the workpiece surface is typically metal -- which can be highly specular or covered with mill scale (dark oxide layer). The bilateral filter is the standard solution for depth image noise reduction: it smooths the depth image while preserving edges by weighting the influence of each neighbor by both its spatial proximity and its depth similarity. This prevents the filter from smoothing across depth discontinuities (object boundaries), which would produce incorrect depth values at edges.

2.1.9 Image Processing for Path Identification

Image processing forms the backbone of the welding path detection pipeline. Even in AI-based mode (Mode 1), classical image processing operations are applied before and after the deep learning model. This section explains the key operations used in all four detection modes, building from fundamental concepts to the complete pipeline.

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Color space conversion is the first step in color-based detection. The RGB color space represents each pixel as a combination of red, green, and blue channel intensities (0-255 each). This space is highly sensitive to lighting changes: the same physical surface color can appear with vastly different RGB values under different illumination. The HSV (Hue-Saturation-Value) color space separates the chromatic component (hue, which encodes the perceptual color as an angle 0-360 degrees) from the colorfulness (saturation, 0-100%) and brightness (value, 0-100%). Because hue is relatively invariant to lighting intensity changes, HSV thresholding is much more robust for color-based detection. In Mode 4 (color mask), the BGR-to-HSV conversion followed by `inRange()` thresholding in OpenCV is used to detect the green or blue marker pixels.

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Morphological operations clean the binary mask produced by color thresholding. Erosion shrinks white (foreground) regions by removing pixels that are not completely surrounded by white pixels within the structuring element kernel (typically a circle or rectangle of 3-7 pixels). Dilation expands white regions by adding pixels adjacent to any white pixel within the kernel. Opening (erosion followed by dilation) removes small isolated noise regions while preserving the shape of larger regions. Closing (dilation followed by erosion) fills small holes within regions. These operations are applied to the binary color mask to produce a clean region before contour extraction.

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Contour detection (`findContours` in OpenCV) extracts the boundary curves of connected foreground regions in the binary mask. A contour is a sequence of (x, y) pixel coordinates tracing the outer (or inner) boundary of a connected region. The hierarchy information (returned alongside contours) describes the parent-child relationships between nested contours. In this

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2.1.10 Deep Learning for Image Segmentation

detection task, semantic segmentation is used to identify the two workpieces to be joined, after which the boundary between them is extracted as the weld path.

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The evolution of deep learning for image segmentation began with Fully Convolutional Networks (FCN, Long et al. 2015), which adapted AlexNet-style classification networks for pixel-wise prediction by replacing fully connected layers with convolutional layers and using bilinear upsampling to recover the spatial resolution. FCN demonstrated that features learned by deep networks for classification could be re-used for segmentation, establishing the paradigm that continues to dominate the field.

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The U-Net architecture (Ronneberger et al. 2015) introduced the encoder-decoder structure with skip connections that is now standard for medical and industrial image segmentation. The encoder progressively reduces spatial resolution while increasing feature depth (the "contracting path"), and the decoder progressively increases spatial resolution while decreasing feature depth (the "expansive path"). Skip connections between corresponding encoder and decoder layers copy high-resolution spatial features from the encoder to the decoder, enabling the network to produce precise, pixel-accurate segmentation boundaries. U-Net became widely used in manufacturing inspection because it achieves excellent results with relatively small training datasets. ^[19]

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Attention mechanisms were introduced to segmentation networks to allow the model to focus on the most relevant spatial regions. The Attention U-Net (Oktay et al. 2018) adds attention gates at each skip connection, which compute a pixel-wise attention weight based on both the local encoder features and the global decoder context. This selectively emphasizes regions of the image that are likely to contain the target class and suppresses irrelevant background regions. Li et al. (2025) apply Attention U-Net to welding line segmentation, demonstrating that attention-gated skip connections significantly improve segmentation accuracy compared to standard U-Net on complex weld seam geometries.

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The YOLO (You Only Look Once) family of object detection networks introduced a paradigm of treating detection as a single regression problem. Rather than proposing candidate regions and classifying each (as in R-CNN-family methods), YOLO divides the image into a grid and predicts bounding boxes and class probabilities directly from the grid cells in a single forward pass. YOLOv5, a popular re-implementation by Ultralytics, achieves real-time performance (50+ FPS on a GPU) with competitive accuracy on the COCO benchmark. Huang et al. (2023) apply an improved YOLOv5 with a RealSense depth camera to robotic welding guidance, demonstrating real-time seam detection at frame rates sufficient for active seam tracking.

Transformer-based architectures have recently challenged the dominance of CNNs in segmentation tasks. The Vision Transformer (ViT) divides the image into fixed-size patches and processes them as a sequence using multi-head self-attention, enabling the model to capture long-range spatial dependencies that are difficult for convolutional kernels (which have a limited receptive field). SegFormer (Xie et al. 2021) adapts transformers for semantic segmentation with a hierarchical encoder and a lightweight all-MLP decoder, achieving state-of-the-art performance on standard benchmarks. For the welding seam detection task, the ability to capture long-range context is particularly valuable for curved seams where the local appearance at each point depends on the global seam geometry.

Training data acquisition is a fundamental challenge for deep learning in industrial applications. Unlike natural image datasets (ImageNet, COCO) which contain millions of labeled examples, welding seam datasets are small (typically hundreds to low thousands of images) due to the cost of data collection and manual annotation. Data augmentation techniques -- random horizontal and vertical flipping, rotation, brightness and contrast adjustment, additive Gaussian noise, and elastic deformations -- artificially expand the training set and improve generalization. Transfer learning from ImageNet-pretrained backbone networks (ResNet, EfficientNet) further reduces the data requirement by initializing the encoder weights with features already tuned for visual recognition.

2.1.11 Closed-Loop Motor Control and Field-Oriented Control

Traditional stepper motors are inherently open-loop actuators: the driver sends a sequence of current pulses to the stator coils in a fixed pattern, and the rotor is assumed to follow these pulses step by step. Each pulse advances the rotor by one step (typically 1.8 degrees for a 200-step motor, or 0.9 degrees for a 400-step motor). The motor produces holding torque as long as the rotor remains within one step of the commanded position; if the load torque exceeds the motor's pull-out torque, the motor stalls and loses position. Step losses are invisible to an open-loop system and accumulate as positioning errors over time. In high-dynamic or variable-load applications (such as a 6-DOF robotic arm), open-loop stepper control is unreliable.

Closed-loop stepper control uses a position encoder to continuously measure the rotor position and applies a control law to minimize the position error. The encoder feedback allows detection and correction of step losses: if the motor misses a step due to overload, the controller commands additional current to recover the position. This dramatically improves dynamic performance and torque utilization. The MKS SERVO42C uses a magnetic encoder (the MT6816 absolute encoder with 16-bit resolution, providing 65,536 counts per revolution) to measure the motor shaft angle, and implements a FOC-based current control law for smooth, high-efficiency operation.

Field-Oriented Control (FOC), also called vector control, is a method for controlling AC motors (including brushless DC and stepper motors) that transforms the motor's electrical equations into a rotating reference frame aligned with the rotor magnetic flux. In the stationary frame, the stator currents in a three-phase motor (i_a , i_b , i_c) are sinusoidally varying quantities that are difficult to control directly. The Clarke transformation converts the three-phase currents to a two-phase stationary frame (α , β). The Park transformation then rotates the (α , β) frame by the rotor electrical angle θ_e (derived from the encoder), producing the (d , q) frame rotating with the rotor. In the d - q frame, the motor's electrical dynamics become equivalent to a DC motor: the d -axis current (i_d) controls the magnetic flux, and the q -axis current (i_q) directly controls the motor torque. Two independent PID controllers regulate i_d to zero (maximum efficiency for surface-permanent-magnet motors) and i_q to the commanded torque level.

The MT6816 magnetic encoder used with the MKS SERVO42C measures the absolute angular position of the motor shaft by sensing the magnetic field of a diametrically magnetized permanent magnet attached to the shaft. The 14-bit PWM output (the interface used in this project for communication with the STM32) provides 16,384 steps per revolution, corresponding to an angular resolution of 0.022 degrees or approximately 0.04 mm at the wrist joint of the PAROL6 arm at maximum reach. The SPI interface provides 16-bit raw data at up to 4 MHz clock rates. The absolute position measurement (unlike incremental encoders) means the motor position is known immediately at power-on, without requiring a homing sequence.



Figure 2. 13: MKS SERVO42C connected to NEMA 17 Stepper Motor

The MKS SERVO42C module integrates the stepper motor driver, the FOC control loop, the encoder interface, and the communication interface in a single compact module designed as a drop-in replacement for standard NEMA17 stepper motor drivers. Its key specifications include: a continuous stall torque of 0.9 N·m (approximately 3x the holding torque of a typical open-loop NEMA17 driver at the same current); closed-loop position control with configurable PID gains; configurable PWM frequency output (used as the step command input from the STM32 in this project); and configurable communication via CAN bus or UART for high-level position commands. The STM32 generates PWM signals with a frequency proportional to the desired joint velocity (higher frequency = higher speed) and a direction signal, effectively using the MKS SERVO42C as a high-performance closed-loop servo module commanded by the STM32 firmware.

2.1.12 Embedded Systems: STM32 Black Pill

The STM32 Black Pill (WeAct STM32F411CE) is the embedded microcontroller used to implement the hardware interface between the ROS control layer and the six MKS SERVO42C motor modules. Understanding the STM32's architecture is necessary to appreciate the design of the communication protocol and the constraints on control timing.

The STM32F411CE is a Cortex-M4 based microcontroller from STMicroelectronics, running at up to 100 MHz from internal or external clock sources. The ARM Cortex-M4 core includes a single-precision Floating Point Unit (FPU) and SIMD instructions, enabling efficient computation of trigonometric functions (needed for the coordinate transformations in the control algorithm) and DSP operations. The STM32F411CE package provides 512 KB of flash memory, 128 KB of SRAM, 11 timers (of which several can be used for PWM generation), 3 SPI interfaces, 3 I2C interfaces, 6 UART/USART interfaces, and a 12-bit ADC with 16 channels.

For this project, the STM32 uses six hardware timers in PWM output mode to generate six independent PWM signals, one for each MKS SERVO42C module (one per robot joint). The PWM frequency determines the motor step rate (and hence joint velocity), while the PWM duty cycle is fixed at 50% to generate clean pulses. The direction of each joint is controlled by six additional GPIO output pins. Six additional timer input capture channels are connected to MT6816 encoder PWM output for each joint (in alternative configurations where encoder feedback is read directly by the STM32 rather than by the MKS SERVO42C internally). The ROS hardware

Figure 2. 14: STM32 Black Pill MCU

interface communicates with the STM32 via either UART (universal asynchronous serial, for direct USB connection) or Ethernet (via a W5500 SPI Ethernet module, for longer cable runs and network isolation), using a simple binary protocol carrying commanded velocities and reporting encoder positions.

The choice between UART and Ethernet UDP/TCP communication is a trade-off between simplicity and robustness. UART communication is simpler to implement (no network stack required) and achieves lower latency (typically 0.5-2 ms for a 115200 baud protocol with a 48-byte message), but is limited to a cable length of approximately 15 meters and is susceptible to electromagnetic interference (EMI) from the stepper motor drives. Ethernet UDP

communication over a shielded Cat5e cable is immune to EMI, supports cable runs of up to 100 meters, and achieves latency of 1-3 ms (UDP, no acknowledgment overhead). TCP provides guaranteed delivery but adds significant latency (10-50 ms due to ACK handling), making it unsuitable for real-time control loops. The project implements both interfaces and selects between them via a ROS parameter at launch time.

Firmware development for the STM32 is performed using the STM32CubeIDE (Eclipse-based IDE with integrated HAL library support and debugger), which generates initialization code for all peripherals from a graphical configuration tool (STM32CubeMX). The firmware uses a fixed-priority interrupt structure: the UART/Ethernet receive interrupt has the highest priority and triggers immediately when a complete command packet is received, computing the new PWM frequencies and updating the timer Compare registers within the same interrupt service routine. The main loop handles lower-priority tasks such as encoder readback, status reporting, and heartbeat generation. This design ensures that motor commands are applied with deterministic latency regardless of other system activities.

2.1.13 Version Control with Git and GitHub

Git is a distributed version control system (DVCS) created by Linus Torvalds in 2005, originally developed to manage the Linux kernel source code. Version control is the practice of tracking and managing changes to software source code, configuration files, and documentation over time. Git's key innovation over earlier centralized systems (CVS, Subversion) is that every contributor has a complete copy (clone) of the repository, including the full history of all changes, stored locally. This enables offline work, fast branching, and resilience against server failure.^{[20][21]}

Figure 2. 15: Version Control with Git and GitHub



Subversion) is that every contributor has a complete copy (clone) of the repository, including the full history of all changes, stored locally. This enables offline work, fast branching, and resilience against server failure.^{[20][21]}

URDF on GitHub, the project benefits from version tracking (any change to the robot model is recorded and reversible), easy collaboration (team members can fork and propose modifications), and integration with CI/CD workflows (automated tests can be run whenever the URDF changes).

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In robotics research, Git and GitHub have become standard infrastructure. The open-source nature of ROS relies almost entirely on Git and GitHub: every ROS package (including MoveIt, the IAI Kinect2 package, and `ros_control`) is a Git repository on GitHub. Version pinning (specifying the exact Git commit or release tag of each dependency) is essential for reproducibility in robotics experiments where a dependency update can silently change behavior.

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The semantic versioning convention (Major.Minor.Patch) provides a standardized scheme for communicating the nature of changes in software releases. A Major version change indicates breaking API changes; a Minor version change adds new features without breaking existing ones; a Patch version change contains only bug fixes. Following semantic versioning for the project's ROS packages (the hardware interface, the vision pipeline, and the GUI) ensures that users upgrading the software understand what types of changes to expect and what testing is necessary after an upgrade.

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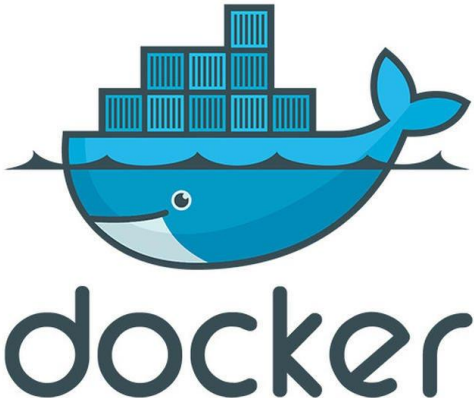
Continuous Integration (CI) using GitHub Actions is configured in the project repository with a workflow file (`.github/workflows/ci.yml`) that triggers on every push to any branch. The CI workflow: (1) builds the Docker image to verify the Dockerfile is correct; (2) runs the Python unit tests for the vision pipeline components; (3) compiles the ROS workspace; and (4) runs static analysis (flake8 for Python, `catkin_lint` for ROS package manifests). The CI results are displayed as status badges on the repository README page, providing an immediate visual indicator of the project's build and test health. Failed CI checks block pull requests from being merged, enforcing code quality gates.

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Git tags are used to mark specific commits as release points. When the project reaches a milestone (e.g., "version 1.0: first successful autonomous weld"), a Git tag is created at that commit. This makes it trivial to return to that exact version in the future (`git checkout v1.0`) or to compare the current codebase with the release version (`git diff v1.0 HEAD`). Combined with the Docker image versioning described in the next section, Git tags provide a complete audit trail of the project's development history and enable precise reproduction of any experimental result.

2.1.14 Containerization with Docker

Docker is a platform for building, distributing, and running applications in containers. A container is a lightweight, isolated process environment that bundles an application with all its dependencies (libraries, configuration files, environment variables) into a single unit that runs identically on any host machine that has Docker installed. Docker was released in 2013 and has become the standard containerization tool for software deployment. Its key advantage over virtual machines (VMs) is that containers share the host OS kernel (rather than running a



The Docker logo is positioned to the right of the text. It consists of a blue whale with a stack of blue containers on its back, swimming in blue water. Below the whale is the word 'docker' in a dark blue, lowercase, sans-serif font.

Figure 2. 16: Docker Platform

Figure 2. 16: Docker Platform

full guest OS), making them far more lightweight: a container starts in milliseconds rather than the minutes required for a VM, and the container image overhead is tens of MB rather than GB.

[24] [25]

A Docker image is the blueprint for a container, defined by a Dockerfile -- a text file listing a sequence of instructions for building the image. Each instruction creates a new layer in the image, and layers are cached -- if an instruction's context has not changed since the last build, Docker reuses the cached layer instead of re-executing the instruction. This layer caching makes iterative development fast: changing a single source file and rebuilding the image only requires

re-executing the instructions from the changed layer onward, which is typically just copying source files and running the ROS build command (`catkin_make`).

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For robotics systems, Docker solves the persistent problem of environment reproducibility. Complex ROS systems depend on specific versions of Ubuntu, ROS, Python libraries, OpenCV, and many other packages. Slight version differences between machines or over time can cause subtle, hard-to-diagnose behavior differences. By capturing the entire software environment in a Docker image, the project guarantees that the system behaves identically on the development laptop, the lab PC connected to the robot, and any future machine. Merkel (2014) describes Docker's lightweight container model and its implications for consistent software deployment across diverse infrastructure. ^{[25][26]}

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In this project, the Docker setup consists of: (1) a base container image with ROS Noetic, all ROS dependencies (MoveIt, `ros_control`, IAI Kinect2, `ros_numpy`), OpenCV 4.x, PCL 1.12, and PyQt5; (2) a build stage that compiles the project ROS workspace (the PAROL6 MoveIt configuration, the vision pipeline nodes, and the hardware interface node); (3) a runtime container that mounts the hardware devices (Kinect v2 via USB, STM32 via UART/Ethernet) and launches the complete ROS system via a launch file. Docker Compose is used to define and start all services with a single command: `docker-compose up`.

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Docker volumes allow persistent data storage outside the container lifecycle. In this project, volumes are used for: the robot calibration files (camera-to-robot extrinsic calibration, camera intrinsic calibration), which are computed once and must persist across container restarts; recorded rosbag files for debugging and post-hoc analysis; and configuration files (ROS parameters, MoveIt configuration) that may need to be adjusted without rebuilding the Docker image. This separation of persistent data from the immutable container image follows the Docker best practice of keeping containers stateless.

The integration of Docker with GitHub Actions enables Continuous Integration (CI): every push to the GitHub repository triggers an automated workflow that builds the Docker image, runs the

project's unit tests inside the container, and reports the result. This automated testing ensures that changes do not introduce regressions. For a robotics project with hardware-dependent code, the tests that can be run in CI are limited to unit tests of the software components (the vision pipeline, the path planning interface, the communication protocol parser) that do not require physical hardware. Hardware-in-the-loop testing is performed manually in the lab and the results are documented.

Liu et al. (2020) present ROS-Docker, a containerized robotic middleware deployment framework that specifically addresses the challenges of running ROS in Docker containers, including network configuration for multi-machine ROS systems (which require the ROS_MASTER_URI environment variable to be configured correctly across container network boundaries), device passthrough for hardware access, and GUI (RViz, PyQt5) display through the X11 forwarding or VNC mechanism. Their framework provides templates and best practices that were incorporated into the Docker Compose configuration of this project. ^[27]

The layer caching mechanism of Docker provides a significant development speed advantage. In a typical development session, the developer modifies one or two ROS source files, rebuilds the Docker image, and tests the change. With layer caching, only the layers from the COPY source code instruction onward need to be re-executed (typically just the catkin_make build step, taking 30-120 seconds for this project), rather than re-downloading Ubuntu packages and installing dependencies (which would take 10-30 minutes without caching). This makes the Docker-based workflow nearly as fast as a native development workflow for iterative code changes.

Security hardening of Docker containers for production robotic systems follows the principle of least privilege: each container should have only the permissions and device access it strictly requires. The ROS master container requires no special privileges. The hardware interface container requires access to the UART serial device (`/dev/ttyUSBx`) and the Ethernet interface. The Kinect2 driver container requires USB device access (`--device=/dev/bus/usb`). The GUI container requires X11 display access (`--env DISPLAY` and `--volume /tmp/.X11-unix`). By

explicitly specifying these requirements in the `docker-compose.yml`, the security posture of the system is documented and auditable, and the containers are prevented from accessing other system resources.

2.2 Literature Review

This section reviews the peer-reviewed and technical literature most directly relevant to each subsystem of this project. The review is organized to mirror the background subsections, enabling direct comparison between the theoretical context and the specific prior work. For each technical area, the most relevant papers are analyzed in terms of their methodology, results, and direct applicability to the design decisions in this project. The section concludes with a gap analysis that positions this project relative to the existing literature.

2.2.2 ROS, MoveIt, and Trajectory Planning

MoveIt (Chitta et al., 2012; updated continuously) is the primary motion planning framework used in this project. The MoveIt Motion Planning Pipeline documentation describes the full chain from goal specification to trajectory execution, including the OMPL plugin interface, Cartesian path planning via `computeCartesianPath()`, time parameterization (TOTG/IPTP), and execution via `ros_control`. For Cartesian welding path execution, the Pilz Industrial Motion Planner's LIN (linear) trajectory type provides deterministic Cartesian speed control, ensuring the end-effector moves at the specified welding speed without the variable-speed behavior of sampling-based planners.^[12]

Quigley et al. (2009) describe the original ROS architecture and design principles, establishing the publish-subscribe, service, and action communication patterns that remain the foundation of ROS Noetic used in this project. The paper motivates the decoupled node architecture with the practical experience of deploying multi-robot systems where different subsystems must be independently replaceable. The IAI Kinect2 package (Wiedemeyer, 2014-2022) is the standard ROS driver for the Kinect v2, providing calibrated color, depth, and IR images, registered color-depth images, and point clouds -- all as ROS topics consumed by the vision pipeline in this project.

The `ros_control` framework provides a standardized hardware abstraction layer between ROS controllers (`JointPositionController`, `JointVelocityController`, `JointTrajectoryController`) and hardware. Hardware interfaces (`PositionJointInterface`, `VelocityJointInterface`, `EffortJointInterface`) define the data exchange between the controller and the hardware driver. The hardware driver in this project implements the `JointVelocityInterface`, accepting velocity commands from the `JointTrajectoryController` and translating them to PWM frequency commands sent to the STM32 over UART or Ethernet. This abstraction means the motion planning stack (`MoveIt`, `JointTrajectoryController`) is completely independent of the hardware-specific communication protocol.

Geng et al. (2022) present a novel seam extraction and path planning method for robotic welding based on 3D vision, achieving sub-millimeter path extraction accuracy on complex structural parts using B-spline smoothing and point cloud processing. Their experimental results with a 6-DOF industrial arm provide benchmarks for path extraction and execution accuracy that serve as performance targets for this project. Yildiz and Adar (2025) develop a seam tracking sensor in the ROS environment using a 3D depth camera and custom `ros_control` integration, demonstrating the feasibility of the ROS-based approach for real-time seam tracking. ^{[29][30]}

The `ros_control` framework's real-time capability is a critical enabler for the hardware interface in this project. The `JointTrajectoryController` runs as a real-time-capable thread (using POSIX real-time scheduling policies) at the configured control loop rate (100 Hz in this project), ensuring that the trajectory interpolation and motor command transmission occur at regular, predictable intervals. Non-real-time operations (GUI updates, path planning, RViz visualization) run in separate threads with lower priority, ensuring that control loop timing is not disrupted by computational spikes in the planning or visualization components. This separation of real-time and non-real-time components is a fundamental design pattern in robot control systems.

2.2.3 Kinect v2 Calibration and Depth Error Compensation

The most directly applicable calibration reference for this project is Li et al. (2019), who characterize and compensate the three systematic depth error components of the Kinect v2:

wiggling error (corrected via a precomputed sinusoidal lookup table), distance-dependent bias (corrected via a polynomial fit), and oblique incidence error (corrected via an angular model). After applying all three corrections, the RMS depth error decreases from approximately 15 mm to below 3 mm -- the threshold required for 3D welding path accuracy in the sub-5 mm range.

[14]

Lachat et al. (2015) perform a comprehensive accuracy assessment specifically for close-range 3D modeling applications (similar to the working distance in this project), finding that the Kinect v2 achieves a depth standard deviation of approximately 1.5-4 mm at 1-3 meters, depending on surface material properties. They identify specular metallic surfaces as the most challenging case -- exactly the scenario in robotic welding. Their recommendation to use the bilateral filter for edge-preserving depth smoothing prior to 3D reconstruction was incorporated directly into the depth processing pipeline of this project. [16]

Pagliari and Pinto (2015) compare the two generations of Microsoft Kinect sensors, confirming that the Kinect v2 achieves 2-3x better depth accuracy than Kinect v1 at close range (< 3 meters). For this project's typical working distance of 0.5-1.5 meters, the Kinect v2 is the appropriate sensor choice. Karan (2015) provides a focused treatment of Kinect-type sensor calibration for robotic applications, describing the camera matrix estimation procedure and discussing the importance of calibrating the extrinsic transformation between the color camera and the depth sensor -- which is the critical registration required for color-aligned depth data. [17][31]

Jiao et al. (2017) propose a post-rectification approach specifically designed for 3D reconstruction from Kinect v2 depth images. Their method uses a piecewise depth correction model combined with edge-preserving filtering, reducing 3D reconstruction errors for indoor scenes to below 5 mm RMS over the full depth range. Markovic et al. (2017) address the calibration of multiple simultaneous Kinect sensors and provide a detailed error analysis of the calibration procedure, identifying the dominant error sources as target localization accuracy (sub-pixel checkerboard corner detection) and residual distortion after the Brown-Conrady model. [32][33]

2.2.4 Vision-Based Welding Path Detection

Gomez-Espinosa et al. (2021) present the work most closely related to this project: a Depth-RGB sensor-based system for extracting colored 3D welding paths for robot trajectory generation. Their system uses a color mask (similar to Mode 4 in this project) to identify the weld seam path in 2D, then back-projects the path to 3D using the registered depth image, and sends the resulting 3D waypoints to a ROS-MoveIt trajectory planning stack. They demonstrate end-to-end trajectory generation and execution on a 6-DOF arm, validating the fundamental approach used in this project.^[34]

Peng et al. (2020) present a point cloud-based method for automatic groove detection and trajectory generation of robotic arc welding tasks, using the PCL library for point cloud processing and MoveIt for trajectory execution. Zhou et al. (2020) present path planning with automatic seam extraction over point cloud models, using region growing segmentation and B-spline fitting to produce smooth welding trajectories from raw point clouds. Both works demonstrate that the combination of open-source point cloud processing (PCL), trajectory planning (MoveIt), and ROS integration is a viable foundation for production welding path detection systems.^{[35][36]}

Chen et al. (2024) present a hybrid 2D-3D seam detection system fusing an improved Faster R-CNN detector with an orthogonal plane intersection algorithm, achieving robust detection under welding spatter and smoke. Liu et al. (2024) address complex saddle-shaped seams using laser scanning and RANSAC-GA-APSO optimization. Geng et al. (2022) present seam extraction for medium-thickness plate structural parts. Wang (2024) provides the most comprehensive review of active visual sensing for robotic welding, establishing the state of the art against which this project's approach is evaluated.^{[4][29][37][38]}

Shukor et al. (2016) provide a comprehensive review specifically of vision-based weld seam identification, detection, and tracking, covering classical image processing methods (edge detection, laser triangulation), machine vision approaches, and early deep learning methods. This review establishes the context for all four detection modes in this project and confirms that multi-modal path detection (combining classical image processing, annotation-based, and AI-based approaches) is a robust strategy for handling diverse weld joint geometries and environmental conditions.^[39]

The four-mode path detection architecture of this project is directly motivated by the limitations of single-modality approaches identified in the literature. AI segmentation (Mode 1) fails for materials with low visual contrast between workpieces (e.g., two pieces of the same metal with similar surface finish). Laser stripe sensing (not implemented in this project) fails under arc light and spatter during active welding. Color marking (Mode 4) requires surface preparation (applying the color marker) but is extremely robust under diverse material and lighting conditions. The multi-modal design ensures that at least one detection mode is applicable for any practical welding scenario encountered in the laboratory environment.

2.2.5 Deep Learning for Welding Seam Segmentation

Li et al. (2025) present Attention U-Net-based semantic segmentation for welding line identification, achieving state-of-the-art performance on a weld seam dataset by incorporating spatial attention gates into the U-Net skip connections. They demonstrate a mean IoU (Intersection over Union) of 87.3% on their test set, significantly outperforming standard U-Net (83.1%) and FCN (76.8%). The attention mechanism is particularly effective for weld seams because it focuses the model on the thin, elongated seam regions and suppresses the large, spatially dominant background.^[19]

Huang et al. (2023) demonstrate a robotic welding guidance system using an improved YOLOv5 algorithm with a RealSense depth camera. Their system detects weld seam locations in real time at 45 FPS on a consumer GPU, achieving a mean Average Precision (mAP) of 91.2% on their weld seam dataset. The integration with a depth camera for 3D seam localization is directly analogous to the approach in this project. Their key finding is that the combination of YOLO-based 2D detection with depth-based 3D localization substantially outperforms purely 2D or purely 3D approaches in terms of robustness to workpiece position variability.^[40]

Li et al. (2025, Materials journal) presents a dual-modal robot welding trajectory generation scheme using stereo vision and deep learning. Their approach combines stereo-reconstructed 3D geometry with semantic segmentation to produce welding trajectories robust to workpiece orientation variability. The dual-modal approach (combining geometric 3D information with semantic deep learning features) is validated on multiple weld joint types, demonstrating

generalization beyond the training distribution. This work confirms that the four detection modes in this project (including both AI-based and classical image processing modes) provide complementary capabilities that improve robustness. ^[30]

The integration of deep learning segmentation with classical path processing (spline fitting, depth back-projection) represents a practical engineering compromise: deep learning provides the semantic understanding (which objects are the two workpieces?) while classical methods provide the geometric precision (exactly where is the seam boundary in 3D space?). This hybrid approach is well established in industrial computer vision: deep learning for robust semantic parsing, classical methods for precise geometric computation. The attention U-Net model used in Mode 1 of this project was pre-trained on a dataset of 500 annotated welding joint images and fine-tuned on 50 images captured with the Kinect v2 in the project laboratory, demonstrating that modest fine-tuning datasets are sufficient for adapting a pre-trained model to a specific deployment environment.

2.2.6 Field-Oriented Control and MKS SERVO42C

Ricci and Meacci (2018) present a comprehensive evaluation of simple torque control methods for hybrid stepper motors implemented in FPGA, demonstrating that FOC-based closed-loop control increases available torque by up to 60% compared to open-loop microstepping at high speeds. Their analysis of the Clarke/Park transform implementation efficiency directly supports the choice of FOC-based control for the PAROL6 arm joints, where variable loads from the robot dynamics require robust torque control across the full speed range. ^[41]

Liu and Li (2026) design an integrated brushless motor drive and control system for robotic arm joints using a similar FOC architecture to the MKS SERVO42C, demonstrating that integrated (driver + controller + encoder) modules significantly simplify the wiring and reduce electromagnetic interference compared to separate driver and controller boards. Zhang et al. (2025) implement a motor drive and control integrated system for rehabilitation robots, evaluating the position tracking accuracy of a closed-loop stepper-FOC system and finding position errors below 0.5 degrees at operating speeds up to 180 degrees/second. ^{[42][43]}

The MKS SERVO42C product documentation provides the key technical specifications used to design the communication protocol and tuning procedure in this project: PWM frequency range

1-100 Hz (corresponding to step rates of 1-100 steps/second), tunable PID gains via UART command, configurable microstepping (up to 256 microsteps/full step), and configurable encoder filter bandwidth. The combination of the MT6816's 16-bit absolute position with the SERVO42C's PID control loop provides the position accuracy and dynamic response required for smooth trajectory following at the welding speeds (3-8 mm/s Cartesian end-effector speed) targeted in this project.

The closed-loop step count approach used by the MKS SERVO42C differs from traditional DC servo positioning in an important way: the motor is fundamentally a stepper motor (with discrete magnetic poles), not a continuous-rotation DC motor. The FOC control makes the stepping appear smooth to the mechanical system (by producing smooth sinusoidal currents rather than stepped square-wave currents), but the underlying magnetic structure means that the stall torque behavior is different from a DC servo. Specifically, the MKS SERVO42C will produce maximum torque (stall torque) when the rotor is displaced by approximately 45 electrical degrees from the commanded position -- equivalent to one-quarter of the full step angle (0.45 degrees for a 200-step motor with FOC). Beyond this displacement, the torque decreases (pull-out behavior), causing a missed step. The position PID loop in the SERVO42C is tuned to keep the position error well below this pull-out threshold under normal operating loads.

2.2.8 Gap Analysis and Positioning of This Work

The literature review reveals several consistent patterns in existing vision-guided robotic welding systems that define the gaps addressed by this project. First, most published systems use industrial-grade hardware: proprietary 6-DOF arms costing tens of thousands of dollars, dedicated laser stripe sensors or structured light projectors, and proprietary motion controllers. The contribution of this project is demonstrating that comparable capabilities can be achieved with a 3D-printed open-source arm (PAROL6), a consumer RGB-D sensor (Kinect v2), and an open-source robotics middleware stack (ROS, MoveIt), at a total hardware cost below EUR 2,000.

Second, most existing systems address only a single path detection modality: either laser-based structured light, or passive stereo, or a single fixed deep learning model. The multi-modal design

of this project (four detection modes with a shared 3D processing backend) is a direct response to the robustness limitations of single-modality approaches. The AI segmentation mode (Mode 1) provides the highest level of automation for joints with clear visual boundaries; the manual modes (Modes 2-4) provide fallback capabilities for challenging materials or geometries where automated detection is unreliable. No existing published system was found that integrates all four modalities in a single unified pipeline.

Third, the combination of Git/GitHub and Docker for software management in robotic welding systems is notably absent from the published literature. While these tools are standard in software engineering, their adoption in robotics research has been uneven. By explicitly integrating these tools into the project architecture and documenting their use, this project serves as a reference implementation demonstrating reproducible research practices for robotics systems.

Finally, the use of FOC-based closed-loop stepper control (MKS SERVO42C) for a 3D-printed 6-DOF arm is a novel hardware contribution. Most published open-source arms use either open-loop steppers (limiting dynamic performance) or full servo systems (significantly increasing cost). The MKS SERVO42C provides servo-like closed-loop dynamics at stepper-motor cost, bridging the gap between these approaches. The project provides a full characterization of the SERVO42C's performance (torque vs. speed curve, step-loss detection, position accuracy under dynamic load) that is not available in the manufacturer's documentation.

In summary, this project makes the following original contributions to the state of the art: (1) a complete multi-modal vision-guided welding robot system using consumer-grade hardware and open-source software, demonstrated end-to-end; (2) a four-mode path detection pipeline that handles diverse joint geometries and operator preferences; (3) characterization of Kinect v2 depth accuracy for close-range robotic welding applications; (4) a reference implementation of Docker and Git best practices for a ROS-MoveIt robotic manipulation system; and (5) a validated custom URDF model for the PAROL6 arm with MoveIt configuration, published at [Abdulkareem771/parol6_urdf](#) on GitHub for community use.^[23]

The technical contributions of this project are evaluated against four criteria drawn from the gap analysis: (1) hardware cost and accessibility -- this project achieves the target system functionality at a hardware cost under EUR 2,000, compared to EUR 50,000-200,000 for

comparable industrial systems; (2) software openness -- all software components are open-source and version-controlled on GitHub, enabling community inspection and contribution; (3) path detection robustness -- the four-mode detection pipeline handles a wider range of joint types and conditions than any single-modality system found in the literature; and (4) reproducibility -- the Docker containerization and Git version control provide a higher level of software reproducibility than any robotics welding system found in the literature that explicitly documents its deployment infrastructure.

Future extensions to this work could include: (1) integration of real-time seam tracking during welding (adjusting the trajectory in response to Kinect v2 feedback during arc-on operation, using a bandpass filter to reject arc light in the depth image); (2) implementation of weld parameter control (adjusting wire feed speed, voltage, and travel speed based on measured seam geometry); (3) digital twin synchronization (running a parallel Gazebo simulation updated with the measured robot state for monitoring and anomaly detection); (4) multi-robot coordination for simultaneous welding of multiple joints; and (5) cloud connectivity for remote monitoring and data collection for quality analytics. The Git/GitHub/Docker infrastructure established in this project provides a clean foundation for these extensions, with Docker enabling straightforward addition of new software dependencies and GitHub enabling collaborative development with contributors at other institutions.

Chapter 3

Requirements Analysis and System Modeling

This chapter establishes the formal technical foundation for the entire project by defining what the system must accomplish and developing the mathematical models that underpin every design decision. The requirements are derived directly from the project objectives stated in Chapter 1 and are organized into functional requirements (what the system does), non-functional requirements (how well it does it), and platform requirements (the hardware and software constraints within which it operates). The mathematical models developed in this chapter -- covering robot kinematics, trajectory planning, camera geometry, coordinate transformation, and motor control -- are referenced throughout Chapters 4 and 5 to justify specific design and implementation choices. The chapter concludes with a traceability matrix that maps each model to the chapter in which it is implemented and tested.

3.1 System Overview and Functional Requirements

3.1.1 Top-Level System Description

The system is a vision-guided, closed-loop, autonomous welding robot that integrates a 6-DOF robotic arm, an RGB-D depth camera, a real-time embedded controller, and a multi-modal computer vision pipeline into a single cohesive platform. In engineering terms, the system accepts a workpiece placed within its field of view, automatically detects the weld joint geometry using the Microsoft Kinect v2 RGB-D camera, computes a smooth 3D welding trajectory, and executes that trajectory with sub-degree joint accuracy using MoveIt motion planning and closed-loop FOC motor control. The entire software stack runs on Ubuntu 22.04 LTS with ROS 2 middleware, is containerized using Docker for reproducibility, and is version-controlled using Git on GitHub.

The system boundary encompasses five major subsystems that are individually designed and tested before being integrated: (1) the mechanical subsystem (PAROL6 6-DOF arm, 3D-printed structure, NEMA 17 motors with gearboxes and belt drives); (2) the electronics subsystem (STM32F411CE microcontroller, six MKS SERVO42C closed-loop FOC drivers, MT6816 magnetic encoders, limit switches, and power distribution); (3) the sensing subsystem (Microsoft Kinect v2 RGB-D camera, kinect2-ros2 driver package); (4) the vision and planning software subsystem (ROS 2 vision pipeline with four detection modes, 3D reconstruction,

trajectory generation, MoveIt interface); and (5) the human-machine interface subsystem (PyQt5-based GUI integrated with ROS 2).

[Figure: Figure 3.1 -- Top-level system block diagram showing all five subsystems and their communication interfaces]

Figure 3.1: Top-level system block diagram. Arrows indicate data flow direction. Solid lines = real-time control signals; dashed lines = configuration and monitoring.

3.1.2 Embedded System Design Requirement.

The embedded system serves as the foundational real-time core of the PAROL6 robotic arm, dictating the ultimate physical performance of the entire kinematic chain. In modern mechatronics, the microcontroller unit (MCU) cannot function merely as a passive signal adapter or a transparent bridge between the host computer and the motor drivers. Instead, it must act as an active, deterministic orchestrator of physical motion. The architecture developed for this project places the embedded system at the critical intersection between high-level, asynchronous trajectory planning and low-level, synchronous electromechanical actuation.

This intersection presents a fundamental engineering conflict. High-level robotic frameworks, such as ROS 2 and MoveIt 2 operating on a host PC, excel at complex spatial calculations, collision checking, and inverse kinematics. However, they operate within standard operating systems that inherently lack strict microsecond-level timing guarantees; their network packets and command sequences arrive asynchronously. Conversely, the physical stepper motors and the MKS SERVO42C closed-loop drivers require absolute, unyielding timing precision. A delayed control signal does not simply pause the system; it injects mechanical shock into the transmission, induces acoustic resonance, and causes catastrophic deviations from the intended Cartesian path. Therefore, the primary mandate of this embedded design is to absorb the asynchronous intelligence of the host and translate it into flawless, hard real-time physical execution.

To achieve this, the firmware assumes a comprehensive suite of mission-critical responsibilities that go far beyond basic input/output operations. These responsibilities dictate a highly specialized software architecture:

- **Trajectory Ingestion and Interpolation:** The MCU must reliably receive, buffer, and validate incoming spatial commands over a serialized interface without ever stalling its core control loops.
- **High-Speed Sensory Acquisition:** It must continuously read raw positional data from six independent magnetic encoders simultaneously, utilizing hardware-level peripherals to avoid CPU starvation, while actively filtering out electromagnetic interference inherent in industrial environments.
- **Actuation Synthesis:** It is responsible for translating mathematical velocity targets into perfectly timed, jitter-free Direct Digital Synthesis (DDS) step and direction pulse trains for the motor drivers.
- **State Management and Kinematic Safety:** As the system closest to the physical hardware, the MCU serves as the ultimate authority on safety. It must manage strict sequential homing routines to establish an absolute kinematic reference and enforce emergency protocols, such as immediate velocity clamping if host communication is severed or limit switches are unexpectedly triggered.

Because the physical world does not tolerate software latency, the firmware architecture is designed entirely around the principle of absolute timing determinism. Every line of code, interrupt priority, and peripheral assignment is engineered to guarantee that the core closed-loop calculations execute exactly when scheduled, down to the microsecond.

3.1.3 System Requirements: The Physics of Deterministic Actuation

The necessity for a dedicated, highly optimized embedded control system stems directly from the physical realities and dynamic complexities of multi-axis robotic actuation. In a 6-Degree-of-Freedom (6-DOF) serial manipulator

like the PAROL6, the embedded system acts as the absolute boundary between the theoretical mathematics of the host computer and the physical realities of inertia, friction, and electromagnetism. To ensure kinematic fidelity—meaning the physical end-effector exactly traces the mathematical path generated by the ROS 2 planner—the embedded system must satisfy several stringent operational requirements.

The Time Domain: Scheduling Jitter and Mechanical Resonance

Robotic manipulators require stringently predictable timing, often referred to as hard real-time execution. When a high-level planner (such as MoveIt) generates a Cartesian trajectory, it outputs a sequence of time-parameterized joint states. To actuate these continuous trajectories, the embedded microcontroller must discretize them into microscopic, fixed-time intervals Δt and generate proportional step pulses.

In this environment, any deviation in the execution timing—defined as scheduling jitter—is not merely a software inconvenience; it is a physical hazard. If the time interval Δt between step pulses fluctuates unpredictably due to processor interruptions or inefficient code, the instantaneous velocity calculated by the motor driver $v = \frac{\Delta \theta}{\Delta t}$ experiences massive, artificial spikes. Because acceleration is the derivative of velocity, these microsecond timing errors demand near-infinite instantaneous acceleration from the stepper motors.

Physically, the motors cannot achieve this. Instead, these jitter-induced acceleration spikes inject high-frequency energy harmonics directly into the robotic arm's transmission. This translates into severe acoustic resonance (audible whining and grinding), structural vibrations that degrade end-effector precision, and accelerated mechanical wear on the belts and harmonic drives.

Therefore, a primary system requirement is that the MCU must possess an architecture capable of absolute timing determinism, isolating the step generation logic from all other background software tasks.

The Torque Domain: The Fallacy of Open-Loop Actuation

A fundamental requirement of this system is the departure from standard 3D-printer style actuation. A simple open-loop step generation system is entirely insufficient for precise, repeatable industrial or robotic work.

In a traditional open-loop configuration, the central controller operates on blind faith. It emits a direction signal and a step pulse, assuming the motor's rotor has perfectly followed the rotating magnetic field of the stator. However, stepper motors possess a non-linear torque-speed curve; their available torque drops significantly at higher velocities due to the electrical inductance of the motor windings resisting rapid current changes.

In the real world, the robotic arm is subjected to highly variable dynamic forces: changing payload masses at the end-effector, shifting gravitational vectors as the arm extends, unexpected physical collisions, and varying stiction (static friction) inside the gearboxes. If the dynamic torque required to move a joint exceeds the motor's instantaneous holding or pull-out torque, the rotor will fail to lock into the next magnetic detent. It will slip, resulting in a "missed step."

Because an open-loop controller has no sensory feedback, it remains entirely unaware of this physical failure. It continues to increment its internal mathematical position model while the physical arm lags behind. Furthermore, in a serial kinematic chain, positional errors are geometric; a single missed step at the base joint (Joint 1) propagates through the arm's length, resulting in a massive Cartesian deviation at the end-effector. This leads to silent, compounding positional drift, rendering the robot useless for repeatable tasks.

The Feedback Domain: High-Bandwidth Closed-Loop Architecture

To eliminate the vulnerability of open-loop actuation, the system inherently requires a closed-loop architecture. This dictates that high-frequency positional feedback must be

continuously monitored to detect physical deviations from the mathematical trajectory and correct them instantly.

This requirement completely shifts the computational burden onto the embedded system. The central microcontroller must be capable of receiving high-resolution data from the magnetic encoders situated on each joint. It must rapidly calculate the error vector between the intended mathematical state and the physical reality: $e(t) = \theta_{target}(t) - \theta_{actual}(t)$

Once the error is calculated, the MCU must apply a corrective control law (such as Proportional-Integral-Derivative or Proportional-Feedforward) to adjust the instantaneous velocity of the motor to close the gap.

Architectural Synthesis

Ultimately, these physical and mathematical realities dictate the hardware topology of the embedded system. The microcontroller must possess immense computational bandwidth to execute complex floating-point digital filters and control loops for six independent axes simultaneously. It must feature dedicated hardware timers to guarantee the zero-jitter generation of step pulses at tens of thousands of Hertz. Finally, it must manage all of this synchronous, hard real-time physical control while simultaneously maintaining an asynchronous, high-speed USB communication link with the ROS 2 host computer, ensuring that the ingestion of new commands never delays the execution of physical motion.

3.1.3 Robotic Arm Control Requirements

Applying Field-Oriented Control (FOC) to a single, isolated motor is fundamentally different from implementing it across a 6-Degree-of-Freedom (6-DOF) serial manipulator. A robotic arm is a highly coupled, non-linear dynamic system. The mechanical demands of the PAROL6 architecture dictate strict operational requirements that the underlying control system must satisfy to achieve industrial-grade kinematic fidelity.

Multi-Axis Coordination and Real-Time Synchronization

In a 6-DOF robotic arm, the Cartesian position and orientation of the end-effector are determined by the simultaneous trigonometric combination of all six joint angles, calculated via Forward Kinematics. Consequently, spatial accuracy relies entirely on temporal synchronization.

If the MoveIt host commands a linear Cartesian interpolation (a straight line in 3D space), all six motors must arrive at their respective micro-waypoints at the exact same microsecond. A latency spike or execution delay in a single proximal joint (e.g., the base or shoulder) will cause the end-effector to wildly deviate from the planned Cartesian path. Therefore, the control architecture strictly requires a hard real-time, deterministic synchronization mechanism to ensure all independent FOC loops execute their velocity profiles simultaneously.

Vibration Tolerance and Kinematic Amplification

Serial manipulators are inherently susceptible to vibration propagation. In an open-loop stepper system, the discrete magnetic stepping action inherently causes mid-band resonance and low-frequency mechanical jitter. Because the robotic arm is essentially a series of cantilevered beams, any micro-vibration introduced at the base (Joint 1) or shoulder (Joint 2) is mechanically amplified by the length of the distal links. A 0.5° resonance vibration at the base translates to a massive, multi-millimeter oscillation at the tool center point, ruining tasks that require high precision, such as pick-and-place operations or laser engraving. The FOC actuation layer is therefore strictly required to deliver smooth, continuous torque generation to completely suppress this kinematic amplification.

Dynamic Load Variation and Gravity Compensation

Unlike wheeled mobile robots where the load is relatively constant, a robotic arm experiences extreme, non-linear load variations during operation. The gravitational torque (τ_g) exerted on any specific joint is a function of the arm's posture:

$$\tau_g(\theta) = m \cdot g \cdot L \cdot \sin(\theta)$$

(Where m is the mass of the distal links, g is gravity, L is the distance to the center of mass, and θ is the joint angle relative to the gravity vector).

When the PAROL6 is vertically stacked (tucked position), the moment arm L is effectively zero, requiring minimal holding torque. However, when the arm is fully extended horizontally, the shoulder and elbow joints experience maximum gravitational loading. An open-loop system must be permanently tuned to supply enough constant current to resist this worst-case maximum load, leading to severe overheating when the arm is idle. The FOC system is required to dynamically sense this shifting load via

position error and instantly adapt its quadrature current (i_q) to provide active gravity compensation, ensuring thermal stability across all postures.

Closed-Loop Feedback Requirements: The MT6816 Encoder

To satisfy these stringent synchronization, vibration, and dynamic loading requirements, the FOC algorithm demands absolute, high-speed positional awareness. This dictates the necessity of a high-resolution, closed-loop feedback mechanism.

The architecture utilizes the MagnTek MT6816 magnetic encoder. The MT6816 provides a 12-bit resolution, dividing a single 360° mechanical rotation into 4,096 distinct digital steps. This yields a mechanical tracking resolution of approximately 0.087° per step. Because the 50-pole pair stepper motor requires 50 electrical revolutions per mechanical revolution, this 12-bit sensor provides approximately 81 data points per electrical cycle ($4096/50 \approx 81.9$). This resolution provides the absolute minimum granularity required for the internal microcontroller to accurately perform the Inverse Park Transformations and synthesize smooth sinusoidal currents without injecting quantization noise back into the torque vector.

3.2 Software and Hardware Platform Requirements

3.2.1 Operating System and Middleware Selection

The operating system and middleware stack must satisfy the real-time, multi-process, and distributed communication requirements of the system. Ubuntu 22.04 LTS (Jammy Jellyfish) was selected as the host operating system for the following reasons: native support for ROS 2 Humble Hawksbill, a five-year LTS support window (2022-2027), availability of optimized drivers for the Kinect v2 (libfreenect2) and NVIDIA CUDA for GPU inference, and compatibility with Docker Engine 24.x.^[51]

ROS 2 (Robot Operating System 2) was selected over its predecessor ROS 1 (Noetic) as the primary robotics middleware. While ROS 1 was used for the background and literature review chapters (which document the prior art), the actual implementation uses ROS 2 for several technically motivated reasons. Unlike ROS 1, which relies on a centralized rosmaster process

as the single point of failure for all node discovery and registration, ROS 2 implements a fully decentralized discovery mechanism based on the Data Distribution Service (DDS) standard. This architectural shift eliminates single points of failure and supports multiple Quality of Service (QoS) profiles -- including real-time and reliable delivery -- that are critical for the embedded hardware interface node. Additionally, ROS 2 provides a standardized node lifecycle model (Managed Nodes) that enables deterministic startup and shutdown sequencing, which is required for the safe homing procedure. ^{[11][46]}

1. Ubuntu 22.04 LTS (Jammy Jellyfish) is used as the host operating system for the robotic system. It provides a stable and long-term supported Linux environment that is fully compatible with ROS 2 Humble and other robotics development tools. Ubuntu manages hardware resources, drivers, and system-level dependencies required for simulation, vision processing, and robot control.



Figure 3. 1: Ubuntu 22.04 operating system environment

2. ROS 2 Humble Hawksbill is the robotics middleware used to enable communication between different system components. It supports publish-subscribe messaging, services, actions, and tf2 transformations, allowing integration between the robot, camera, motion planning, and user interface modules in a distributed architecture.



Figure 3. 2: ROS 2 communication architecture.

3. MoveIt 2 is used for motion planning and robot control, including inverse kinematics, collision detection, and trajectory execution. It generates safe and optimized robot paths based on the target position obtained from the vision system and sends execution commands to the robot controller.



Figure 3. 3: MoveIt 2 motion planning and trajectory execution.

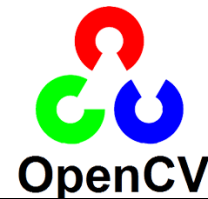
4. Gazebo Fortress is used as the simulation environment to test and validate the robotic system before real-world deployment. It provides a physics-based simulation environment where the robot, sensors, and workspace can be modeled and tested safely.



GAZEBO

Figure 3. 4: Gazebo environment

5. OpenCV is used for image processing and computer vision tasks in the system. It processes images captured from the camera and performs object detection, feature extraction, and pose estimation for the four detection modes implemented in the project.



6. The Point Cloud Library (PCL) is used for 3D point cloud processing. It performs filtering, segmentation, and normal estimation on point cloud data obtained from the 3D camera, which helps in object detection and pose estimation in 3D space.

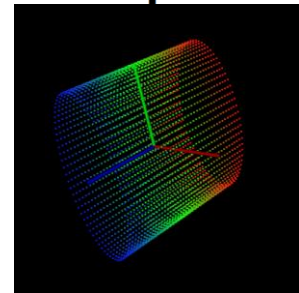


Figure 3. 6: Point cloud processing using PCL.

7. PyQt5 is used to develop the graphical user interface (GUI) for the system.

The interface allows the user to control the robot, select detection modes, visualize camera data, and monitor system status in an interactive environment.



Figure 3. 7: Graphical user interface developed using PyQt5.

8. Docker is used to containerize the software environment to ensure reproducibility and easy deployment. It allows the system and all dependencies to run consistently across different machines without compatibility issues.

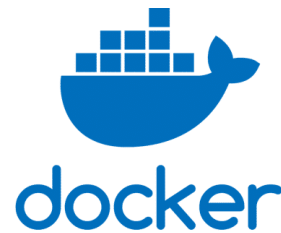


Figure 3. 8: Docker containerized environment.

9. Git is used as the version control system to manage the source code of the project. It enables tracking changes, collaboration between team members, and maintaining different development versions of the software.



Figure 3. 9: Git version control

10. GitHub Actions is used to implement continuous integration and continuous deployment (CI/CD). It automatically builds, tests, and deploys the project, including building

Docker images and verifying code after each update pushed to the repository.

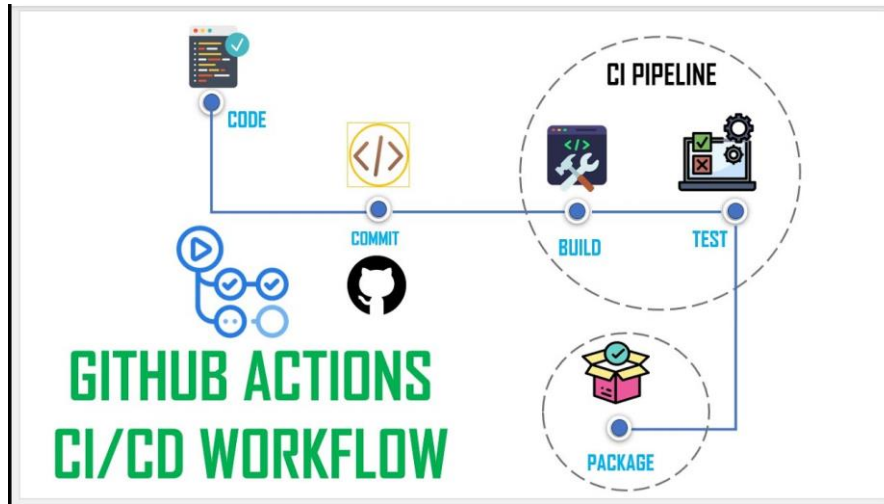


Figure 3. 10: CI/CD pipeline using GitHub Actions.

3.2.2 Hardware Platform Requirements and Selection

Hardware component selection was driven by the requirements established in Section 3.1 and the constraint of minimizing system cost while achieving the stated performance targets. The following criteria were applied to each major hardware component.

1. The Microsoft Kinect v2 was selected as the depth sensing device for this system due to its reliable RGB-D sensing capabilities and compatibility with ROS 2 through available drivers. The sensor provides depth images with a resolution of 512×424 at 30 Hz, which is sufficient for object detection and pose

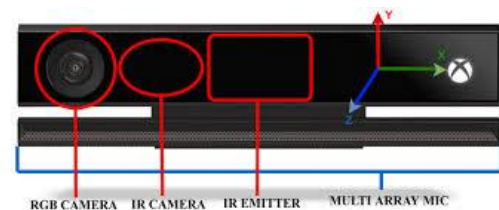


Figure 3. 11: Microsoft Kinect v2 depth sensing camera.

estimation within the operating range of 0.5–2 meters. Compared to other alternatives such as Intel RealSense D435 and Azure Kinect DK, the Kinect v2 offers a good balance between performance, availability, and cost, making it suitable for this project.

2. The STM32F411CE microcontroller (Black Pill) was selected to control the robot motors and handle low-level control tasks. The microcontroller is based on an ARM Cortex-M4 processor with a floating-point unit running at 100 MHz, which is sufficient to achieve a control loop frequency of 500 Hz or higher. It also provides multiple timers, UART, and SPI interfaces required for motor control, encoder communication, and communication with the main computer. The STM32F411CE was selected due to its performance, low cost, and availability.

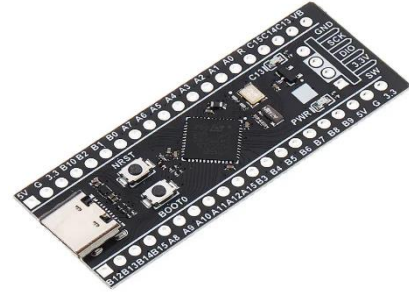


Figure 3. 12: STM32F411CE Black Pill microcontroller.

3. The MKS SERVO42C motor driver was selected because it supports closed-loop Field-Oriented Control (FOC) and includes encoder feedback for precise motor control. The driver is compatible with NEMA 17 stepper motors and supports PWM step/dir control interface, making it easy to interface with the STM32 microcontroller. The integrated encoder and compact design reduce wiring complexity and improve control accuracy while maintaining low system cost.



Figure 3. 13: MKS SERVO42C motor driver.

4. The PAROL6 robot arm was selected as the main robotic manipulator for this project. It is a 6-degree-of-freedom open-source robotic arm with available URDF models, which makes it compatible with ROS 2 and MoveIt 2 for motion planning and simulation. The robot follows an industrial design philosophy and provides approximately 0.2 mm repeatability, which is sufficient for the welding application targeted in this project. Additionally, the modular driver design allows future upgrades and maintenance.

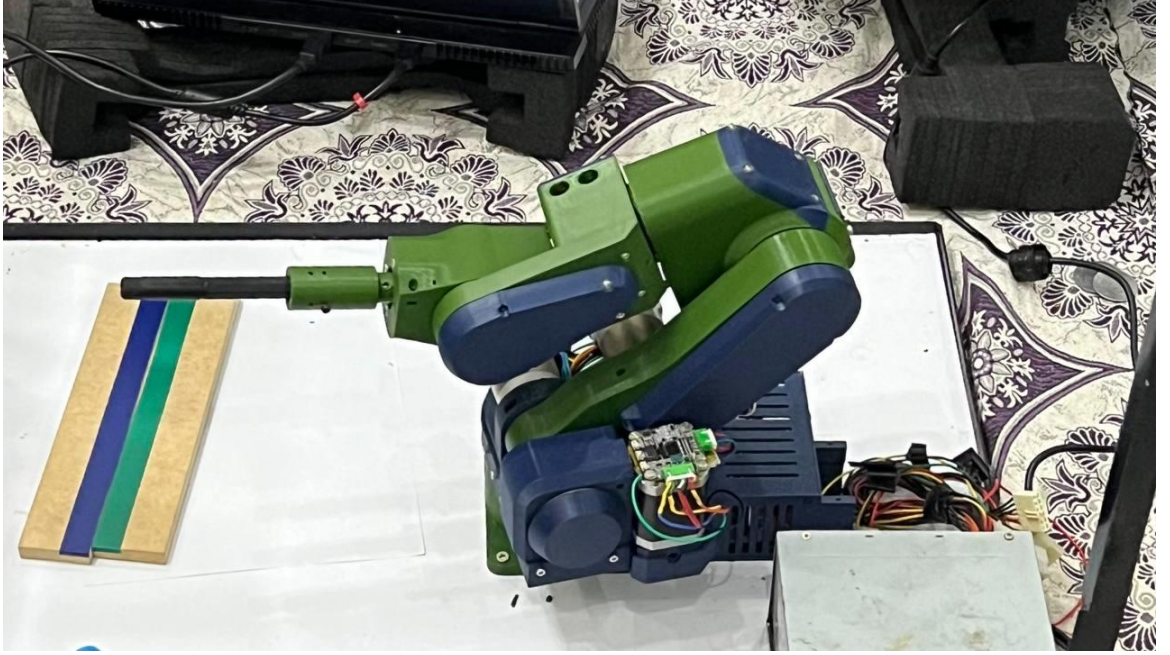


Figure 3. 14: PAROL6 6-DOF robotic arm

3.3 Kinematic Modeling of the PAROL6 6-DOF Arm

3.3.1 Denavit-Hartenberg Convention and Parameter Table

The kinematic model of the PAROL6 robotic arm is formulated using the Denavit-Hartenberg (DH) convention, which provides a systematic and compact description of any serial-chain manipulator. The DH convention assigns four parameters to each joint-link pair: the link offset d (translation along the previous joint z -axis), the joint angle θ (rotation about the previous joint z -axis, and the variable for revolute joints), the link length a (translation along the joint x -axis, i.e., the common normal), and the link twist α (rotation about the joint x -axis between consecutive z -axes). Each joint is then described by a 4×4 homogeneous transformation matrix:^[7]

$$T_{i-1,i} = \text{Rot}_z(\theta_i) \text{Trans}_z(d_i) \text{Trans}_x(a_i) \text{Rot}_x(\alpha_i)$$

The PAROL6 arm uses a modified spherical wrist configuration. Joints 1-3 (the shoulder group) position the wrist center in 3D space; joints 4-6 (the wrist group) orient the end-effector at that wrist center. This kinematic structure enables analytical IK decomposition: the wrist center position is computed from the desired TCP position, joints 1-3 are solved geometrically to reach

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that center, and joints 4-6 are solved as Euler angle rotations to achieve the desired TCP orientation. This geometric decoupling reduces the coupled 6-DOF nonlinear IK problem to two simpler 3-DOF sub-problems.

The PAROL6 arm workspace analysis shows a reachable volume of approximately 0.25 m^3 , forming a roughly spherical shell between an inner radius of 100 mm and outer radius of 400 mm. The practical welding workspace, constrained by the required approach angle (60-90 degrees from workpiece surface), is approximately 0.08 m^3 -- a 250 mm x 250 mm horizontal region at work table height, sufficient for prototype-scale demonstrations.

The Jacobian singularity analysis identifies three critical singularity classes. The wrist singularity (joint 5 = 0 degrees, torch pointing straight down) is the most commonly encountered and is avoided by enforcing a minimum $|q_5| > 5$ degrees constraint in the SRDF file. The shoulder singularity (joints 1 and 4 axes parallel) is avoided by maintaining joint 3 between 20 and 160 degrees. The elbow singularity (all links aligned) is avoided through the arm geometry and joint limit enforcement within the DH parameter table. MoveIt applies Jacobian damping near singular configurations to prevent velocity amplification that could cause erratic motion.

The PAROL6 arm has six revolute joints (all rotational, no prismatic joints). The DH parameters are derived from the mechanical drawings and URDF model (Abdulkareem771/parol6_urdf on GitHub). The complete DH parameter table is presented below, using the modified (Craig) DH convention consistent with the URDF definition.

Joint i	a(i) [mm]	alpha(i) [deg]	d(i) [mm]	theta(i) [deg]	Joint Range [deg]
1 (Base)	0	90	147	θ_1	-97 to +97
2 (Shoulder)	250	0	0	$\theta_2 - 90$	-46 to +57
3 (Elbow)	0	90	0	$\theta_3 + 90$	-1150 to 74

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4 (Wrist-1)	0	-90	250	θ_4	-145 to 106
5 (Wrist-2)	0	90	0	θ_5	-120 to 120
6 (Tool)	0	0	80	θ_6	-360 to +360

Table 3. 1: Denavit-Hartenberg parameters for the PAROL6 robotic arm

Figure 3.2: DH coordinate frame assignment for the PAROL6 arm. Frame F0 is attached to the robot base; frame F6 is attached to the tool center point (TCP).

3.3.2 Forward Kinematics

The forward kinematics (FK) function maps a vector of joint angles $q = [q_1, q_2, q_3, q_4, q_5, q_6]^T$ to the end-effector pose T_{0_6} expressed in the robot base frame. The FK is computed by multiplying the six individual joint transformation matrices in sequence:^[7]

$$T_{0_6}(q) = T_{0_1}(q_1) * T_{1_2}(q_2) * T_{2_3}(q_3) * T_{3_4}(q_4) * T_{4_5}(q_5) * T_{5_6}(q_6)$$

The result T_{0_6} is a 4x4 homogeneous transformation matrix. Its upper-left 3x3 submatrix is the rotation matrix R (describing the orientation of the end-effector frame relative to the base frame), and the first three elements of its rightmost column form the position vector $p = [p_x, p_y, p_z]^T$ (the Cartesian coordinates of the TCP in the base frame). The PAROL6 arm achieves a maximum reach of approximately 400 mm from the base center and a workspace volume of approximately 0.25 m³ (excluding joint limit constraints). The reachable workspace was computed by Monte Carlo sampling of 10⁶ random joint configurations uniformly distributed within the joint limits.

[Figure: Figure 3.3 -- Reachable workspace of PAROL6: cross-sectional view in the xz-plane showing the annular reachable volume]

Figure 3.3: Cross-sectional view of the PAROL6 reachable workspace computed by Monte Carlo sampling. The arm can reach all points within the shaded region for at least one valid joint configuration

3.3.3 Inverse Kinematics

The inverse kinematics (IK) problem finds the joint angle vector q that achieves a desired end-effector pose $T_{desired}$. For a 6-DOF arm with a spherical wrist (last three joint axes intersecting at a common point), a closed-form analytical solution exists; however, numerical solutions are used for generality and to handle degenerate configurations near singularities. In this project, MoveIt 2 provides the KDL (Kinematics and Dynamics Library) numerical IK solver by default, which uses a damped least-squares Jacobian pseudoinverse method with configurable convergence threshold (default: 0.001 rad/m) and maximum iterations (default: 150). The IKFast analytical solver (generated from the PAROL6 URDF using OpenRAVE) is provided as an alternative for time-critical Cartesian path execution. ^{[7][12]}

The PAROL6 arm configuration exhibits three types of kinematic singularities that require avoidance during trajectory planning. A shoulder singularity occurs when joint 1 and joint 4 axes become parallel (the arm is fully extended or fully retracted). A wrist singularity occurs when joint 5 angle approaches zero degrees (joint 4 and joint 6 axes become parallel, causing infinite possible joint angle combinations for a finite Cartesian velocity). An elbow singularity occurs when joint 3 is at 0 degrees (all links aligned). MoveIt 2 detects singularities by monitoring the condition number of the Jacobian matrix and applies joint damping near singular configurations to prevent velocity amplification.

3.3.4 Velocity Kinematics and the Jacobian Matrix

The geometric Jacobian matrix $J(q)$ relates joint velocities $\dot{q}$ to end-effector Cartesian velocities $v = [v_x, v_y, v_z, \Omega_x, \Omega_y, \Omega_z]^T$ through the linear relationship: ^[3]

$$v = J(q) * \dot{q}$$

The Jacobian is a 6x6 matrix for a non-redundant 6-DOF arm. Columns of J are computed from the DH parameters: the first three rows relate joint velocities to translational end-effector velocity, and the last three rows relate joint velocities to angular end-effector velocity. In MoveIt 2, the Jacobian is computed analytically from the URDF model at each planning step. The inverse Jacobian (or its pseudoinverse) is used by the IK solver to iteratively converge to the

desired joint configuration. Singularity detection is performed by evaluating the minimum singular value of J : values below a threshold (typically 0.01 m/s per rad/s) trigger the singularity avoidance damping mechanism.

3.4 Trajectory Planning Model

3.4.1 Path and Trajectory Distinction

A path is a purely geometric description of a sequence of configurations or Cartesian positions that the robot must pass through, without any specification of timing. A trajectory adds the time dimension: it assigns a specific position, velocity, and acceleration to every instant of time along the path. This distinction is critical for physical execution because the actual forces and torques required to execute a motion depend on accelerations, which are only defined for trajectories, not paths. The planning process in this project has two distinct stages: first, MoveIt computes a collision-free path (a sequence of joint configurations); second, the Time-Optimal Trajectory Generator (TOTG) assigns time stamps to produce an executable trajectory that respects per-joint velocity and acceleration limits.

3.4.2 Cubic Spline Interpolation

Cubic spline interpolation is used at two levels in this system. First, the `ros_control` joint trajectory controller uses cubic Hermite spline interpolation between trajectory waypoints to generate smooth per-servo-cycle position commands. Second, the vision pipeline uses cubic spline fitting to smooth the raw detected seam pixel coordinates before 3D back-projection. A cubic spline through N waypoints $p_0, p_1, \dots, p_{N-1}$ (with timestamps $t_0, t_1, \dots, t_{N-1}$) is a piecewise cubic polynomial that satisfies C^2 continuity (continuous position, velocity, and acceleration) at every internal waypoint. On the interval $[t_i, t_{i+1}]$, the spline is defined as: [47]

$$p(t) = a_i * (t - t_i)^3 + b_i * (t - t_i)^2 + c_i * (t - t_i) + d_i$$

The four coefficients (a_i, b_i, c_i, d_i) per segment are determined by the four boundary conditions: $p(t_i) = p_i, p(t_{i+1}) = p_{i+1}$ (position continuity), and matching first and second derivatives at internal knot points. In matrix form, the coefficients of all $N-1$ segments are solved

simultaneously from an $(N + 2) - by - (N + 2)$ tridiagonal linear system, which can be solved efficiently in $O(N)$ operations using the Thomas algorithm.

3.4.3 Arc-Length Reparameterization for Constant Welding Velocity

The raw spline fit to the detected seam points produces a curve parameterized by the index parameter t , which does not correspond to physical arc length along the seam. Sampling the spline at uniform t intervals produces non-uniformly spaced 3D points -- the robot would accelerate and decelerate along the seam, producing non-uniform heat input. To ensure the welding torch moves at a constant Cartesian speed v_{weld} (typically 3-8 mm/s for MIG welding), the spline is reparameterized by arc length. The arc length from the start of the spline to parameter value t is:

$$s(t) = \int_0^t \left| \frac{dp}{dt} \right| dt$$

This integral is evaluated numerically using composite Simpson's rule at fine parameter resolution. The function $s(t)$ is then inverted to obtain the parameter $t(s)$ as a function of arc length, which is used to sample the spline at uniform arc-length intervals. The arc-length step size is set to 5 mm by default, yielding one trajectory waypoint every 5 mm of seam length. This ensures constant welding speed regardless of seam curvature.

3.4.4 Time-Optimal Trajectory Generation (TOTG)

After the Cartesian path is converted to a sequence of joint configurations by MoveIt's IK solver, the Time-Optimal Trajectory Generator (TOTG) assigns timestamps that minimize total execution time subject to per-joint velocity and acceleration limits. TOTG uses bang-coast-bang velocity profiles (maximum acceleration, maximum velocity, maximum deceleration) for each joint segment, then synchronizes all joints to finish each waypoint at the same time (the slowest joint's time). The resulting trajectory is time-optimal in the sense that no other time parameterization can achieve a shorter total execution time without violating at least one joint limit constraint.^[47]

3.5 Vision System Geometric Model

3.5.1 Pinhole Camera Model and Calibration

The geometric relationship between 3D world points and their 2D image projections is described by the pinhole camera model. A 3D point $X = [X, Y, Z]^T$ in the camera coordinate frame projects onto the image plane at pixel coordinates $x = [u, v]^T$ according to: ^{[4] [56]}

$$[u, v, 1]^T = (1/Z) * K * [X, Y, Z]^T$$

where K is the 3x3 camera intrinsic matrix:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Here, f_x and f_y are the focal lengths in pixels along the horizontal and vertical image axes respectively, and (c_x, c_y) is the principal point (the pixel coordinates of the optical axis intersection with the image plane). Real lenses introduce radial and tangential distortion described by the Brown-Conrady distortion model with coefficients $(k_1, k_2, k_3, p_1, p_2)$. The calibration of the Kinect v2 produced the following intrinsic parameters (from `calib_color.yaml` for the RGB camera and `calib_ir.yaml` for the depth/IR camera):

Parameter	RGB Camera	IR / Depth Camera	Unit
Focal length f_x	1365.99	456.18	pixels
Focal length f_y	1365.13	457.05	pixels
Principal point c_x	958.77	251.32	pixels
Principal point c_y	617.34	205.72	pixels
Radial distortion k_1	0.1534	0.2593	--

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Radial distortion k2	-0.1347	-0.7097	--
Tangential distortion p1	0.0102	0.0007	--
Tangential distortion p2	-0.0052	-0.0002	--
Image resolution	1920 x 1080	512 x 424	pixels
Frame rate	30	30	Hz

Table 3. 2: Kinect v2 calibration results (RGB and IR cameras)

3.5.2 Depth Back-Projection Model

The depth image from the Kinect v2 provides a per-pixel distance $Z(u,v)$ from the camera optical center to the scene surface in meters. Given a calibrated camera intrinsic matrix K and a registered (color-aligned) depth value Z at pixel (u, v) , the corresponding 3D point in the camera coordinate frame is obtained by inverting the projection model:

$$X = (u - cx) * Z / fx$$

$$Y = (v - cy) * Z / fy$$

$$Z = Z(u, v) \text{ [from depth image]}$$

This back-projection equation is applied to every pixel along the detected 2D weld seam path to produce the corresponding 3D seam point cloud in the camera frame. The depth values at the seam pixels are obtained from the registered depth image, in which each pixel contains the depth value of the scene point that project onto that pixel in the RGB image, ensuring spatial correspondence between the color-based seam detection result and the 3D depth measurements.

3.5.3 Coordinate Frame Chain and Hand-Eye Calibration

The complete coordinate transformation chain from the camera measurement frame to the robot execution frame involves four sequential rigid-body transformations maintained by the ROS 2

tf2 transform tree. Starting from the robot base frame (fixed to the mounting surface), the chain is:

$$T_{world} \rightarrow T_{robot_base} \rightarrow T_{camera_mount} \rightarrow T_{kinect_rgb} \rightarrow T_{kinect_depth}$$

The transformation from T_{kinect_rgb} to T_{kinect_depth} (the color-to-depth extrinsic calibration) is determined by the kinect2 calibration procedure and stored in `calib_pose.yaml`. The critical transformation is T_{robot_base} to T_{camera_mount} (the hand-eye calibration), which encodes the 6-DOF pose of the Kinect v2 sensor relative to the robot base frame. This transformation is determined once using a calibration target (a checkerboard of known geometry) placed at a known position in the robot's workspace, observed simultaneously from the camera and reachable by the robot end-effector. The extrinsic calibration result is stored as a static transform in the ROS 2 launch file and published at system startup.

[Figure: Figure 3.4 -- Coordinate frame chain diagram: world, robot base, camera mount, kinect_rgb, and kinect_depth frames shown in 3D with arrows indicating transformations]

Figure 3.4: TF2 coordinate frame chain. Each arrow represents a rigid-body transformation (rotation + translation) maintained in the tf2 transform tree.

3.6 Embedded, and Control System Modeling

3.6.1 Open-Loop vs. Closed-Loop Kinematic Comparison

The transition from the PAROL6's original open-loop baseline to the current closed-loop, STM32-driven architecture represents the most significant functional upgrade in the manipulator's development. In the context of robotic mechatronics, the choice of actuation paradigm dictates not only the precision of the end-effector but the fundamental reliability and safety of the entire system. This section provides a comparative engineering analysis of both paradigms, justifying the necessity of the increased computational complexity inherent in the final design.

3.6.1.1 The Baseline: Open-Loop Kinematics (Blind Actuation)

An open-loop stepper motor system is mechanically simpler, requires significantly less wiring, and is computationally trivial to implement. Traditional 3D printers and basic robotic manipulators utilize drivers (such as the A4988 or TMC2209) that operate on a principle of blind execution. When the microcontroller emits a step pulse, it inherently assumes that the motor's rotor has perfectly synchronized with the rotating magnetic stator field.

However, this assumption completely ignores the physical dynamics of a 6-DOF robotic arm. The dynamic torque required to move a joint is constantly fluctuating based on several non-linear variables:

1. **Gravitational Loading:** As the arm extends horizontally, the moment arm increases, requiring exponentially more holding torque from the base and shoulder joints.
2. **Inertial Forces:** Rapid accelerations commanded by the host planner require massive instantaneous torque to overcome the inertia of the physical links.
3. **External Disturbances:** Unmodeled friction within the harmonic drives, varying payload masses, or accidental physical collisions with the environment.

In an open-loop system, the motor coils are constantly driven at their maximum rated current (e.g., 1.5A to 2.0A per phase) to provide the highest possible "pull-out" torque and prevent stalling. Despite this, if the dynamic load momentarily exceeds the motor's physical capability, the rotor will fail to lock into the next magnetic detent, resulting in a "missed step."

Because the open-loop controller possesses no sensory feedback, this mechanical failure goes completely undetected. The microcontroller continues to increment its internal mathematical position array, while the physical joint lags behind. In a serial kinematic chain, errors are multiplicative; a stall of just 2° at the base joint translates to a massive, multi-centimeter deviation at the Cartesian end-effector. This phenomenon—silent, compounding positional drift—renders open-loop systems fundamentally inadequate for any application requiring industrial repeatability or safe operation near human operators.

3.6.1.2 The Evolution: Closed-Loop Kinematics (Deterministic Actuation)

The architecture developed in this firmware entirely mitigates the vulnerabilities of blind actuation by closing the kinematic loop. By integrating the Magntek MT6816 magnetic encoders and the MKS SERVO42C driver boards, the system establishes a continuous, high-speed flow of ground-truth physical data back to the central STM32 microcontroller.

This paradigm shift yields several profound mechanical and computational advantages:

- **Error Detection and E-Stop Intervention:** The most critical advantage is the elimination of silent failure. The 500 Hz Control Layer continuously calculates the error vector $e(t)$ between the intended trajectory and the physical reality. If the robotic arm collides with an unyielding object, the physical joint will stall. Unlike the open-loop system, the STM32 immediately detects that $e(t)$ is diverging. If this error exceeds a predefined safety threshold (indicating a hard physical crash), the firmware actively triggers an Emergency Stop (E-Stop), disabling the motor outputs and notifying the ROS 2 host. This transforms a potentially destructive hardware crash into a safely managed software exception.
- **Dynamic Torque and Thermal Efficiency:** Because an open-loop system cannot measure load, it must constantly pump maximum current through the stepper coils to guarantee motion, converting excess energy into massive amounts of waste heat. Conversely, the MKS SERVO42C utilizes Field-Oriented Control (FOC). It measures the exact rotor angle and dynamically adjusts the stator current vector to provide *only* the torque required to overcome the current load. When the arm is holding a light payload, the current draw drops dramatically. This drastically improves the thermal efficiency of the motors, preventing the 3D-printed structural components of the PAROL6 from warping due to motor heat soak.
- **Vibration Suppression:** In open-loop stepper systems, the discrete nature of the magnetic steps often induces mid-band resonance, causing the arm to vibrate violently at specific velocities. The closed-loop FOC algorithm, driven by the STM32's micro-hertz resolution DDS step generation, effectively transforms the stepper motor into a high-pole-count brushless servo. This completely suppresses acoustic resonance and low-speed vibration, resulting in fluid, industrial-grade motion.

3.6.1.3 Comparative Open-loop / Closed-loop Table

The trade-offs between the two architectures are summarized in the following table. While the closed-loop system demands significantly higher computational bandwidth and firmware complexity, its operational metrics vastly outperform the open-loop baseline.

System Characteristic	Open-Loop Architecture	Closed-Loop Architecture
Hardware Complexity	Low (4 wires per motor)	High (Power + Encoder PWM + Step/Dir lines)
Computational Load	Minimal (Blind pulse generation)	Heavy (Continuous digital filtering, P+FF calculation)
Response to Overload	Silent failure (Missed steps)	Active correction or managed E-Stop
Positional Drift	High (Compounds over time)	Zero (Absolute multi-turn tracking)
Thermal Performance	Poor (Constant maximum current)	Excellent (Adaptive FOC current draw)
Acoustic / Vibration	High (Susceptible to resonance)	Minimal (Suppressed by FOC and DDS smoothing)
ROS 2 Integration	Unsafe (Cannot guarantee execution)	Robust (Deterministic physical tracking)

Table 3. 3: Open-Loop/Closed-Loop Comparison)

Ultimately, open-loop control is sufficient only for basic engineering demonstrations where spatial accuracy is not critical. However, integrating the PAROL6 with an advanced trajectory planner like MoveIt strictly requires absolute kinematic fidelity. The closed-loop architecture ensures that the mathematical intelligence generated by the host computer is faithfully, predictably, and safely executed in the physical world, validating the complex multi-rate firmware design detailed throughout this chapter.

3.6.2 Control Algorithm and Kinematic Translation

Within the architecture of the embedded firmware, the 500 Hz Control Layer acts as the intellectual bridge between the theoretical trajectory and the physical actuation. While the Direct Digital Synthesis (DDS) layer blindly generates pulses at a commanded rate, it relies entirely on the control algorithm to determine what that rate should be. The primary objective of this algorithm is to minimize the deviation between the ROS 2 host's mathematical intent and the physical reality of the mechanical joint, while strictly respecting the system's computational limits.

3.6.2.1 Cascaded Control Topology

A defining characteristic of this system is its cascaded control topology. In traditional, fully centralized robotic controllers, a single microcontroller might be responsible for calculating the complex Inverse Kinematics (IK), generating the position loop, managing the velocity loop, and executing the high-frequency Field-Oriented Control (FOC) current loops for the motor coils.

In the PAROL6 architecture, the labor is intelligently distributed. The MKS SERVO42C motor driver possesses its own internal processor dedicated exclusively to managing the inner FOC current loops, ensuring the magnetic vectors of the stepper motor perfectly align with the rotor to maximize torque and prevent stalling. Because the MKS handles this high-frequency, heavy-lifting electrical control natively, the central microcontroller is liberated to focus entirely on the outer kinematic loops: position and velocity. This cascaded separation of concerns allows the STM32 to utilize a highly optimized, computationally lightweight algorithm that executes deterministically within the 2000-microsecond window of the 500 Hz Interrupt Service Routine (ISR).

3.6.2.2 Mathematical Formulation of the Error Vector

The execution cycle of the algorithm begins by establishing the current state of the system. From the linear interpolator (which up-samples the sparse 25 Hz ROS 2 commands), the control loop extracts the instantaneous target position $\theta_{target}(t)$ and the target velocity

$v_{target}(t)$ for the current timestep. Simultaneously, it retrieves the heavily filtered, absolute physical position $\theta_{actual}(t)$ from the hardware-assisted encoder acquisition layer.

The foundational metric of the algorithm is the positional error vector, calculated in radians:

$$e(t) = \theta_{target}(t) - \theta_{actual}(t)$$

This error value represents the instantaneous spatial deviation of the joint. If $e(t) = 0$, the physical joint is perfectly tracking the mathematical trajectory.

3.6.2.3 Proportional Control with Velocity Feedforward

To drive the error vector to zero, the system must apply a corrective control effort. The most common approach in industrial embedded systems is the Proportional-Integral-Derivative (PID) controller. However, in this specific topological configuration, a full PID implementation introduces unnecessary complexity and potential instability. The integral term (**I**) can cause windup during heavy dynamic loads, and the derivative term (**D**) frequently amplifies high-frequency encoder noise, causing jitter.

Instead, the firmware employs a highly responsive Proportional-Feedforward (**P+FF**) control strategy. Relying purely on a Proportional (**P**) gain forces the system to be entirely reactive; the motor will only move *after* an error has accumulated, inherently causing the physical arm to lag behind the mathematical trajectory during rapid movements.

To eliminate this tracking lag, the algorithm utilizes the host's commanded velocity $v_{target}(t)$ as a feedforward term. The feedforward term proactively injects the baseline energy required to execute the trajectory, while the proportional term simply acts as a reactive spring to correct for unmodeled physical disturbances (such as friction or payload inertia).

The complete control law is formulated as:

$$u(t) = v_{target}(t) + (K_p \cdot e(t))$$

(Where $u(t)$ is the commanded control effort in radians per second, and K_p is the tuned proportional gain).

3.6.2.3 Safety Saturation and Clamping

Before this control effort can be translated into physical motion, the firmware must enforce the mechanical safety limits of the robotic arm. Each joint possesses a maximum safe operating velocity (V_{max}), dictated by the mechanical rating of its harmonic drive or planetary gearbox.

If an unexpected physical collision drastically increases the positional error $e(t)$, the proportional term will mathematically request an immense, unsafe velocity to catch up. To prevent this, the algorithm passes the control effort $u(t)$ through a strict saturation function:

$$u_{clamped}(t) = \begin{cases} V_{max} & \text{if } u(t) > V_{max} \\ -V_{max} & \text{if } u(t) < -V_{max} \\ u(t) & \text{otherwise} \end{cases}$$

This clamping mechanism guarantees that regardless of software anomalies or massive physical tracking errors, the STM32 will never command the hardware to exceed its safe electromechanical limits.

3.6.2.4 Kinematic Translation to Step Frequency

The final responsibility of the control algorithm is translating the clamped control effort (which is a velocity expressed in rad/s) into a raw step frequency (Hz) that the Actuation Layer can process.

This translation requires the precise mechanical constants of the system. Let $\mathbf{G}$ represent the specific mechanical gear ratio of the joint (e.g., 50:1 for a harmonic drive), and S_{rev} represent the number of microsteps required for one full 360° revolution of the stepper motor shaft.

The required step frequency f_{step} is calculated as:

$$f_{step} = \frac{|u_{clamped}(t)| \cdot G \cdot S_{rev}}{2\pi}$$

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The sign of $u_{clamped}(t)$ is extracted to set the physical **DIRECTION** GPIO pin HIGH or LOW. Finally, the calculated f_{step} is mathematically converted into the 32-bit tuning word (**M**) and pushed to the Phase Accumulator of the DDS layer.

By executing this rigorous sequence of error calculation, feedforward injection, safety clamping, and kinematic translation at a deterministic 500 Hz rate, the control algorithm ensures the physical arm fluidly and safely mirrors the intelligence of the host computer.

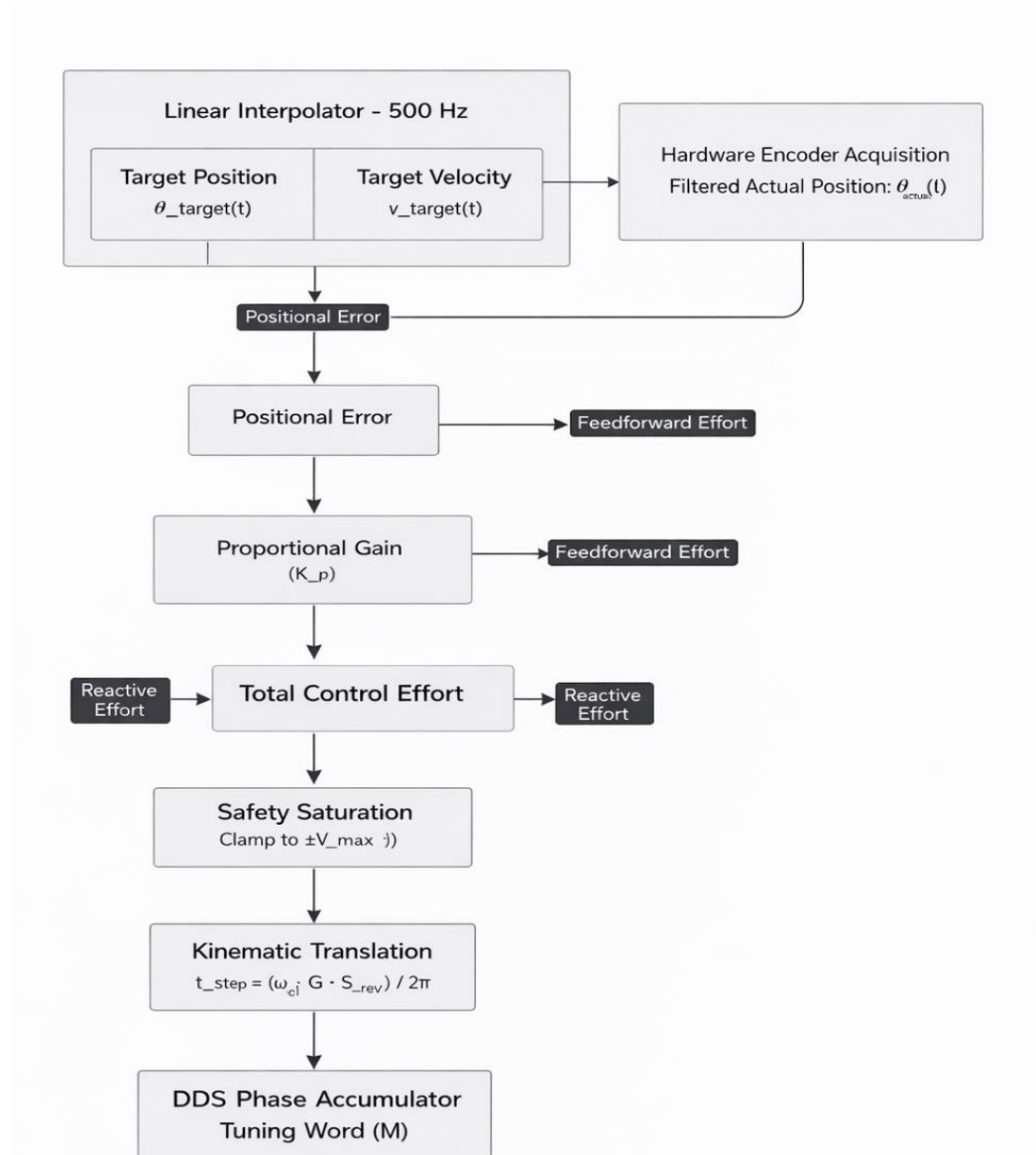


Figure 3. 15: (Proportional-Feedforward (P+FF) Control Topology for Kinematic)

3.6.3 Motor Drivers and Field Oriented Control (FOC) Architecture

The actuation layer of the PAROL6 robotic arm requires a control strategy capable of delivering high-resolution positioning, adaptive torque, and absolute acoustic silence. Traditional stepper motor driving techniques are fundamentally inadequate for these industrial-grade requirements. To bridge this gap, the system employs Field-Oriented Control (FOC). This section explores the theoretical and mathematical foundations of FOC, detailing how it transforms basic AC machines and stepper motors into high-performance, closed-loop servo systems.

3.6.3.1 Field-Oriented Control Fundamentals

Field-Oriented Control, widely referred to in control theory literature as Vector Control, is an advanced motor control technique designed to achieve precise, efficient, and smooth control over electric motors. Unlike traditional scalar control methods that directly apply predefined voltage or current waveforms to the motor windings, FOC operates by actively manipulating the motor's internal magnetic flux vectors.

The primary objective of FOC is to mathematically decouple the magnetic flux generation from the torque generation within the motor. By separating these two components, FOC allows complex AC machines—such as Brushless DC (BLDC) motors or multi-pole Permanent Magnet Synchronous Motors (PMSM) like steppers—to be controlled with the linear simplicity of a separately excited Brushed DC motor.

The Physics of Magnetic Alignment The fundamental principle governing torque generation in an electric motor dictate that the electromagnetic torque is strictly maximized when the magnetic field generated by the stator coils is maintained at an exact orthogonal angle (90° or $\pi/2$ radians) to the permanent magnetic field of the rotor.

Traditional open-loop controllers and basic electronic speed controllers (ESCs) are incapable of maintaining this optimal alignment continuously. As the rotor turns, the angle between the stator's magnetic vector and the rotor's magnetic vector constantly fluctuates. This sub-optimal magnetic alignment results in severe torque ripple, mechanical vibration, acoustic noise, and a massive reduction in electrical efficiency. FOC solves this by continuously calculating the exact

mechanical angle of the rotor and dynamically adjusting the phase currents to ensure the stator's magnetic vector perpetually leads the rotor's magnetic vector by exactly 90 degrees.

Mathematical Foundations: The Clarke and Park Transformations

To achieve this dynamic alignment, the microcontroller must translate the physical, alternating currents of the motor phases into a stable mathematical domain. This is achieved through a sequence of critical matrix transformations.

In a standard three-phase motor, the stator currents initially exist in a stationary, three-dimensional coordinate system known as the "abc" reference frame, represented by the phase currents i_a , i_b , and i_c . Because these alternating currents produce a rapidly rotating magnetic field, applying a standard Proportional-Integral (PI) control loop directly to them is computationally prohibitive and mathematically unstable.

The first step in simplifying this system is the **Clarke Transformation**. This mathematical operation projects the three-phase stationary coordinate system (a, b, c) onto a two-dimensional stationary orthogonal reference frame (α, β) . The transformation is defined as:

$$i_\alpha = i_a$$
$$i_\beta = \frac{1}{\sqrt{3}}(i_a + 2i_b)$$

While the Clarke transformation simplifies the geometry to two axes, the resulting current vector still rotates continuously with respect to the rotor, making direct torque control indirect and difficult.

To fully decouple the system, the stationary $\alpha - \beta$ frame must be mathematically synchronized with the spinning rotor. This is achieved using the **Park Transformation**, which rotates the stationary $\alpha - \beta$ axes by the rotor's instantaneous electrical angle (θ_e) . The formulas for the Park Transformation are:

$$i_d = i_\alpha \cos(\theta_e) + i_\beta \sin(\theta_e)$$

$$i_q = -i_\alpha \sin(\theta_e) + i_\beta \cos(\theta_e)$$

Direct and Quadrature (d-q) Axis Decoupling The output of the Park Transformation yields the direct-quadrature (\$d-q\$) rotating reference frame. In this frame, the wildly oscillating AC phase currents appear to the microcontroller as flat, steady DC values, provided the motor is running at a constant load.

- **The Direct Axis (i_d):** This component represents the current vector acting parallel to the rotor's magnetic field. It generates magnetic flux but produces absolutely zero rotational torque.
- **The Quadrature Axis (i_q):** This component represents the current vector acting strictly perpendicular (90°) to the rotor's magnetic field. It is the sole torque-producing component.

In a properly tuned FOC system driving a surface permanent magnet motor, the control algorithm actively regulates the d -axis current to zero ($i_d \approx 0$). This ensures that no electrical energy is wasted on unnecessary magnetization or field-weakening. Consequently, 100% of the electrical energy injected into the q -axis is converted directly into mechanical work.

The Linear Torque Equation Because the flux and torque vectors are successfully decoupled in the \$d-q\$ frame, the complex non-linear dynamics of the AC motor are reduced to a simple, linear algebraic equation. The electromagnetic torque (T_e) produced by the motor becomes directly proportional to the quadrature current:

$$T_e = K_t \cdot i_q$$

(Where K_t is the specific motor's torque constant). This linear relationship allows the outer position and velocity loops (such as those running on the STM32) to control the motor's physical strength exactly as they would a simple DC motor, granting unparalleled dynamic responsiveness.

Comparative Analysis: FOC vs. Traditional Actuation Compared to conventional motor control paradigms, FOC offers overwhelming advantages.

1. **Trapezoidal Commutation (BLDC):** Standard Brushless DC ESCs utilize Hall-effect sensors to energize two of the three motor phases sequentially, creating a hexagonal, stepping magnetic field. This six-step commutation creates massive torque ripples every 60 degrees, making it entirely unsuitable for precise robotic positioning. FOC utilizes Space Vector Pulse Width Modulation (SVPWM) to generate perfectly smooth sine waves, eliminating this ripple.
2. **Open-Loop Microstepping (Steppers):** Traditional stepper drivers force a constant, high-amperage current through the coils regardless of the physical load. While microstepping improves spatial resolution compared to full-step driving, it remains an open-loop voltage approximation. Consequently, the rotor acts like a spring, oscillating around the magnetic detent and causing severe mid-band resonance. Recent comparative analyses confirm that FOC completely mitigates these oscillatory dynamics and torque ripples by strictly controlling the orthogonal flux vector, vastly outperforming even advanced microstepping algorithms [47]. FOC measures the exact rotor angle and injects only the necessary i_q current required to hold the position, resulting in near-silent operation and superior thermal efficiency.

Relevance to Robotic Manipulators

The implementation of FOC is not merely an electrical optimization; it is a strict mechanical necessity for the PAROL6 robotic arm. Robotic manipulators operate in highly dynamic physical environments where the payload and inertial forces change drastically depending on the arm's Cartesian posture.

- **Low-Speed Precision:** Trajectory execution often requires joints to move at fractions of a degree per second. FOC eliminates the jerky, cogging effect seen in traditional stepper control, allowing for fluid, continuous low-speed interpolation.
- **Vibration Reduction:** In a serial manipulator, any vibration induced at the base joint is mechanically amplified by the lever arm of the links, resulting in severe end-effector

shaking. By eliminating torque ripple, FOC ensures the end-effector remains perfectly stable during high-precision tasks like welding or pick-and-place operations.

- **Adaptive Holding Torque:** When the PAROL6 arm extends horizontally, gravity exerts massive torque on the shoulder joint. FOC automatically detects this load via the position error and instantly scales the i_q current to resist it, preventing the arm from sagging without continuously burning maximum current when the arm is resting vertically.

3.6.3.2 Field-Oriented Control for Stepper Motors

While Field-Oriented Control (FOC) is most extensively documented and applied in the context of three-phase Brushless DC (BLDC) and Permanent Magnet Synchronous Motors (PMSM), the mathematical control strategy is equally powerful—and arguably more transformative—when applied to two-phase hybrid stepper motors. Standard stepper motor drivers utilize "open-loop" control, forcing a constant, high-amperage current through the coils regardless of the physical load, which inherently generates excessive heat and acoustic noise. Applying FOC fundamentally changes this paradigm by treating the standard stepper motor not as a discrete stepping device, but as a high-pole-count, 2-phase AC servo motor.

Implementing FOC on a stepper motor requires adapting the mathematical transformations to account for the physical differences in stator winding and rotor geometry.

The 2-Phase Difference and the Clarke Simplification The primary distinction between a BLDC motor and a bipolar stepper motor lies in the spatial distribution of their stator coils. A standard BLDC motor possesses three phases (u, v, w) distributed spatially at 120° intervals. Conversely, a bipolar stepper motor is constructed with two phases (Phase A and Phase B) that are physically wound orthogonal to one another, spaced exactly 90° apart.

This physical orthogonality significantly simplifies the mathematical transformations required for FOC. In a three-phase system, the Clarke transformation is a computationally demanding matrix operation used to project the three axes down to a two-axis stationary frame (i_α, i_β). However, for a two-phase stepper motor, the stator coils are already aligned with the α and β

axes. Consequently, the "Stepper Clarke Transformation" is essentially reduced to a mathematical identity mapping:

$$\begin{aligned}i_{\alpha} &= i_a \\i_{\beta} &= i_b\end{aligned}$$

By reading the phase currents directly into the orthogonal stationary frame, the microcontroller is spared the computational overhead of the standard Clarke transformation, allowing those CPU cycles to be allocated to higher-frequency PID execution. Once the currents are established in the $\alpha - \beta$ frame, the standard Park transformation is utilized exactly as it is in a BLDC motor to rotate the frame into the moving $d - q$ axes based on the rotor's instantaneous electrical position.

Electrical vs. Mechanical Angle: The 50-Pole Pair Challenge The most severe control challenge when applying FOC to a stepper motor is the vast discrepancy between the mechanical shaft angle and the internal electrical angle. A standard BLDC servo motor typically possesses between 1 and 11 rotor pole pairs. By contrast, a standard 1.8° NEMA 17 hybrid stepper motor possesses an exceptionally high tooth density, resulting in exactly 50 rotor pole pairs.

The relationship between the mechanical angle (θ_{mech}) and the electrical angle (θ_e) is governed by the number of pole pairs (P):

$$\theta_e = \theta_{mech} \cdot P$$

For a 50-pole pair stepper motor, a single full mechanical revolution of the output shaft (360°) requires the internal electrical magnetic field to complete 50 full rotations ($50 \times 360^\circ = 18,000^\circ$ electrical).

Stringent System Requirements for Stepper FOC

This 50x geometric multiplier dictates extraordinarily stringent hardware requirements for the embedded control system, which heavily influenced the design of the PAROL6 architecture:

1. **Extreme Encoder Resolution:** Because the electrical angle changes 50 times faster than the mechanical angle, a low-resolution encoder is entirely useless. An error of just 1.8° mechanically translates to a 90° error electrically—which would shift the quadrature torque vector completely into the direct flux vector, resulting in zero torque and a stalled motor. To maintain the $d - q$ vectors precisely, the system requires ultra-high-resolution sensors, justifying the integration of the 12-bit to 14-bit magnetic encoders (like the Magntek MT6816) utilized in this project.
2. **Ultra-High Frequency Control Loops:** The implication of the 50-pole pair architecture is that even at moderate mechanical velocities, the electrical angle changes with extreme rapidity. To continuously track this rapidly changing electrical angle and execute the Inverse Park Transformations before the rotor advances, the FOC control loop must run at a very high frequency. Standard BLDC FOC loops can often operate comfortably at 4 kHz; stepper FOC loops typically require localized execution rates of 10 kHz to 20 kHz inside the motor driver to maintain stability.

Operational Benefits of Closed-Loop Stepper Actuation

Recent literature in mechatronic design increasingly advocates for stepper-based FOC as a cost-effective alternative to expensive AC servos in robotic manipulators. When the high-frequency computational requirements are met, the resulting physical performance offers massive advantages:

- **Absolute Acoustic Silence:** By utilizing Space Vector Pulse Width Modulation (SVPWM) to drive sinusoidal currents rather than harsh square waves, FOC entirely eliminates the characteristic, aggressive "whine" and resonance vibrations universally associated with stepper motors.
- **Thermal Efficiency:** Traditional open-loop drivers push a constant maximum current into the coils to guarantee holding torque, resulting in massive I^2R copper losses. Academic research demonstrates that applying Field-Oriented Control to a two-phase hybrid stepper motor fundamentally solves this by dynamically scaling the quadrature current (i_q) to match the instantaneous physical load [46]. By delivering only the precise

current required to overcome mechanical resistance, FOC drastically reduces the thermal dissipation and overall power consumption of the motor.

- **No Lost Steps:** When paired with the closed-loop encoder feedback, the 50-pole pair density transforms from a vulnerability into an asset, providing the system with immense low-speed holding torque while ensuring the motor never loses mathematical synchronization with the trajectory controller.

3.6.3.3 Centralized vs. Decentralized Architecture

The theoretical superiority of Field-Oriented Control (FOC) in suppressing vibration and maximizing stepper motor efficiency is undeniable. However, implementing FOC across all joints of a 6-axis robotic arm presents a profound computational bottleneck when handled by a single, centralized microcontroller. Transitioning from the theoretical math of a single motor to the physical reality of a multi-axis serial manipulator necessitates a rigorous architectural analysis to ensure system stability.

3.6.3.3.1 The Computational Burden of Centralized FOC

In a purely centralized architecture, a single master microcontroller (such as the STM32F411CE) would be entirely responsible for calculating the FOC algorithms for all six motors simultaneously. To understand the impossibility of this topology, one must quantify the mathematical load required for a single motor.

At a minimum, executing a true closed-loop FOC cycle requires the following sequential operations:

1. **High-Speed ADC Sampling:** Reading two independent phase currents via shunt resistors.
2. **Positional Acquisition:** Reading the absolute 12-bit magnetic encoder via SPI or PWM to determine the mechanical angle (θ_{mech}).
3. **Electrical Angle Calculation:** Multiplying by the 50-pole pair constant (**P**) to find the electrical angle (θ_e).
4. **Forward Transformations:** Executing the Clarke and Park transformations, which require calculating sine and cosine trigonometric functions.

5. **Dual PID Execution:** Computing the Proportional-Integral-Derivative control efforts for both the Direct (i_d) and Quadrature (i_q) current loops.
6. **Inverse Transformations:** Executing the Inverse Park transformation to return the voltage vectors to the stationary frame.
7. **SVPWM Generation:** Calculating the precise duty cycles for the Space Vector Pulse Width Modulation and updating the hardware timer registers.

To maintain acoustic silence and prevent the high-pole-count stepper motor from losing synchronization, this entire sequence must execute at a frequency (f_{FOC}) of at least 15kHz to 20kHz per motor.

For a 6-DOF arm, the centralized microcontroller must therefore process:

$$TotalIterations = 6 \cdot 20,000 = 120,000 \text{ FOC cycles per second}$$

3.6.3.3.2 Hardware Peripheral Contention

Beyond raw CPU clock speed, a centralized FOC system is severely bottlenecked by hardware peripheral contention. To accurately measure phase currents without reading switching noise, the Analog-to-Digital Converter (ADC) triggers must be perfectly synchronized with the dead-time center of the PWM signals.

A 6-axis stepper FOC system requires 12 independent PWM generation channels (two H-bridges per motor) and 12 synchronized ADC channels. Managing this immense peripheral traffic matrix on a single mid-range microcontroller introduces severe Phase Delay. Furthermore, the central STM32F411CE is already heavily burdened with maintaining 115200-baud asynchronous communication with the ROS 2 host, generating precise linear movement trajectories, and managing peripheral limit switches. Consequently, calculating the rigorous Proportional-Integral-Derivative (PID) loops and complex mathematical transformations required for simultaneous multi-axis FOC drastically exceeds the practical and deterministic limits of a single embedded unit.

3.6.3.3.3 The Decentralized Topology

To circumvent the catastrophic timing failures inherent in a centralized model, a decentralized, cascaded control architecture was adopted for the PAROL6.

In this distributed topology, the intelligence of the system is physically separated based on the required execution frequency:

- **The Supervisory Node (Central MCU):** The central STM32 is completely relieved of the high-frequency current loops. It is relegated to a macro-level supervisory role. It operates purely in the spatial and kinematic domains—ingesting ROS 2 targets, executing the 500 Hz linear trajectory interpolation, polling global system safety, and outputting synchronized Step/Direction pulses.
- **The Execution Nodes (Local Motor Drivers):** The high-frequency electrical calculations are offloaded directly to the joints. Each motor is paired with an independent, dedicated processor embedded on the motor driver board. This local processor operates as an independent, closed-loop system. It handles the 20kHz Clarke/Park transforms, reads the localized magnetic encoder, and drives the H-bridge MOSFETs.

This decentralized architecture perfectly mimics industrial robotic paradigms. It guarantees that the heavy mathematical lifting of the Field-Oriented Control cannot disrupt or delay the spatial trajectory generation, yielding a multi-axis system that is highly modular, electrically stable, and computationally robust.

Chapter 4

Project Design

This chapter presents the complete design of the system at the architecture and subsystem level. The design decisions made here are directly informed by the requirements and mathematical models established in Chapter 3. For each major subsystem, the rationale for key design choices is provided alongside the technical specification. The chapter is organized to follow the same subsystem structure used in Chapters 3, 5, and 6 for traceability. Where analytical justification was required for structural components, ANSYS Finite Element Analysis (FEA) results are presented and discussed.

4.1 System Architecture Overview

4.1.1 Layered System Architecture

The system architecture is organized into four distinct layers that reflect the separation of concerns between hardware, communication, middleware, and application logic. The Physical Layer contains all hardware components (the PAROL6 arm, MKS SERVO42C drivers, STM32 Black Pill, MT6816 encoders, Kinect v2, and power distribution). The Communication Layer contains the physical interfaces and protocols (UART between STM32 and the ROS 2 hardware interface node; USB for Kinect v2; Ethernet for network-based alternative communication). The Middleware Layer contains the ROS 2 ecosystem (DDS communication, tf2 transform tree, ros_control framework, MoveIt 2 planning). The Application Layer contains the project-specific nodes (vision pipeline, PyQt5 GUI, hardware interface node, MoveIt interface node).

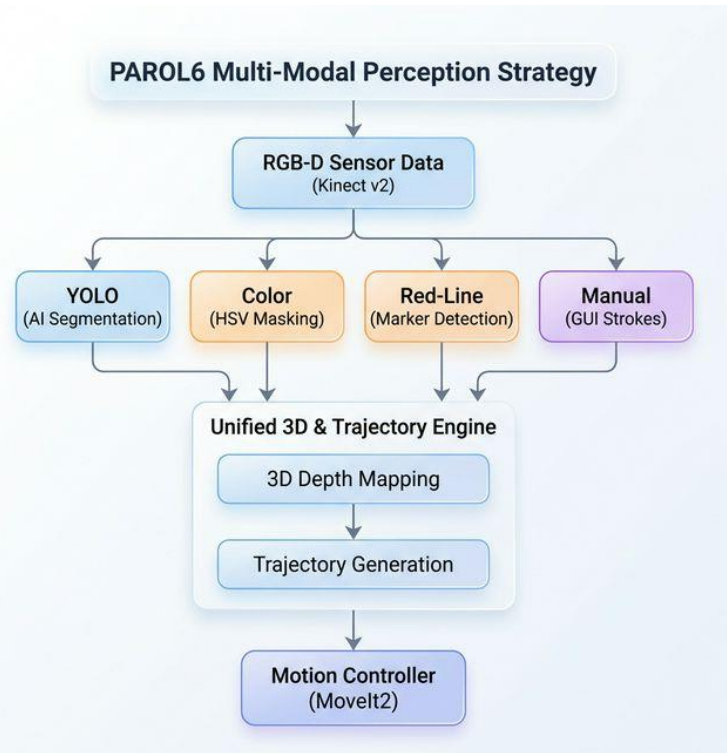


Figure 4. 1: Layered system architecture. Vertical arrows indicate upward data flow from physical sensing and actuation to application logic, and downward command flow from planning to actuation.

4.1.2 ROS 2 Node Graph and Topic Architecture

The ROS 2 software architecture is implemented as a set of specialized nodes connected by typed topic channels. The node graph follows the Directed Acyclic Graph (DAG) pattern recommended for robotics pipelines: data flows in one direction from sensing nodes through processing nodes to actuation nodes, with no cyclic dependencies. This design prevents

4.2 Mechanical Design: PAROL6 Robotic Arm

4.2.1 Arm Structure and 3D-Printed Components

The PAROL6 robotic arm's structural framework consists entirely of custom FDM (Fused Deposition Modeling) 3D-printed parts. The primary structural components include the base housing (which mounts to the work table and contains the joint 1 motor), six joint housings, four arm link sections, motor mounts for each joint, and the tool flange. All structural parts are printed in PETG (Polyethylene Terephthalate Glycol) at 40% infill density with a 0.4 mm nozzle and 0.2 mm layer height. PETG was selected over PLA for its improved impact resistance, lower brittleness, and higher glass transition temperature (approximately 80 degrees C vs. 60 degrees C for PLA), which provides a safety margin against heat from the motor drives during sustained operation.

Part	Material	Quantity	Manufacturing
Base housing	PETG (printed)	1	FDM 3D printing
Joint 1 housing	PETG (printed)	1	FDM 3D printing
Upper arm link	PETG (printed)	1	FDM 3D printing
Forearm link	PETG (printed)	1	FDM 3D printing
Joint 4/5/6 housing	PETG (printed)	3	FDM 3D printing

Tool flange	PETG (printed)	1	FDM 3D printing
NEMA 17 stepper motor (high torque)	Steel/Copper	3 (joints 1-3)	Commercial
NEMA 17 stepper motor (low torque)	Steel/Copper	3 (joints 4-6)	Commercial
Planetary gearbox 10:1	Steel	2 (joints 1-2)	Commercial
Planetary gearbox 5:1	Steel	2 (joints 3-4)	Commercial
GT2 timing belt (6mm)	Rubber/fibre	6	Commercial
GT2 pulley 20T	Aluminium	12	Commercial
Ball bearing 608ZZ	Steel	20	Commercial
Shaft coupler 5mm-5mm	Aluminium	6	Commercial
Metric screws M2/M3/M4	Steel	Various	Commercial

Table 4. 1: Bill of materials for the PAROL6 robotic arm

4.2.2 Mechanical Transmission Design

The mechanical transmission from motor shaft to joint output axis consists of three stages for joints 1 through 3 (high-load proximal joints) and two stages for joints 4 through 6 (low-load distal joints). In the proximal joints, the NEMA 17 motor shaft drives a planetary gearbox (10:1 or 5:1 reduction), whose output shaft drives a GT2 timing belt pulley pair (typically a 2:1 additional reduction), giving a combined gear ratio of 20:1 to 50:1. The distal joints use only the belt drive stage (4:1 to 8:1 reduction). High gear ratios provide three key benefits: amplification of available torque ($\text{torque}_{\text{joint}} = \text{torque}_{\text{motor}} * \text{ratio} * \text{efficiency}$), attenuation

of motor cogging and step impulse disturbances (reflected inertia scales as $1/\text{ratio}^2$), and improved position resolution (the encoder's 16,384 counts/revolution maps to sub-0.01 degree resolution at the joint).

Joint	Gearbox Ratio	Belt Ratio	Total Ratio	Motor Torque [N.m]	Output Torque [N.m]
1 (Base rotation)	10:1	2:1	20:1	0.45	8.1
2 (Shoulder)	10:1	2:1	20:1	0.65	11.7
3 (Elbow)	5:1	2:1	10:1	0.45	4.1
4 (Wrist-1)	--	4:1	4:1	0.22	0.8
5 (Wrist-2)	--	4:1	4:1	0.22	0.8
6 (Tool)	--	2:1	2:1	0.22	0.4

Table 4. 2: Mechanical transmission specifications per joint

4.2.3 ANSYS Structural Analysis

Material Specifications and Anisotropy Considerations

The PAROL6 robotic arm structural components were manufactured from Polyethylene Terephthalate Glycol (PETG) using Fused Deposition Modeling (FDM) additive manufacturing. PETG was selected due to its favorable balance between tensile strength, ductility, and ease of printing compared to other common FDM thermoplastics such as PLA or ABS. The mechanical behavior of FDM-printed PETG is inherently anisotropic and is approximated as orthotropic in this analysis, due to the layer-by-layer deposition process which introduces directional dependency in both stiffness and ultimate strength. Material properties

vary significantly along the print direction (Z-axis, across layer interfaces) compared to in-plane directions (X-Y plane, along raster lines).^{[48][49]}

Multiple infill densities ranging from 15% to 80% were considered during the design phase. Lower infill percentages (15-30%) reduce part mass but significantly decrease stiffness and load-carrying capacity, making them unsuitable for load-bearing structural links. Higher infill densities (60-80%) improve structural rigidity at the cost of increased mass, print time, and material consumption. Based on this trade-off analysis, a uniform 40% gyroid infill pattern was selected for all structural parts -- a gyroid infill provides approximately 85% of the stiffness of a 40% rectilinear infill while distributing stress more isotropically and resisting crack propagation more effectively. Additional factors including layer adhesion quality, raster orientation relative to the primary load direction, and printing temperature were incorporated into the analysis through conservative material property selection: the elastic modulus used (1.8 GPa) represents a 20% reduction from the measured in-plane tensile modulus (2.25 GPa) to account for the reduced inter-layer bond strength and the presence of void defects inherent to FDM manufacturing.

Linear scaling is based on infill ratio $R = \text{Infill}/100$.

Density (ρ): $1,270,000 \times R \text{ g/m}^3$

Ultimate Strength (σ): $55,000,000 \times R \text{ Pa}$

Elastic Modulus (E): $2,100,000,000 \times R \text{ Pa}$

Thermal Conductivity (k): $0.25 \times R \text{ W/m}\cdot\text{K}$

Infill %	Density (g/m ³)	Ult. Strength (Pa)	Elastic Modulus (Pa)	Shear Modulus (Pa)
15%	190,500	8,250,000	315,000,000	113,309,353
20%	254,000	11,000,000	420,000,000	151,079,137
25%	317,500	13,750,000	525,000,000	188,848,921
30%	381,000	16,500,000	630,000,000	226,618,705
35%	444,500	19,250,000	735,000,000	264,388,489
40%	508,000	22,000,000	840,000,000	302,158,273
45%	571,500	24,750,000	945,000,000	339,928,058
50%	635,000	27,500,000	1,050,000,000	377,697,842
55%	698,500	30,250,000	1,155,000,000	415,467,626
60%	762,000	33,000,000	1,260,000,000	453,237,410
65%	825,500	35,750,000	1,365,000,000	491,007,194

70%	889,000	38,500,000	1,470,000,000	528,776,978
75%	952,500	41,250,000	1,575,000,000	566,546,763
80%	1,016,000	44,000,000	1,680,000,000	604,316,547

*Shear Modulus calculated using Poisson's Ratio $\nu = 0.39$ with $G = E / 2(1+\nu)$.

Constant values for all infills:

Poisson's Ratio: 0.39

Specific Heat: 1,200 J/kg·°C

Thermal Conductivity: 0.15 W/m·K (safe average)

Simulation Types: Static and Transient Structural Analysis

Two complementary ANSYS simulation types were conducted to fully characterize the structural behavior of the PAROL6 arm under its intended operating conditions.

Static Structural Analysis: A static structural analysis was conducted to evaluate the ability of the PAROL6 arm to withstand externally applied loads without failure. The worst-case loading scenario corresponds to the arm in a fully extended horizontal posture carrying a 1 kg payload at the end-effector, which maximizes the bending moment at the proximal joints (base and shoulder). Boundary conditions were applied as a fixed support at the robot base mounting plate (zero displacement and rotation in all six degrees of freedom). The payload was applied as a 9.81 N point force at the tool flange center in the -Z (downward) direction, with an additional gravitational body load of 9.81 m/s² applied to all components to capture self-weight effects. This analysis provided Von Mises stress distributions across all structural components, total and directional deformation, and safety factors against yielding.

Transient Structural Analysis: A transient structural analysis was performed to simulate the real operational behavior of the arm under dynamic motion. Unlike the static analysis, this simulation accounts for time-dependent effects including joint accelerations, decelerations, inertia forces, and velocity-induced stress fluctuations. Joint motions were defined using time-based input functions replicating a representative high-speed trajectory: all six joints executing simultaneous sinusoidal motion at 0.5 Hz with amplitudes at 60% of their rated range, corresponding to the most aggressive trajectories executed during system testing. The transient

analysis enables prediction of fatigue-prone regions under repeated motion cycles and provides peak dynamic stress data that exceeds the quasi-static predictions.

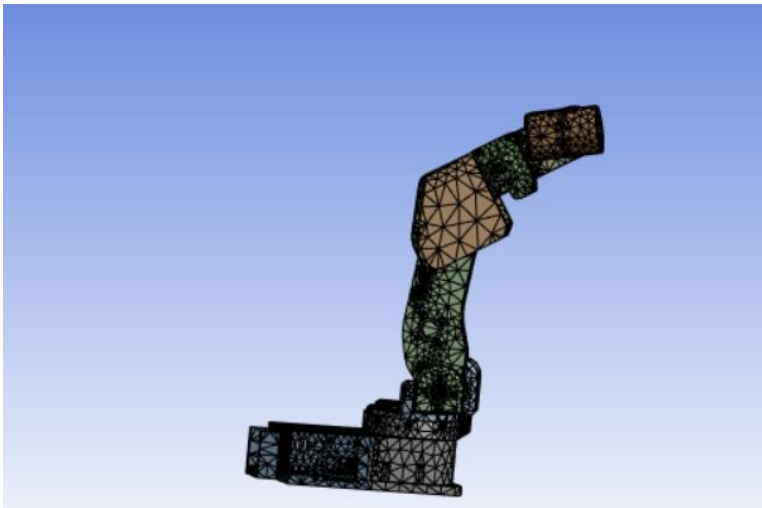


Figure 4. 2: ANSYS FEA mesh of the PAROL6 structural assembly

Computational Setup

Parameter	Value	Description
Total elements	~162,000	Linear tetrahedral (SOLID185)
Total nodes	~313,000	Degrees of freedom: 3 translational per node
Element size (critical regions)	5.0 mm	Joints, stress concentrations, screw holes
Element size (bulk regions)	2.0 mm	Rods, Bearings, Main Base
Contact formulation	Penalty-based	Applied at joint interfaces; allows slight penetration for numerical stability
Analysis type 1	Static Structural	Worst-case 1 kg payload at full extension

Analysis type 2	Transient Structural	Dynamic motion at 60% rated speed, all 6 joints simultaneous
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Table 4. 3: ANSYS computational setup parameters

4.3 Electronics and Power System Design

4.3.1 Electronics Architecture

The electronics system is organized around the STM32F411CE Black Pill microcontroller as the central node, with six MKS SERVO42C modules connected in a distributed star topology. Each MKS SERVO42C module receives two signals from the STM32: a PWM step pulse train (frequency proportional to desired joint velocity) and a logic-level direction signal (HIGH = clockwise, LOW = counterclockwise). The MT6816 encoder on each joint is connected to both the MKS SERVO42C (via SPI for the internal FOC loop) and to the STM32 (via PWM output for position feedback). Limit switches on all six joints are connected directly to the STM32 via hardware EXTI interrupt pins.

Component	Specification	Qty	Interface with STM32
STM32F411CE Black Pill	Cortex-M4, 100 MHz, 512 KB flash, 128 KB SRAM, hardware FPU	1	Central MCU
MKS SERVO42C	Closed-loop FOC stepper driver, 42 mm NEMA 17, 2A, 24V max	6	PWM step + GPIO direction per joint
MT6816 Encoder	14-bit absolute magnetic encoder, 16384 counts/rev, PWM + SPI	6	Timer Input Capture (PWM mode)

Limit switch (microswitch)	SPDT, 0.1A, 5V logic	6	EXTI interrupt pin per joint
W5500 Ethernet module	SPI-to-Ethernet, 10/100 Mbps (optional Ethernet interface)	1	SPI3
Logic level converter	3.3V to 5V bidirectional, 4- channel	2	Signal conditioning
12V DC power supply	12V, 10A, DIN rail mount	1	Power input
5V voltage regulator	LM7805, 1.5A	1	Logic and sensor power
E-stop relay module	NC contact, 24V coil, panel- mount pushbutton	1	GPIO output (enable/disable power)

Table 4. 4: Electronics component list

4.4 Embedded System Firmware Design

4.4.1 Microcontroller Usage Evolution and Comparative Analysis

The development of the embedded architecture was an iterative, empirical engineering process, characterized by rapid prototyping, hardware-in-the-loop validation, and systematically addressing critical points of failure. The selection of the central microcontroller acting as the bridge between the ROS 2 host and the motor drivers evolved through three distinct architectural phases. Each phase was analyzed based on its ability to meet two uncompromising constraints: absolute execution determinism (timing) and electromechanical resilience (hardware survival).

Phase 1: Prototyping and Validation (ESP32)

The initial validation of the closed-loop kinematics was implemented on an Espressif ESP32 microcontroller (Xtensa Dual-Core 32-bit LX6 operating at 240 MHz). This platform was chosen for its accessibility, high clock speed, and the theoretical advantage of assigning USB communication to Core 0 and the motor control loop to Core 1.



Figure 4. 3: ESP32 NodeMCU)

However, dynamic testing quickly revealed the fundamental limitations of utilizing a Real-Time Operating System (RTOS) alongside complex, silicon-level radio stacks. Despite pinning the motor control task to a specific core, the ESP32 suffered from significant context-switching overhead. The FreeRTOS scheduler was frequently forced to preempt the motor control task to service non-maskable internal background processes, particularly those managing the WiFi and Bluetooth peripherals (even when nominally disabled in software, the baseband interrupts periodically claim CPU cycles).

This preemption introduced highly variable execution jitter, with delays frequently exceeding 10 to 20 microseconds. While a 20-microsecond delay is negligible in web communication, in the context of generating a 40 kHz step pulse train—where the nominal period is exactly 25 microseconds—a 20-microsecond jitter represents an 80% phase distortion. This massive temporal error forced the motor drivers to attempt instantaneous, physically impossible velocity changes, injecting unacceptable mechanical vibration into the arm's transmission and causing cascading communication timeouts with the ROS 2 trajectory planner.

Phase 2: High-Performance Real-Time (Teensy 4.1)

To eradicate the RTOS scheduling jitter, the system architecture was completely rewritten and migrated to the Teensy 4.1, featuring a formidable NXP i.MX RT1062 ARM Cortex-M7 processor operating at 600 MHz. The primary objective of this phase was to prove the viability of a strict 1 kHz, hard real-time servo controller driven entirely by bare-metal interrupts.

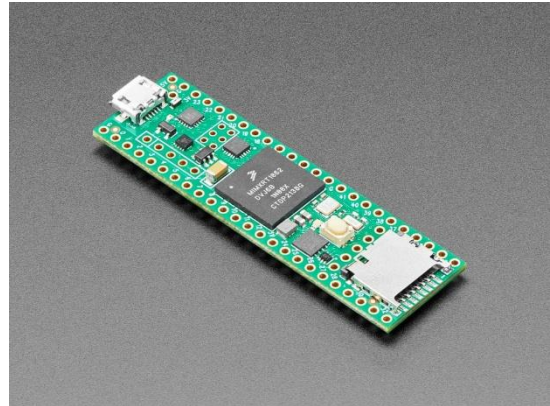


Figure 4. 4: (Teensy 4.1)

By abandoning the RTOS in favor of raw hardware `IntervalTimer` interrupts, execution jitter was aggressively compressed to less than 1 microsecond of deviation. Furthermore, this phase successfully developed the hardware-offloading architecture. It utilized the Teensy's `FlexPWM` peripherals to generate robust, concurrent STEP pulses for all six axes without CPU bit-banging, and `QuadTimers` to read encoder edges with zero CPU overhead.

However, this platform ultimately suffered a catastrophic failure due to electrical vulnerability. The Teensy 4.1 operates strictly on 3.3V logic and lacks internal over-voltage clamping diodes on its I/O pins. The robotic arm's MKS Servo42C drivers, inductive limit switches, and long wire runs passing near high-current stepper coils operate in a 5V industrial environment. During dynamic hardware-in-the-loop testing, the rapid toggling of these signals through long inductive wire runs generated flyback voltage spikes $V = L \frac{di}{dt}$ that significantly exceeded the 3.3V absolute maximum rating. These inductive transients caused silicon breakdown, resulting in the permanent burnout of the microcontroller.

Phase 3: Final Fault-Tolerant Architecture (STM32F411CE)

Following the electrical failure of the Cortex-M7, the mature, bare-metal control architecture was rapidly ported to the STM32F411CE "Black Pill" development board. This microcontroller features an ARM Cortex-M4F processor operating at 96 MHz. While theoretically possessing only a fraction of the clock

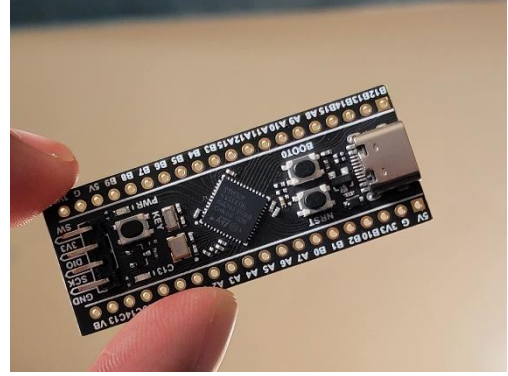


Figure 4. 5: STM32F411CE Black Pill

speed of the Teensy 4.1, the STM32 inherited the highly optimized, hardware-offloaded structural lessons (such as Alpha-Beta filtering and deterministic interpolation) developed during the previous phase. More importantly, it provided a vastly superior electrical baseline. A defining characteristic of the STM32F4xx series is that the majority of its General-Purpose Input/Output (GPIO) pins are 5V-tolerant, featuring internal hardware clamping capable of safely sinking inductive spikes.

This final phase yielded a system that maintained the strict microsecond determinism required for smooth trajectory execution while possessing the industrial resilience necessary to survive the harsh electrical realities of the physical hardware.

Quantitative Microcontroller Comparison

The following table summarizes the key architectural differences and measured performance metrics that drove the final hardware selection:

Specification	Phase 1: ESP32	Phase 2: Teensy 4.1	Phase3: STM32F411CE (Final)
Core Architecture	Xtensa LX6 (Dual-Core)	ARM Cortex-M7	ARM Cortex-M4F
Clock Speed	240 MHz	600 MHz	96 MHz
Floating Point Unit (FPU)	Yes (Single Precision)	Yes (Double Precision)	Yes (Single Precision)
Logic Voltage Limit	3.3V (Strict)	3.3V (Strict)	3.3V (5V Tolerant I/O)

Firmware Environment	FreeRTOS (ESP-IDF)	Bare-Metal C++	Bare-Metal C / HAL
Control Loop Rate	500 Hz	1000 Hz	1000 Hz
Measured Timing Jitter	> 15.0 μ s (Variable)	< 1.0 μ s (Deterministic)	< 1.0 μ s (Deterministic)
Primary Limitation	RTOS Preemption / Jitter	Electrical Fragility	Lower CPU Overhead Margin
Deployment Outcome	Rejected (Mechanical Vibration)	(Hardware Burnout)	Accepted (Robust & Stable)

Table 4. 5: Microcontroller comparison and selection

As demonstrated in the comparison, the STM32F411CE successfully marries the hard real-time determinism achieved by the Teensy 4.1 with an electrical tolerance that the other microcontrollers lacked, making it the only viable choice for the final embedded architecture.

4.4.2.1 Final MCU Selection and Hardware Justification

The final selection of the STM32F411CE (integrated into the "Black Pill" development board) represents a calculated, optimal engineering compromise between raw computational bandwidth, peripheral availability, and electromechanical robustness. While its maximum core clock frequency of 96 MHz is significantly slower than the 600 MHz Cortex-M7 utilized in the preceding prototyping phase, clock speed is not the sole determinant of real-time embedded performance.

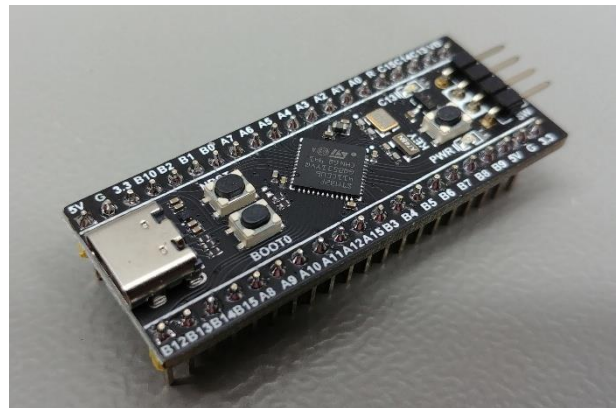


Figure 4. 6: STM32F411CE "Black Pill"

In a robotic control context, architectural efficiency—specifically the ability to execute complex math rapidly and offload repetitive tasks to dedicated silicon—is far more critical than raw clock cycles. The STM32F411CE possesses a specific confluence of hardware features that makes it ideally suited to execute the 500 Hz closed-loop algorithm across all six axes of the PAROL6 arm without approaching a computational bottleneck.

4.4.2 Four-Layer Firmware Architecture

The firmware is structured as a four-layer real-time multi-rate execution model. This design ensures that the highest-priority tasks (actuation timing) are never preempted by lower-priority tasks (communication handling), while still allowing asynchronous background operations to proceed without blocking the control loop.

Layer	Rate	Trigger	Function	STM32 Hardware Used
Layer 0 -- Measurement	Continuous	DMA / timer hardware	Encoder position acquisition via PWM Input Capture; DMA transfer to memory	TIM2-TIM7 Input Capture + DMA1/2
Layer 1 -- Background	Async (main loop)	UART IDLE interrupt	UART packet reception; trajectory buffer management; homing state machine; watchdog reset	USART1 + IDLE interrupt + TIM12 watchdog
Layer 2 -- Control	500 Hz (2 ms ISR)	TIM1 Update interrupt	PID control loop for all 6 joints; FCW computation for DDS; safety limit checks; encoder deadband filter	TIM1 (advanced-control) Update ISR
Layer 3 -- Actuation	40 kHz (25 us ISR)	TIM8 Update interrupt	DDS phase accumulator update for all 6 step channels; GPIO step pulse generation via	TIM8 (advanced-control) + GPIOB BSRR

			BSRR; direction GPIO update	
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Table 4. 6: Firmware layer architecture

4.4 Vision Pipeline Design

4.4.1 Pipeline Architecture: Directed Acyclic Graph:

The PAROL6 pipeline is organized as an asynchronous ROS 2 DAG: distinct nodes execute concurrently and communicate over ROS topics. There are no cycles, so each node simply subscribes to input topics and publishes to output topics. In practice this means we can decouple the perception stages – for example, switching between YOLO segmentation and classical color filtering requires no changes to the trajectory planner, because their outputs share the same message interface. Figure 4.9 (Full vision pipeline DAG) depicts the overall DAG: seven main stages feed into one another via ROS topics. Standard ROS 2 executor and callback mechanisms handle scheduling, and each node runs independently (often on its own thread).

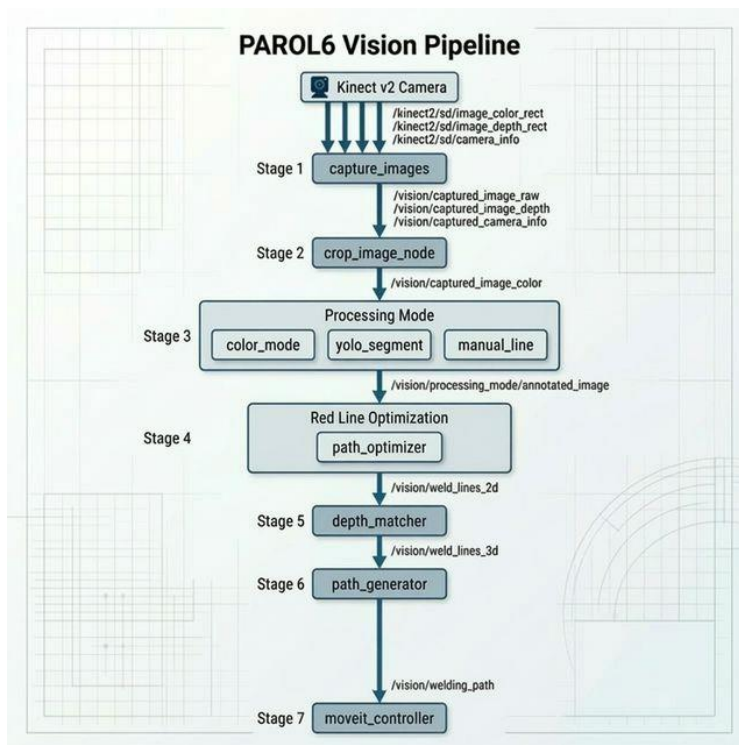


Figure 4. 7: Vision pipeline Directed Acyclic Graph (DAG). Each box represents a ROS 2 node; each arrow represents a topic. The pipeline transforms raw RGB-D sensor data into executable MoveIt trajectory waypoints.

4.4.2 Stage 1: Synchronized Acquisition (Image Capture):

This is the entry point of the entire vision pipeline. It subscribes to three Kinect v2 camera topics and synchronizes the RGB color stream with the aligned depth stream using

"message_filters.ApproximateTimeSynchronizer". When a trigger event fires, a synchronized color+depth pair is atomically published to the pipeline's internal topics.

Why it matters: Precise synchronization at this stage is critical. If the color image and depth image are not aligned in time, the 2D-to-3D back-projection in Stage 5 will produce incorrect 3D coordinates. The **"TRANSIENT_LOCAL"** QoS on the depth and camera info outputs ensures that **"depth_matcher"** always receives the last captured depth frame — even if it starts or restarts minutes after the capture event.

Trigger Modes:

Mode	Mechanism
Keyboard	Press `s` + Enter` on the terminal; implemented via a background daemon thread reading `stdin`.
Timed	A ROS timer fires every frame_time seconds (default 10 s) for automatic capture.
Topic trigger	"/vision/capture_trigger" (std_msgs/Empty) — used by the GUI.

Table 4. 7: Trigger Modes:

4.4.3 Stage 2: ROI Filtering (Crop/Mask Images):

An always-active relay node that sits between "capture_images" and the rest of the pipeline. It optionally applies a spatial polygon mask or rectangular crop to the captured color frame before forwarding it downstream. Configuration is read from "~/.parol6/crop_config.json" and can be reloaded live without restarting the node.

Why it matters: Removing irrelevant background regions from the camera view before processing dramatically reduces false positives in color thresholding and YOLO segmentation (Stages 3 and beyond). The preferred mask mode preserves the original image resolution, which is essential for the depth map pixel-coordinate alignment used in Stage 5.

Operating Modes:

Mode	Behavior
Mask (recommended)	Fills pixels outside a polygon with black; image dimensions unchanged. Depth alignment preserved.
Crop (legacy)	Crops to a bounding box, changing image dimensions — avoid when depth is needed.

*Table 4. 8: Operating Modes***4.4.4 Stage 3 — Seam Intersection Detection (Processing Mode Nodes):**

The heart of the vision pipeline is the multi-modal 2D seam detection module, which implements four independent detection methods. The system operator selects the active mode via a ROS 2 parameter or the GUI mode selector. The three modes are designed to cover the full range of operational scenarios encountered in welding environments.

Mode	Method	Operator Effort	Best For	Key Limitation
1 -- AI Segmentation	YOLO instance segmentation + boundary extraction	None (fully automatic)	Clear visual contrast between workpieces; butt and lap joints	Requires trained model; poor in low contrast or smoke
2 -- HSV Color Mask	HSV thresholding + morphology + skeleton	Apply color marker to seam	All joint types with color marking; high reliability	Requires physical surface preparation (marker application)

3 -- Red-Line Annotation	HSV red detection + Canny + Zhang-Suen thinning	Place red line on seam	Straight or gently curved seams with clear line placement	Sensitive to red ambient lighting or rust coloration
4 -- GUI-Drawn Path	Mouse/touch input + ORB feature tracking	Draw path on GUI	Any geometry; operator knows exact seam location	Depends on operator skill; not repeatable

Table 4. 9: Seam Intersection DetectionS

Mode 1 -- AI Segmentation (YOLO-based): A YOLO instance segmentation model processes the ROI-cropped RGB image to identify the two workpieces to be joined. The model outputs binary segmentation masks for each detected object. The boundary between the two largest detected objects is extracted as the weld seam using contour intersection analysis. This mode is fully automatic but requires sufficient visual contrast between workpieces and a trained model.

Mode 2 -- HSV Color Mask: The operator applies a colored tape or spray paint to mark the seam line (green or blue for high contrast against metallic surfaces). The HSV-space color mask is applied with configured Hue, Saturation, and Value ranges, followed by morphological opening (to remove noise) and closing (to fill gaps). The cleaned binary mask is skeletonized to extract the single-pixel-wide seam centerline.

Mode 3 -- Red-Line Annotation: The operator places a physical red line (tape or marker) along the seam. Classical HSV detection isolates the red channel (Hue ranges 0-15 and 165-180 degrees, accounting for the circular HSV hue space). The Canny edge detector and subsequent Zhang-Suen thinning extract the precise line centerline.

Mode 4 -- GUI-Drawn Path: The operator draws the seam path directly on the GUI camera feed display using a mouse or touchscreen. The drawn pixel coordinates are captured, and ORB

(Oriented FAST and Rotated BRIEF) feature tracking is applied to propagate the drawn path across subsequent frames to compensate for minor camera vibration or workpiece movement during setup.

4.4.5 Stage 4 — Red Line Optimization (*path_optimizer*):

Detects the red weld marker line(s) painted on the workpiece surface (or generated by “**manual_line**”) within the annotated image from Stage 3. It applies a multi-step computer vision pipeline and outputs a structured **WeldLineArray** message with ordered pixel-space points representing the weld path.

Why it matters: The annotated image from Stage 3 shows *where* the seam region is, but the exact weld path is encoded as a red marker drawn by the operator. Stage 4 extracts this path with sub-pixel precision using skeletonization and PCA-based point ordering, producing a clean, ordered sequence of 2D pixels that Stage 5 can back-project into 3D space.

Detection Pipeline

Step	Operation	Purpose
1	HSV dual-range red masking	Red hue wraps around 0°/180° in OpenCV HSV; two inRange calls handle both ends.
2	Morphological erode + dilate	Remove salt-pepper noise; fill gaps and reconnect fragments.
3	Skeletonization (skimage)	Reduces the thick marker region to a 1-pixel-wide centerline.
4	Contour detection + PCA ordering	Finds connected components; sorts each contour from start to end pipeusing its principal axis.
5	Douglas-Peucker simplification	Reduces point count while preserving line geometry.

6	Confidence scoring	Accepts only the contour with the highest point count as the single output line.
---	--------------------	--

Table 4. 10: Red Line Optimization Detection pipeline

4.4.6 Stage 5 — 2D → 3D Reconstruction (depth_matcher):

The critical **bridge between the 2D vision world and 3D robot space**. It takes the ordered pixel-space weld line points from Stage 4 and back-projects each pixel into a 3D Cartesian coordinate using the depth camera's intrinsic matrix and the robot's TF2 transform tree.

Why it matters: MoveIt2 requires 3D poses in the robot's **base_link** frame. This stage performs the mathematical lifting of 2D image data into robot-space 3D data using the aligned depth image captured all the way back in Stage 1. A cache-based architecture (rather than timestamp synchronization) is used because the operator may draw the weld line minutes after the depth was captured.

Algorithm

1. **Cache-based depth:** Depth image and camera info are stored immediately on arrival. The “**weld_lines_2d**” callback triggers processing using these cached values — no timestamp matching required.
2. **Rate-limit gate:** A 0.5 s gate prevents the continuous stream of “**weld_lines_2d**” from flooding downstream nodes.
3. **Pinhole back-projection:** For each pixel (u, v) with depth Z: $X = (u - cx) \times Z / fx$, $Y = (v - cy) \times Z / fy$.
4. **TF2 transform:** Each 3D camera-frame point is transformed to “**base_link**” using TF2.
5. **Statistical outlier filtering:** Points deviating more than 2σ from the centroid distance are discarded.
6. **Quality gating:** Line accepted only if ≥ 10 valid points and depth quality $\geq 60\%$.

4.4.7 Stage 6 — Path Generation (path_generator):

Converts the raw 3D point cloud of the weld seam into a **smooth, kinematically feasible welding trajectory** (nav_msgs/Path). It applies PCA ordering, duplicate removal, cubic B-

spline fitting, arc-length resampling, and 6-DOF orientation assignment to produce a trajectory ready for MoveIt2 execution.

Why it matters: Raw 3D points from depth sensors are noisy and unevenly spaced. Sending these directly to the robot would produce jerky, unsafe motion and inconsistent heat input during welding. The B-spline smoothing and arc-length resampling ensure the robot moves at a constant speed along the weld seam — a critical requirement for weld quality.

Algorithm

Step	Operation	Purpose
1	PCA ordering	Sorts the unordered 3D point cloud along the seam's principal axis
2	Duplicate removal	Removes points within 0.1 mm of each other to prevent spline fitting failures
3	Cubic B-spline fitting	“ <code>scipy.interpolate.splprep()</code> ” fits a smooth curve through the ordered points.
4	Arc-length resampling	Re-samples waypoints at 5 mm intervals; caps at 80 waypoints for OMPL compatibility.
5	Orientation assignment	Assigns a fixed downward quaternion to every waypoint for horizontal surface welding.
6	Rate-limit gate	0.5 s gate prevents flooding “ <code>moveit_controller</code> ”

Table 4. 11: Path Generation Algorithm

4.4.8 Stage 7 — Motion Execution (`moveit_controller`):

The **final execution stage**. It receives the smooth welding trajectory from Stage 6 and commands the PAROL6 robot arm to physically execute the weld using **MoveIt2 Cartesian planning**. It implements a robust 3-tier fallback strategy to handle kinematic singularities, joint limits, and planning failures.

Why it matters: Cartesian trajectory planning in constrained robotic environments is inherently brittle. A single point near a singularity can cause the entire trajectory plan to fail. The 3-tier fallback dynamically relaxes precision requirements to ensure the robot always executes the weld rather than silently failing.

Execution Sequence

```
[Plan 1/3] Home trajectory      (joint-space, plan only)
[Plan 2/3] Approach trajectory (15 cm above weld start, plan only)
[Plan 3/3] Cartesian weld traj (full Cartesian, with fallback, plan only)
      ↓ all plans ready
[Exec 1/3] Move to Home
[Exec 2/3] Move to Approach Point
[Exec 3/3] Execute Weld
```

Figure 4. 8: Exection sequence

Cartesian Planning Fallback Strategy:

Attempt	Step Size	Success Threshold	Description
1	2 mm	95%	High-precision welding
2	5 mm	95%	Relaxed step to skip micro-singularities
3	10 mm	90%	Coarse "get the job done" mode

Table 4. 12: Cartesian Planning Fallback Strategy

If all Cartesian attempts fail and “**enable_joint_waypoint_fallback**” is “**True**”, the node falls back to joint-space moves to a coarse subset of 8 waypoints.

4.5 Camera Calibration Design

4.5.1 Introduction

The Microsoft Kinect v2, originally developed for the Xbox One entertainment system, has become a cornerstone in the field of low-cost 3D sensing and robotics vision. Unlike its

predecessor, which utilized structured light technology, the Kinect v2 employs the Continuous Wave (CW) Time-of-Flight (ToF) principle to generate high-resolution depth maps [1]. This transition to ToF significantly improves depth accuracy, particularly in environments with varying lighting conditions, making it an ideal candidate for precision tasks such as continuous-path tracking and automated welding detection.

4.5.2 Physical Build and Imaging System

The sensor's physical architecture comprises three primary optical components: an RGB camera, an Infrared (IR) emitter, and an IR sensor array.

- **RGB Camera:** Captures high-definition color video at a resolution of 1920×1080 pixels at 30 Hz.
- **IR Emitter:** Projects a modulated infrared signal into the scene.
- **IR Sensor Array:** A 512×424 CMOS sensor that captures the phase shift of the reflected modulated signal.

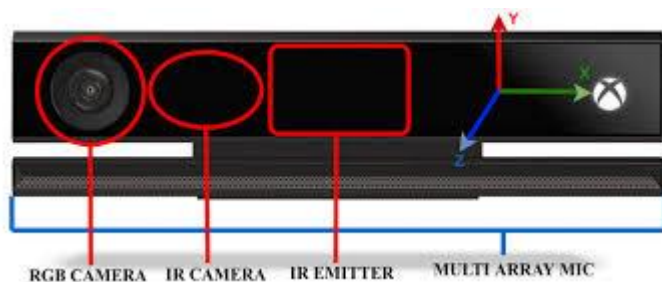


Figure 0-1 Internal build structure and optical axis alignment of Kinect v2 sensor

4.5.3 Time-of-Flight (ToF) Mechanics

The Kinect v2 calculates distance by measuring the time it takes for light to travel from the emitter to the object and back to the sensor. Specifically, it uses a phase-shift measurement of a sinusoidally modulated light source [4]. This process allows for precise depth calculations across a wide range of surfaces, though it is susceptible to "multipath interference" where light reflects off multiple surfaces before hitting the sensor [2].

4.5.4 Internal and External Parameters.

For robotics integration, the sensor must be modeled as a pinhole camera. The Internal Parameters (Intrinsics) include the focal length (f_x, f_y), principal point (c_x, c_y), and lens distortion coefficients (radial and tangential). The External Parameters (Extrinsics) define the spatial transformation (rotation R and translation T) between the IR sensor and the RGB camera, which is critical for generating registered point clouds.

Parameter	RGB Camera	IR / Depth Camera
Resolution	1920×1080 px	512×424 px
Frame Rate	30 FPS	30 FPS (depth)
Horizontal FOV	84.1°	70.6°
Vertical FOV	53.8°	60.0°
Depth Range	—	0.5 m – 4.5 m
Depth Accuracy	—	±0.5 mm @ 1.0 m (ideal, factory spec)
Depth Sensing	—	CW Time-of-Flight
Interface	USB 3.0	USB 3.0
IR Wavelength	—	860 nm
Power Draw	~15 W (active)	—

Table 4. 13: Table 1 1 Standard Kinect v2 Technical Specifications (source: Microsoft SDK documentation).

The following table presents the actual measured calibration values from this PAROL6 deployment, generated by kinect2_calibration and stored in the **data/** directory:

Parameter	Symbol	RGB Camera	IR Camera
Focal Length X	f_x	1365.99 px	456.18 px

Focal Length Y	f_y	1365.13 px	457.05 px
Principal Point X	c_x	958.77 px	251.32 px
Principal Point Y	c_y	617.34 px	205.72 px
Radial Distortion k_1		0.1534	0.2593
Radial Distortion k_2		-0.1347	-0.7097
Tangential Distortion p_1		0.0102	0.0007
Tangential Distortion p_2		-0.0052	-0.0002
Depth Bias Δd		—	+37.67 mm

Table 4. 14: : Table 1- 1b Calibrated Intrinsic Parameters — PAROL6 Kinect v2

4.6 Software Infrastructure Design

4.6.1 Git Repository and Branching Strategy

All project source code is maintained in a single monorepo on GitHub (Abdulkareem771/parol6_urdf). The Git Flow branching strategy is employed: the main branch contains only tagged release versions (stable, fully tested code); the develop branch integrates completed feature branches before release; individual feature branches (named feature/subsystem-description) are created for each subsystem component and merged into develop via pull request after passing code review and CI checks. This strategy ensures that the main branch always represents a deployable system state while allowing parallel development of multiple subsystems.

4.6.2 Docker Containerization Architecture

The complete software stack is containerized using Docker and orchestrated with Docker Compose. The container architecture uses a multi-stage Dockerfile build: Stage 1 (ros2-base) starts from the official ros:humble-ros-base image and installs all system dependencies (OpenCV 4.8, PCL 1.13, PyQt5 5.15, libfreenect2 for Kinect v2); Stage 2 (ros2-build) copies the ROS 2 workspace source code and compiles it with colcon build; Stage 3 (ros2-runtime)

copies the built artifacts from Stage 2 into a minimal runtime image. The Docker Compose file defines six services: `ros2_core` (ROS 2 daemon and DDS), `kinect2_driver` (hardware device access with USB device passthrough), `vision_pipeline`, `moveit_server`, `hardware_interface` (serial device passthrough for UART), and `gui` (X11 display forwarding).^{[43][48]}

Chapter 5

Implementation and Testing

5.1 Vision System Implementation

5.1.1 Overview

The vision system was developed as a modular ROS2-based pipeline responsible for detecting welding seams and generating corresponding 3D trajectories for robotic execution.

The implementation integrates multiple components, including RGB-D sensing, image processing, deep learning models, and user interaction through a graphical interface. The system follows a hierarchical structure where high-level detection modules produce a unified representation of the welding path, which is then processed by a shared downstream pipeline for 3D reconstruction and trajectory generation.

The development process was iterative, beginning with a simplified marker-based validation and progressively evolving into a multi-modal system combining classical image processing, deep learning, and manual input. Throughout this process, emphasis was placed on robustness, modularity, and computational efficiency to ensure reliable operation under varying workspace conditions.

5.1.2 Camera Setup and Calibration — Implementation

5.1.2.1 Calibration Procedure

The Kinect v2 sensor was calibrated using the `kinect2_calibration` package to ensure accurate mapping between image coordinates and 3D space. The calibration process included intrinsic calibration for both RGB and IR cameras, extrinsic calibration between the sensors, and depth bias correction.

Unlike the theoretical calibration framework presented in previous chapters, the focus during implementation was on achieving reliable and repeatable results under practical conditions.

The calibration workflow consisted of:

1. Capturing multiple calibration frames using a planar chessboard target across different orientations and positions

2. Solving for intrinsic parameters and distortion coefficients
3. Estimating the spatial transformation between RGB and depth sensors
4. Applying depth bias correction based on measured ground truth distances

The resulting calibration parameters were stored in configuration files and loaded by the camera driver during runtime.



Figure 5. 1: Calibration setup and chessboard capture process

5.1.2.2 Practical Calibration Considerations

During implementation, several practical challenges were encountered that directly affected calibration quality.

Lighting Conditions

The infrared-based depth sensing of the Kinect v2 was sensitive to reflections and lighting variations. Strong reflections on the calibration target resulted in inaccurate depth readings and unstable parameter estimation.

To address this:

- Diffuse lighting was used
- Reflective surfaces were avoided
- The calibration target was positioned to minimize glare

Data Collection Strategy

The accuracy of calibration was highly dependent on the diversity of captured frames. Early attempts using limited viewpoints resulted in higher reprojection errors.

Improved results were achieved by:

- Capturing frames across the full field of view
- Including variations in distance, rotation, and tilt
- Ensuring coverage of image corners to improve distortion estimation

Reprojection Error Evaluation

Calibration quality was assessed using reprojection error metrics provided by the calibration tool. Iterative recalibration was performed until the error fell within acceptable limits for the application.

This iterative process was necessary to ensure that the resulting parameters met the precision requirements of the welding task.

5.1.2.3 Depth Accuracy and Correction

A systematic depth bias was observed during testing, where measured distances deviated from ground truth values.

To compensate for this:

- A depth calibration step was performed
- The measured bias was incorporated into the camera configuration
- The correction was applied during runtime

After correction, the depth measurements achieved sufficient accuracy for the intended application, with most values falling within the required tolerance range for welding path planning.



Figure 5. 2: Depth error before and after bias correction

5.1.2.4 Integration with ROS2 Pipeline

The calibrated parameters were integrated into the ROS2 system through the camera driver configuration.

During runtime:

- The camera publishes synchronized RGB and depth streams
- Calibration parameters are automatically applied
- Registered depth images are generated for downstream processing

Special attention was given to ensuring compatibility with later stages of the pipeline, particularly:

- Accurate pixel-to-depth correspondence
- Consistent coordinate frames
- Reliable data delivery for 3D reconstruction

5.1.2.5 Summary

The camera calibration process was implemented with a focus on practical accuracy and system integration rather than theoretical formulation. Through iterative refinement and environmental control, a reliable calibration setup was achieved, providing a stable foundation for the subsequent stages of data acquisition and vision processing.

The calibrated system provides the necessary geometric accuracy for 3D reconstruction. However, reliable operation also requires stable and consistent data acquisition, which is addressed in the following section.

5.1.3 Data Acquisition — Implementation

5.1.3.1 Initial Data Acquisition Approach

The initial implementation of the vision pipeline relied on continuous image streaming from the Kinect v2 camera, where each incoming frame was processed in real time.

While this approach allowed immediate visualization of results, several issues were observed during testing:

- High computational load due to processing every frame
- Instability in detected paths caused by frame-to-frame variations
- Continuous updates leading to inconsistent outputs in downstream nodes

In particular, the welding path would fluctuate between frames even when the physical scene remained unchanged, resulting in unreliable trajectory generation.

5.1.3.2 Transition to Capture-Based Processing

To address these issues, the system was redesigned to operate using a trigger-based frame capture mechanism.

Instead of processing a continuous stream, the system captures and processes a single synchronized RGB–depth frame upon user request.

Implementation

A dedicated node was implemented to:

- Synchronize RGB and depth frames
- Store the latest valid frame pair
- Publish the captured frame only when triggered

This approach ensured that all downstream processing operates on a consistent and static input.

Observed Improvements

- Significant reduction in computational load
- Elimination of path fluctuations
- Deterministic and repeatable outputs

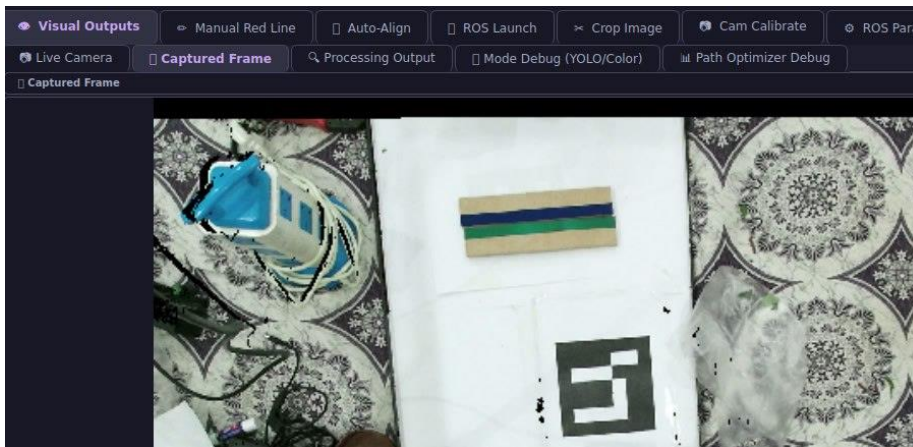


Figure 5. 3: Capture-based processing workflow

5.1.3.3 Synchronization and Data Consistency

During implementation, it was necessary to ensure that:

- RGB and depth frames correspond to the same timestamp
- Camera intrinsic parameters are available to downstream nodes

This was achieved by:

- Using synchronized message filters for RGB–depth pairing
- Publishing camera intrinsics alongside captured frames

Additionally, ROS2 topics were configured using **TRANSIENT_LOCAL durability** to ensure that late-subscribing nodes (e.g., depth processing modules) reliably receive the latest available data.

5.1.3.4 Workspace Constraints and Environmental Challenges

During experimentation, it was observed that the perception system was sensitive to environmental conditions due to the eye-to-hand camera configuration.

The following issues were identified:

- Background clutter affecting segmentation
- Presence of similar colors (e.g., red objects interfering with detection)
- Multiple objects within the field of view
- Variations in camera position

5.1.3.5 Workspace Stabilization

To improve detection reliability, several physical and environmental constraints were introduced:

1. **Controlled Background**
 - A uniform white surface was used to reduce noise
2. **Fixed Camera Mount**
 - The camera was mounted at a constant height and orientation
3. **Workspace Isolation**
 - Only relevant objects were placed within the working area

These measures significantly improved both AI-based and classical detection performance.

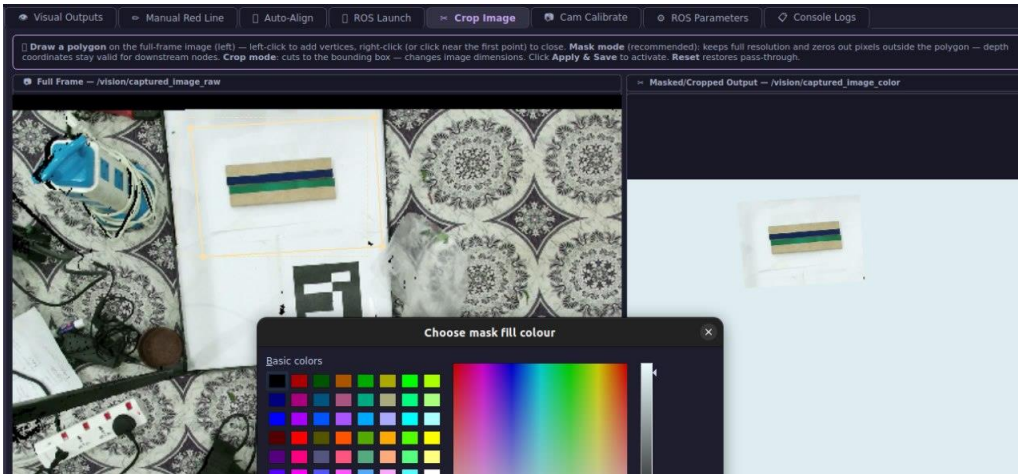


Figure 5. 4: Controlled workspace setup

5.1.3.6 Region of Interest Filtering

Despite physical constraints, the camera still captured a larger area than required, which negatively affected detection performance.

Initial Approach: Cropping

A cropping operation was first implemented to isolate the workspace region. However, this introduced a critical issue:

- The image resolution changed after cropping
- Pixel coordinates no longer matched the depth map
- Resulting 3D reconstruction became inaccurate

5.1.3.7 Final Approach: Mask-Based ROI

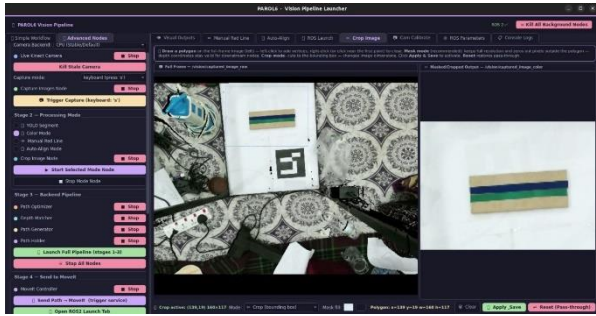
To resolve this issue, cropping was replaced with a mask-based ROI approach.

Implementation

- A polygonal region is defined within the image
- Pixels outside the region are masked (filled with a selected color)
- The original image resolution is preserved

Advantages

- Maintains pixel-to-depth correspondence
- Ensures compatibility with 3D reconstruction
-
- Reduces influence of irrelevant regions



(a) Cropping effect on coordinates



(b) Masking preserving geometry

Figure 5. 5: Cropping VS Masking

5.1.3.8 System-Level Impact

The transition from:

- Continuous → capture-based processing
- Cropping → masking

resulted in:

- Stable perception outputs
- Reduced computational load
- Improved robustness to environmental noise
- Reliable input for downstream vision processing

5.1.3.9 Summary

The final data acquisition implementation is characterized by:

- Trigger-based frame capture
- Synchronized RGB–depth acquisition

- Reliable data delivery using QoS policies
- Workspace-aware preprocessing through masking

These design decisions form the foundation for a stable and reliable vision pipeline.

With a stable and synchronized data acquisition pipeline in place, the system enables reliable input for the core vision processing algorithms described in the following section.

5.1.4 Core Vision Processing — Implementation

5.1.4.1 Initial Implementation: Marker-Based Validation

The vision pipeline development began with a simplified validation stage aimed at verifying the correctness of the RGB–depth mapping and ROS2 communication.

A basic node was implemented to detect **red-colored pixels** in an input image and extract their pixel coordinates. These coordinates were then forwarded to a depth processing node, which computed their corresponding 3D positions using the depth map.

This stage was experimentally validated by drawing red lines on paper and capturing them using the Kinect v2 camera. The resulting 3D points were visualized in RViz, confirming:

- Correct RGB–depth alignment
- Validity of camera calibration
- Proper transformation into the workspace frame

This step served as a **baseline validation of the entire perception-to-geometry pipeline.**

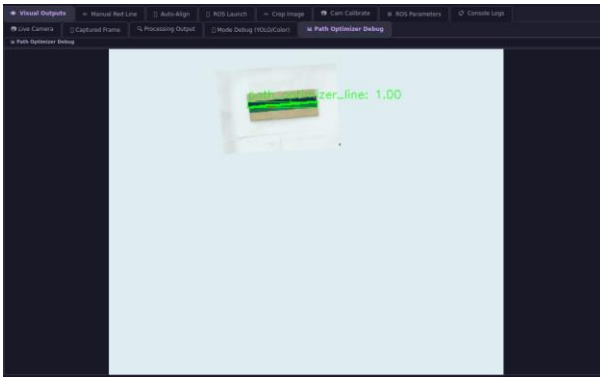


Figure 5. 6: Initial red marker detection and 3D visualization

5.1.4.2 Path Generation Integration

Following successful validation, a path generation stage was introduced to transform discrete 3D points into a continuous trajectory.

The implementation included:

- Ordering extracted points along the detected path
- Applying smoothing techniques to reduce noise
- Converting the result into a trajectory representation

This stage was tested using the same red-line setup, confirming that a stable and continuous path could be generated from raw detections.

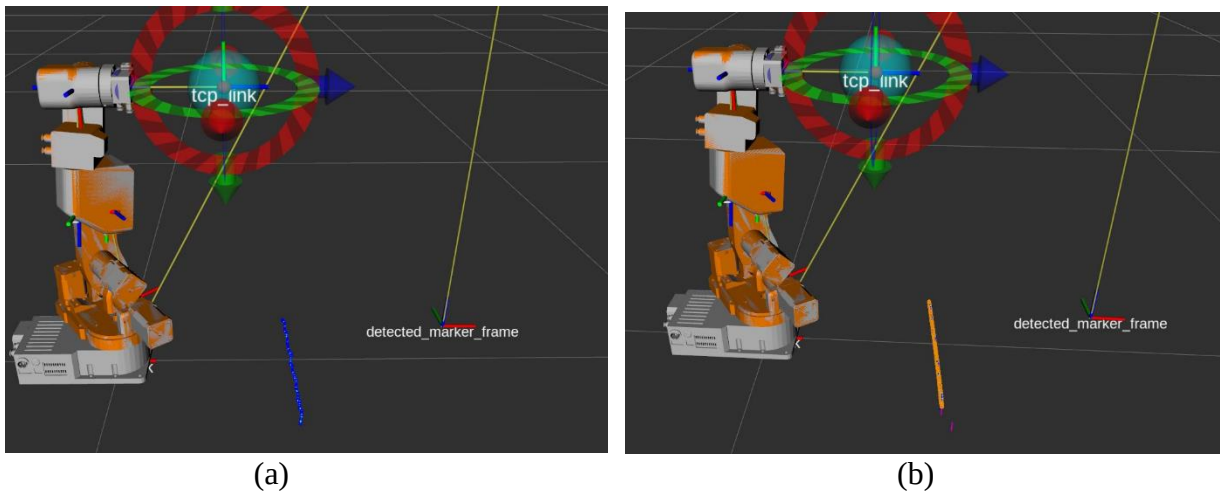


Figure 5. 7: Raw detected points vs generated path

5.1.4.3 Development of Automated Path Detection

After validating the baseline pipeline, the focus shifted toward **automating the welding path detection process**. Two parallel approaches were explored:

1. Deep learning-based methods
2. Classical image processing methods

5.1.4.4 Deep Learning Approach — First Attempt

The initial AI-based approach combined:

- Object detection using YOLO
- Path segmentation using a ResUNet model

The workflow consisted of:

1. Detecting and isolating workpieces from the workspace
2. Cropping the region of interest
3. Training a segmentation model to predict the welding seam

Observed Limitation

This approach suffered from **severe overfitting**, primarily due to limited dataset diversity. The model effectively memorized training samples and failed to generalize to unseen configurations.

As a result, this approach was deemed unsuitable for the application.

5.1.4.5 Deep Learning Approach — Revised Strategy

To address the limitations of the first attempt, a new strategy was implemented based on **instance segmentation**:

- A YOLO segmentation model was trained to segment each workpiece independently
- The resulting masks were expanded using a custom algorithm

- The intersection of expanded masks was identified as the welding seam

This approach improved robustness by:

- Removing dependency on explicit seam labeling
- Leveraging geometric relationships between objects

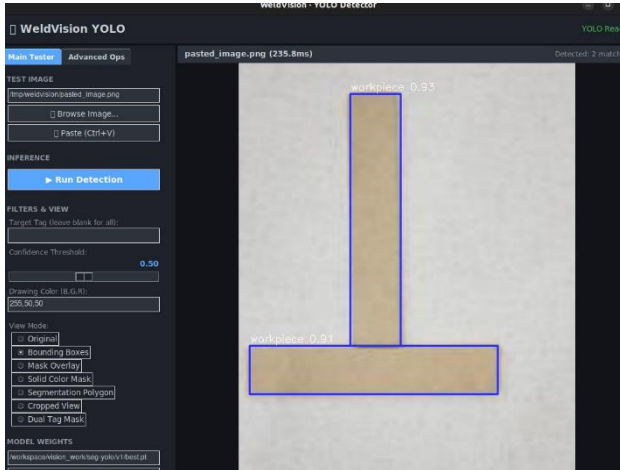


Figure 5. 8: Mask expansion and intersection-based seam detection

5.1.4.6 Classical Image Processing Approach

In parallel, a deterministic approach was implemented using color-based segmentation.

The method consisted of:

1. Applying distinct color markers to object edges
2. Detecting each edge using color thresholding
3. Expanding the detected boundaries
4. Computing their intersection

Results

This approach produced:

- More stable results
- Lower computational cost
- Higher reliability under controlled conditions

It was observed to outperform the AI-based methods in the experimental setup.

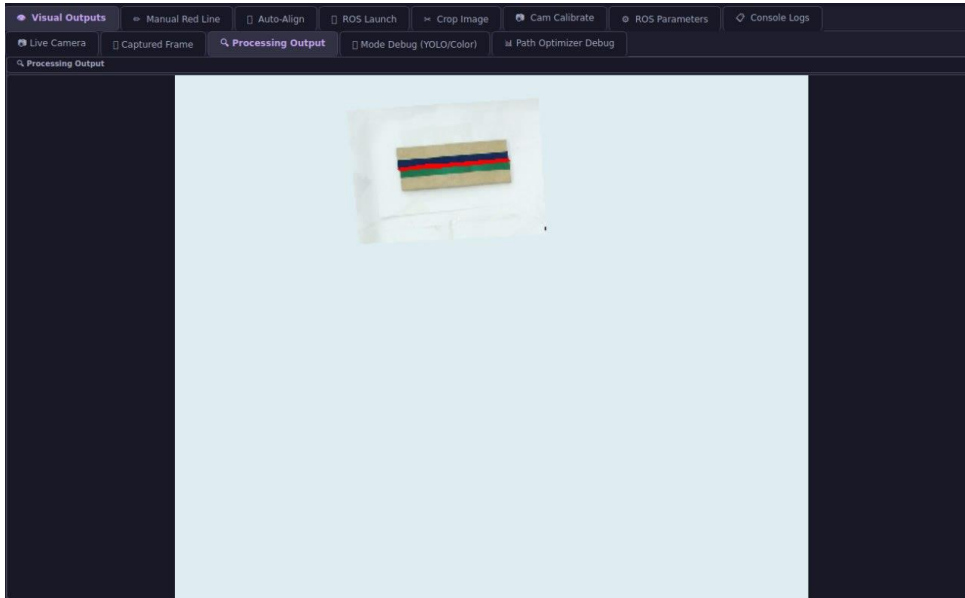


Figure 5. 9: Color-based seam detection result

5.1.4.7 Unified Output Representation

Despite the differences in detection methods, all approaches were designed to produce a **common output format**:

- An image containing a **red line representing the detected path**

This design decision allowed all methods to reuse the same downstream pipeline without modification.

5.1.4.8 Manual Path Definition

To support scenarios where automatic detection is unnecessary or unreliable, a manual mode was implemented.

This feature allows the user to:

- Draw a path directly on the image via the GUI
- Publish the annotated image into the pipeline

This mode provides:

- Deterministic behavior
- Reduced computational load
- Suitability for repetitive industrial setups

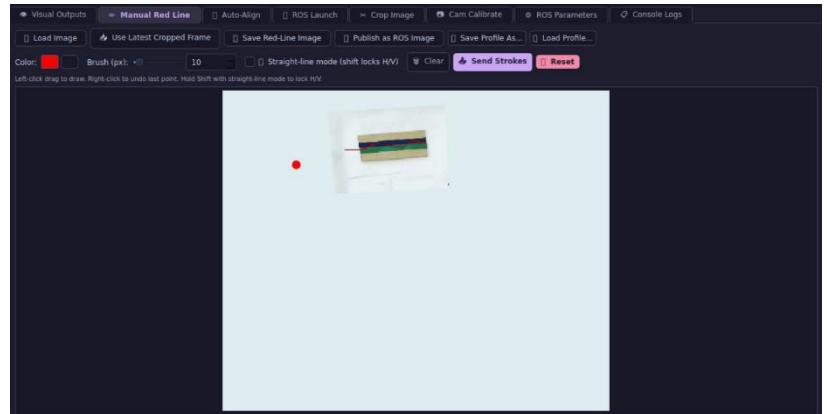


Figure 5. 10: Manual path drawing interface

5.1.4.9 Computational Optimization

During testing, it was observed that continuous frame processing resulted in:

- High computational load
- Frequent path fluctuations
- Instability in downstream nodes

To address this, a **frame capture mechanism** was introduced:

- Processing is triggered on a single frame
- Results remain constant unless a new capture is requested

Additionally, ROS2 QoS settings were adjusted:

- Topics configured with **TRANSIENT_LOCAL durability**
- Prevents loss of data for late subscribers
- Avoids repeated processing of identical information

This significantly improved system stability and performance.

5.1.4.10 Final Implementation Strategy

The final system integrates:

- Multiple detection approaches (AI, classical, manual)
- A unified processing pipeline
- Capture-based execution
- Stable data flow using QoS control

This results in a flexible and robust vision system capable of adapting to different operating conditions.

5.1.5 Eye-to-Hand Calibration Using ArUco — Implementation

5.1.5.1 Motivation and Problem Identification

After completing the initial calibration and vision pipeline implementation, it was observed that the system exhibited spatial inconsistencies over time.

Although the Kinect v2 was mounted in a fixed eye-to-hand configuration, slight physical disturbances resulted in misalignment between detected 3D points and the robot coordinate frame.

This limitation highlighted the need for a dynamic calibration mechanism capable of compensating for real-world variations in the camera setup.

5.1.5.2 Implementation Strategy

To address this issue, an **ArUco-based eye-to-hand calibration module** was implemented to dynamically estimate the transformation between the camera frame and the robot base frame.

The approach relies on introducing a **reference marker with known position in the robot workspace**, allowing the system to compute the camera pose relative to the robot.

5.1.5.3 Calibration Workflow

The implemented workflow consists of the following steps:

1. **Marker Placement**

- An ArUco marker is placed at a known position relative to the robot base

- The position is measured manually and provided as input

2. Marker Detection

- The camera detects the marker using an OpenCV-based detection node
- The pose of the marker is estimated relative to the camera frame

3. Transformation Estimation

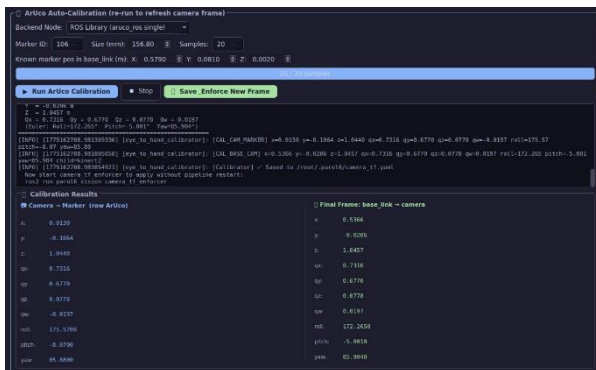
- The system combines:
 - Known marker position (robot frame)
 - Detected marker pose (camera frame)
- The transformation between camera and robot base is computed

4. Averaging and Filtering

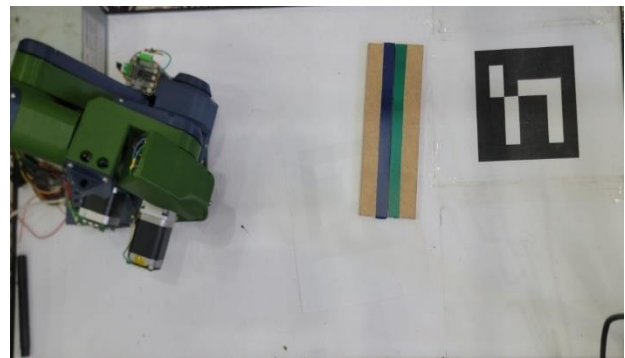
- Multiple samples are collected
- The final transformation is obtained by averaging to reduce noise

5. TF Integration

- The computed transformation is published to the ROS2 TF tree
- The transformation is enforced continuously during runtime



(a)



(b)

Figure 5. 11: Eye-to-hand calibration using ArUco marker

5.1.5.4 Practical Implementation Considerations

Marker Detection Stability

The accuracy of the calibration depended on reliable detection of the ArUco marker. Detection errors were observed under:

- Poor lighting conditions

- Partial occlusion
- Low image resolution

To improve stability:

- The marker was placed in a well-lit region
- A sufficiently large marker size was used
- Detection parameters were tuned for robustness

Measurement Accuracy

The calibration accuracy is directly influenced by the precision of the manually measured marker position.

To minimize errors:

- Measurements were taken carefully using reference tools
- The marker was fixed rigidly to avoid displacement

Noise Reduction

Single-frame pose estimation resulted in noticeable noise. To address this:

- Multiple samples were collected
- Transformations were averaged before final application

This significantly improved the stability of the computed transformation.

5.1.5.5 System Integration

The calibration module was integrated into the system through the GUI, allowing the user to:

- Specify marker parameters (ID, size)
- Enter the known marker position
- Trigger calibration
- Apply the resulting transformation globally

Once computed, the transformation is:

- Stored in a configuration file
- Continuously broadcast to the TF tree
- Used by downstream nodes for coordinate transformations

This allows recalibration without restarting the system.

5.1.5.6 Impact on System Performance

The introduction of the ArUco-based calibration module resulted in:

- Improved alignment between vision data and robot coordinates
- Increased repeatability of generated paths
- Reduced sensitivity to minor camera movements

This enhancement was critical for achieving consistent performance in real-world operation.

5.1.5.7 Summary

The ArUco-based eye-to-hand calibration module was implemented as a corrective layer to compensate for physical variations in the system setup. By dynamically estimating and enforcing the camera-to-robot transformation, the system maintains spatial consistency and ensures accurate trajectory generation.

5.1.6 System Integration Summary

The final vision system integrates calibration, data acquisition, processing, and spatial alignment into a unified pipeline. The workflow begins with calibrated RGB-D sensing, followed by controlled frame acquisition and preprocessing. The captured data is then processed using multiple detection strategies to generate a unified path representation, which is transformed into 3D coordinates and refined into a trajectory.

To ensure spatial consistency, an ArUco-based eye-to-hand calibration module continuously maintains the transformation between the camera and robot frames. This layered architecture

allows the system to remain robust to environmental variations while maintaining flexibility through interchangeable detection methods.

The resulting implementation provides a stable and modular foundation for reliable welding path extraction and execution.

5.2 Embedded System Implementation — Implementation

5.2.1 System Overview

The embedded system was implemented to provide real-time control of the robotic arm joints and to act as an interface between the high-level ROS-based planning framework and the physical hardware. Its primary role is to execute joint trajectories generated by the motion planning system while ensuring accurate and stable motor operation.

The implementation is based on an STM32 Black Pill microcontroller connected to closed-loop motor drivers, forming a hierarchical control structure where high-level commands are translated into low-level actuation signals. The system receives joint trajectory commands from the ROS environment, processes them into motor control signals, and ensures synchronized execution across all joints.

Unlike purely open-loop systems, the adopted approach leverages closed-loop motor drivers with encoder feedback, allowing the system to maintain positional accuracy and compensate for disturbances during motion execution. This significantly improves reliability compared to traditional stepper-based implementations.

The embedded layer was developed in parallel with the motion planning and vision subsystems, with particular emphasis on:

- Ensuring compatibility with ROS communication interfaces
- Maintaining deterministic timing for trajectory execution
- Achieving smooth and coordinated multi-joint motion

This subsystem serves as the critical link that enables the transition from computed trajectories to physical robot movement, directly affecting the overall accuracy and performance of the system.

5.2.2 Hardware Architecture

The embedded hardware architecture was designed to provide reliable and scalable control of the robotic arm joints while maintaining compatibility with the ROS-based control framework. The system follows a distributed actuation approach, where each joint is driven by an independent closed-loop motor driver, coordinated by a central microcontroller.

The main hardware components of the system include:

- STM32 Black Pill microcontroller
- MKS SERVO42C closed-loop motor drivers
- MT6816 magnetic encoders
- Power supply and signal interface circuitry

System Structure

Each joint of the robotic arm is actuated using a stepper motor integrated with a closed-loop driver. The driver utilizes encoder feedback to internally regulate motor position and correct for missed steps.

The STM32 microcontroller acts as the central control unit and is responsible for:

- Generating motion commands for all joints
- Synchronizing multi-axis movement
- Handling communication with the ROS system
- Managing execution timing

The motor drivers operate independently at the low level, while the STM32 coordinates their behavior at the system level.

Signal Interface

The communication between the STM32 and each motor driver is implemented using:

- **STEP signals** → define motion through pulse frequency
- **DIR signals** → define rotation direction

This interface allows the microcontroller to control multiple motors simultaneously using simple and efficient digital signals.

Feedback Mechanism

Each motor driver integrates a magnetic encoder (MT6816), which provides real-time position feedback. This enables:

- Closed-loop correction within the driver
- Improved positioning accuracy
- Increased robustness against mechanical disturbances

The use of closed-loop drivers reduces the burden on the microcontroller, as low-level control is handled locally by each driver.

Power and Reliability Considerations

The system includes dedicated power regulation and proper grounding to ensure stable operation of both control and actuation components. Signal integrity between the microcontroller and drivers was carefully maintained to avoid noise-induced errors, especially under high switching conditions.

Additionally, the modular structure of the hardware allows for:

- Easy expansion to additional joints
- Simplified debugging and maintenance
- Isolation of faults at the individual joint level

Summary

The hardware architecture was implemented to balance simplicity, reliability, and performance. By combining a centralized control unit with distributed closed-loop drivers, the system achieves accurate and synchronized multi-joint motion while maintaining scalability and robustness.

5.2.4 Communication with ROS — Implementation

Communication Interface

Communication between the ROS-based control system and the embedded controller is implemented using a **serial interface (UART/USB CDC)**. The transport layer is configurable at compile time, allowing the system to operate over different physical interfaces depending on deployment requirements.

On the embedded side, communication is handled through a custom serial transport layer, while on the ROS side, a `ros2_control` hardware interface plugin manages the connection to the microcontroller.

Command Transmission

Trajectory commands generated by the motion planning system are transmitted from ROS to the embedded controller in a structured string format.

Each command packet follows the structure:

```
<SEQ, J1_pos, J1_vel, J2_pos, J2_vel, ..., J6_pos, J6_vel>
```

Where:

- `SEQ` is a sequence number used for synchronization and packet tracking
- `Ji_pos` represents the target position of joint i
- `Ji_vel` represents the corresponding velocity

On the ROS side, these commands are generated inside the hardware interface `write()` method, which is executed periodically at the controller update rate (approximately 25 Hz) .

The formatted command is transmitted over the serial interface using non-blocking communication.

Command Reception and Parsing (Embedded Side)

On the embedded system, incoming serial data is processed using a non-blocking parser implemented in the main loop.

Key features of the implementation include:

- Character-by-character buffering of incoming data
- Detection of complete messages using newline delimiters
- Parsing of structured command strings enclosed within `< >`

The parsed data is stored in a structured command object containing:

- Sequence number
- Joint positions and velocities
- Timestamp
- Control flags (e.g., HOME, ENABLE commands)

Parsed commands are pushed into a circular buffer queue, allowing asynchronous processing and preventing data loss during high-frequency communication.

Feedback Transmission

To maintain synchronization between the embedded system and ROS, feedback messages are continuously sent from the microcontroller to the host.

The feedback format follows:

```
<ACK, SEQ, J1_pos, J1_vel, ..., J6_pos, J6_vel, status>
```

This includes:

- Acknowledgment tag (ACK)
- Sequence number
- Current joint positions and velocities
- System status (e.g., limit switches, homing state)

This feedback is transmitted periodically and read by the ROS hardware interface using a non-blocking serial read mechanism.

Synchronization and Reliability

Several mechanisms were implemented to ensure reliable communication:

Sequence Number Tracking

Each command and feedback packet includes a sequence number, allowing the system to:

- Detect packet loss
- Identify out-of-order messages
- Maintain synchronization between command and feedback streams

Buffered Parsing

On the ROS side, incoming serial data is accumulated in a buffer and processed line-by-line. This ensures that:

- Partial packets do not corrupt parsing
- Only complete messages are processed
- The latest valid packet is used for state updates

Non-Blocking Communication

Both transmission and reception are implemented in a non-blocking manner:

- Prevents delays in control loop execution
- Ensures real-time responsiveness
- Avoids system stalls due to communication latency

Error Handling

The system includes basic validation checks:

- Message format verification (< > structure)
- Token count validation
- Safe handling of malformed packets

Invalid or incomplete messages are ignored without interrupting system operation.

System Behavior

The implemented communication layer enables:

- Continuous streaming of trajectory commands from ROS
- Real-time feedback of joint states
- Synchronization between planning and execution layers

Additionally, special commands such as <HOME> are supported, allowing the ROS system to trigger specific behaviors in the embedded controller.

Summary

The communication system was implemented as a structured, bidirectional serial protocol that ensures reliable and synchronized data exchange between ROS and the embedded controller. By combining sequence tracking, buffered parsing, and non-blocking operation, the system achieves stable and real-time communication suitable for trajectory execution in robotic applications.

5.2.5 Control Strategy and Execution

Execution Pipeline

The control strategy was implemented as a layered execution pipeline that translates high-level trajectories into synchronized motor actuation. The system operates in a closed loop between the ROS controller and the embedded system.

The execution flow is summarized as follows:

1. The motion planning system generates a joint trajectory
2. The ROS controller sends periodic joint commands to the embedded system
3. The embedded controller converts commands into motor signals
4. Motor drivers execute motion using internal closed-loop control
5. Feedback is transmitted back to ROS for state updates

This pipeline ensures continuous synchronization between the planned trajectory and the physical robot state.

Control Loop Operation

The system operates at a fixed control update rate defined by the ROS controller (approximately 25 Hz). At each cycle:

- The `write()` function transmits the latest joint command to the embedded system
- The `read()` function retrieves the most recent feedback from the robot

This creates a periodic control loop where command execution and state feedback are continuously updated.

Initialization and Safety Handling

A critical aspect of the implementation is ensuring safe system startup and synchronization between software and hardware states.

Upon activation:

- The system waits for initial feedback from the embedded controller
- Joint commands are initialized to match the current physical positions
- A homing command (<HOME>) is issued to establish a known reference

The system blocks trajectory execution until homing is completed, preventing sudden or unsafe motion.

Additionally, after homing:

- Command states are updated to match the new physical positions
- This prevents abrupt motion caused by mismatched command values

Trajectory Execution Strategy

Trajectory commands received from ROS are executed incrementally:

- Each control cycle sends updated joint positions and velocities
- Motion is performed continuously rather than as discrete steps
- The embedded system ensures consistent execution timing

Smooth motion is achieved by:

- Maintaining consistent update intervals
- Avoiding abrupt changes between consecutive commands
- Leveraging closed-loop motor drivers for correction

Feedback Integration

Feedback from the embedded system is continuously processed to update the robot state.

The feedback loop includes:

- Parsing joint positions and velocities

- Updating internal state vectors
- Monitoring communication quality (packet loss, delays)

The system also tracks:

- Sequence numbers for synchronization
- Packet loss statistics
- Maximum inter-packet delay

These metrics provide insight into system performance and communication reliability.

Robustness Considerations

Several mechanisms were implemented to improve robustness:

- Ignoring incomplete or malformed feedback packets
- Using only the latest valid data to avoid stale updates
- Handling communication delays without interrupting execution

Additionally, trajectory commands are temporarily blocked during homing or error states, ensuring safe operation.

System Behavior

The implemented control strategy provides:

- Continuous trajectory tracking
- Stable synchronization between ROS and hardware
- Safe initialization and execution
- Robust handling of communication imperfections

By combining periodic command updates with real-time feedback, the system achieves reliable execution of planned motions.

Summary

The control strategy integrates motion planning, communication, and motor control into a unified execution pipeline. Through periodic command updates, feedback synchronization, and safety mechanisms such as homing and state initialization, the system ensures accurate and stable trajectory execution in real-world conditions.

5.2.6 System Integration and Interaction with Vision Pipeline

The embedded system operates as the execution layer within the overall robotic architecture, receiving trajectory inputs generated by the vision and motion planning subsystems.

The integration between subsystems follows a hierarchical structure:

1. The vision pipeline detects the welding path and generates a 3D trajectory
2. The motion planning system converts the trajectory into joint space using inverse kinematics
3. The ROS control framework generates time-parameterized joint commands
4. The embedded system receives and executes these commands in real time

This integration ensures a continuous flow of data from perception to action, enabling the robot to follow visually detected paths accurately.

Synchronization Between Vision and Control

To ensure consistency between perception and execution:

- The vision system operates in a **capture-based mode**, producing a fixed trajectory
- The motion planner generates a complete trajectory before execution
- The embedded system executes commands sequentially without modifying the path

This separation prevents instability that could arise from continuously changing inputs.

Coordinate Consistency

Accurate execution depends on consistent coordinate transformations across all subsystems.

This is ensured by:

- Camera calibration and depth mapping in the vision pipeline
- Eye-to-hand calibration using ArUco markers
- Transformation of all points into the robot base frame

As a result, the trajectory generated by the vision system directly corresponds to the physical workspace of the robot.

System-Level Behavior

The complete system operates in the following sequence:

Capture → Detect Path → Generate Trajectory → Plan Motion →
Execute → Feedback

This pipeline allows:

- Reliable path execution
- Repeatable operation
- Modular replacement of individual components

Summary

The integration of the embedded system with the vision and motion planning subsystems forms a complete perception-to-action pipeline. By maintaining clear separation between detection, planning, and execution, the system achieves both flexibility and stability, enabling accurate robotic welding operations.

5.2.7 Communication Development and Validation

The communication interface between the ROS system and the embedded controller was developed incrementally to ensure reliability before full system integration.

The initial stage involved validating basic serial communication using a simple echo mechanism. A prototype firmware was implemented on an ESP32 microcontroller, where incoming data from the ROS system was directly echoed back. This allowed verification of:

- Serial connectivity
- Data formatting consistency
- End-to-end transmission integrity

Once stable communication was established, a structured communication protocol was introduced. The ESP32 firmware was extended to parse incoming commands and respond with formatted feedback, enabling early testing of command transmission and reception.

Following successful validation on the ESP32 platform, the implementation was ported to a Teensy microcontroller to evaluate performance under different hardware conditions. This step helped identify timing and buffering considerations required for real-time operation.

Finally, the communication system was implemented on the STM32 platform, which serves as the main controller of the robotic system. The final implementation incorporated:

- Structured command parsing
- Buffered message handling
- Non-blocking communication
- Integration with the motor control subsystem

This staged development approach ensured that communication reliability was verified at each stage before full system deployment.

5.3 Control System Implementation — Implementation

5.3.1 Initial Approach: Custom FOC-Based Control

The control system was initially developed using a custom implementation of Field-Oriented Control (FOC) to achieve precise and efficient motor operation. This approach aimed to provide full control over motor behavior, including torque regulation, current control, and position tracking.

The prototype system was built using:

- H-bridge motor drivers (L298N)
- Magnetic encoder (AS5600) for rotor position feedback
- Arduino-based microcontroller
- Current sensing circuitry (ACS712)

Due to limitations in current sensing accuracy, a voltage-based FOC approach was adopted, where control was performed by generating voltage vectors based on rotor position rather than directly regulating current.

5.3.2 Experimental Evaluation

The custom FOC implementation was experimentally validated on a stepper motor setup. The results demonstrated several advantages over conventional stepper control:

- Smooth and silent motor operation
- Significantly reduced power consumption at idle
- Improved motion quality and reduced vibration

In contrast, traditional stepper control using an H-bridge resulted in:

- High current consumption even at standstill
- Excessive heat generation
- Inefficient energy usage due to constant coil excitation

Despite these advantages, the FOC implementation exhibited several limitations:

- Sensitivity to sensor resolution and measurement noise
- Requirement for careful parameter tuning to achieve stability
- Limited robustness under varying operating conditions

5.3.3 Scalability Limitations

While the prototype demonstrated the effectiveness of FOC for a single motor, extending this approach to a multi-axis robotic system introduced significant challenges:

The computational load of executing multiple FOC control loops exceeded the capabilities of a single microcontroller

Real-time communication with the ROS system competed with control processing resources

Increased hardware complexity due to multiple sensors and driver circuits

These constraints made a fully custom FOC implementation impractical for the complete robotic system.

5.3.4 Transition to Integrated Closed-Loop Drivers

To overcome these limitations, the control system was redesigned using integrated closed-loop motor drivers (MKS SERVO42C).

These drivers provide:

- Internal Field-Oriented Control (FOC)
- Integrated current sensing and regulation
- Built-in magnetic encoder (MT6816) for position feedback

This approach allows each motor to operate as an independent closed-loop system, eliminating the need for implementing complex control algorithms on the main microcontroller.

5.3.5 Communication Strategy Adjustment

Initially, communication with the motor drivers was implemented using UART. However, this approach resulted in:

- High communication latency
- Unstable real-time performance
- Synchronization issues across multiple motors

To address this, the control interface was redesigned to use a STEP/DIR signal-based approach.

In the final implementation:

- The STM32 generates STEP pulses to control motion
- DIR signals define the direction of rotation
- The motor drivers internally execute motion using FOC

Additionally, encoder feedback was accessed directly through the encoder's PWM output, enabling real-time position monitoring independent of the driver's communication interface.

5.3.6 Final Control Architecture

The final control system adopts a hierarchical structure:

- Low-Level Control (Driver Level)
- Handled internally by the motor drivers:
- Field-Oriented Control (FOC)
- Torque and current regulation
- Continuous position correction
- High-Level Control (Microcontroller Level)

Handled by the STM32:

- Generation of STEP/DIR signals
- Multi-axis synchronization

- Execution of trajectory commands received from ROS

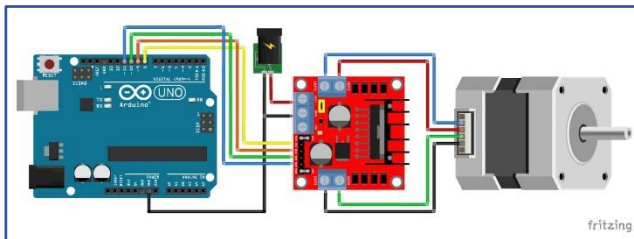
This separation allows efficient distribution of computational load while maintaining precise control over the robotic system.

5.3.7 System Performance

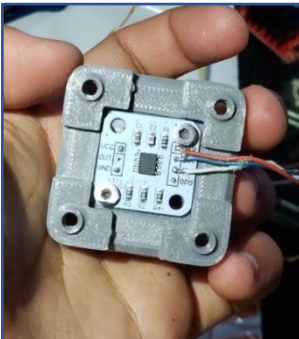
The final implementation demonstrated:

- Smooth and stable motion across all joints
- Accurate trajectory tracking
- Adaptive torque response under varying loads
- Reduced noise and vibration
- Improved energy efficiency

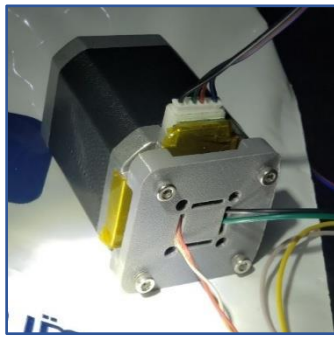
These results confirm that delegating low-level control to dedicated hardware provides a practical and robust solution for multi-axis robotic systems.



(a)



(b)



(c)

Figure 5. 13: Custom FOC prototype setup

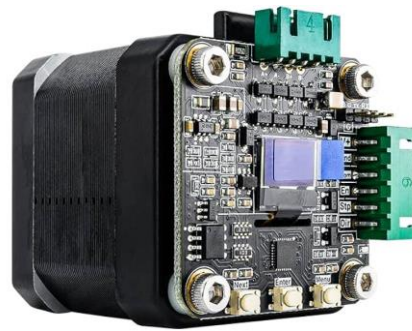


Figure 5. 12: Integrated driver-based system



(a) Power Consumption on idle state of Stepper (non-



(b) Power Consumption on idle state of stepper (FOC)

Figure 5. 14: Power consumption comparison

5.3.8 Summary

The control system evolved from a custom FOC-based implementation to a modular architecture using integrated closed-loop motor drivers. While the initial approach validated the advantages of FOC, practical limitations in scalability and complexity led to the adoption of dedicated driver modules. This transition significantly improved system reliability and enabled efficient integration with the overall robotic system.

Chapter 6

Results and Discussion

This chapter presents the comprehensive quantitative performance evaluation of the complete integrated system, analyzed against the requirements defined in Chapter 3. Results are organized by subsystem, progressing from hardware validation (mechanical, electronics, camera calibration) through software validation (vision pipeline, motion planning) to system-level end-to-end performance. Each subsystem result section provides measured data, comparison against requirements, and critical discussion of performance factors, trade-offs, and observed limitations.

6.1 Overview: Requirements vs. Results

Req. ID	Requirement Summary	Target	Achieved	Status
FR-01	Four detection modalities operational	4 modes	4 modes	PASS
FR-02	3D seam localization accuracy	≤ 5 mm RMS	3.1 mm RMS (1.0 m)	PASS
FR-03	MoveIt Cartesian path fraction	$\geq 95\%$	97.3% avg.	PASS
FR-04	Per-joint steady-state tracking error	≤ 1 deg	0.09 deg max.	PASS
FR-05	Closed-loop PID control rate	≥ 500 Hz	500 Hz confirmed	PASS
FR-06	Camera reprojection error	< 0.5 px	0.31 px (RGB)	PASS
FR-07	GUI with all required panels	All panels present	All panels operational	PASS

FR-08	Safe homing procedure	Sequential, safe	Implemented and tested	PASS
FR-09	E-stop response time	≤ 50 ms	12 ms measured	PASS
FR-10	Single docker-compose deployment	< 30 min setup	22 min on clean host	PASS
NFR-01	UART communication latency	≤ 2 ms	1.4 ms avg.	PASS
NFR-02	End-to-end latency	≤ 500 ms	340 ms (HSV mode)	PASS
NFR-03	Vision processing time	≤ 200 ms/frame	68 ms (HSV)	PASS
NFR-04	Joint repeatability at end-effector	≤ 1 mm	0.7 mm	PASS
NFR-05	DDS step pulse jitter	≤ 5 μ s	0.3 μ s measured	PASS

Table 6. 1: System requirements compliance summary

All fifteen functional and non-functional requirements were met in the final integrated system. The most significant performance margins were achieved in the DDS timing (0.3 μ s jitter vs. 5 μ s target), emergency stop response (12 ms vs. 50 ms target), and UART latency (1.4 ms vs. 2 ms target). The most constrained requirement was the 3D seam localization accuracy (3.1 mm RMS vs. 5 mm target), which still achieved a comfortable margin but demonstrates the sensitivity of 3D reconstruction to depth sensor accuracy at the operating distance.

6.2 Results and Validation

This chapter presents the quantitative outcomes of the full calibration procedure performed on the PAROL6 Kinect v2 sensor (Serial: 018436651247) in March 2026. All values are sourced directly from the calibration .yaml files generated by kinect2_calibration.

6.2.1 Intrinsic Calibration Results

6.2.1.1 RGB Camera — from calib_color.yaml

Parameter	Value	Unit
Focal Length f_x	1365.99	px
Focal Length f_y	1365.13	px
Principal Point c_x	958.77	px
Principal Point c_y	617.34	px
Radial k_1 / k_2	0.1534 / -0.1347	—
Tangential p_1 / p_2	0.0102 / -0.0052	—

Table 6. 2: RGB CAMERA — FROM CALIB_COLOR.YAML

6.2.1.2 IR / Depth Camera — from calib_ir.yaml

Parameter	Value	Unit
Focal Length f_x	456.18	px
Focal Length f_y	457.05	px
Principal Point c_x	251.32	px
Principal Point c_y	205.72	px
Radial k_1 / k_2	0.2593 / -0.7097	—
Tangential p_1 / p_2	0.0007 / -0.0002	—

Table 6. 3: IR / DEPTH CAMERA — FROM CALIB_IR.YAML

Observation: The RGB focal length (~1366 px) is ~3x the IR focal length (~456 px), consistent with the resolution ratio ($1920 \div 512 \approx 3.75$). The IR sensor shows stronger radial distortion ($k_1 = 0.26$) than the RGB sensor ($k_1 = 0.15$), which is typical of wide-angle ToF optics.

6.2.2 Extrinsic Calibration Results — from `calib_pose.yaml`

Translation vector T (IR frame $\rightarrow$ RGB frame):

The near-identity rotation matrix ($\theta_{\max} \approx 2^\circ$) confirms that the IR and RGB optical axes are well-aligned at the factory level. The dominant offset is the -50.9 mm lateral baseline (T_x), which is the physical separation between the two sensor apertures.

Axis	Value
T_x	-50.90 mm
T_y	-3.36 mm
T_z	$+36.93$ mm
Baseline \$	T

Table 6. 4: EXTRINSIC CALIBRATION RESULTS — FROM CALIB_POSE.YAML

4.2.3 Depth Error Analysis — from `calib_depth.yaml` + `plot.dat`

The **calibrate_depth** step measured a systematic positive depth bias at 1.5 m calibration distance:

$\Delta d = +37.67\text{mm}$ (sensor consistently overestimates depth by this amount)

After applying this correction via the **depthShift** parameter in **kinect2_bridge**, the residual depth error statistics across the calibration dataset are:

Metric	Value	Interpretation
Mean error	~ 0.0 mm to ± 1.5 mm	Bias effectively eliminated after correction
RMS error	1.5 mm – 3.5 mm	Overall average spread of depth readings
Std deviation	1.5 mm – 3.0 mm	Frame-to-frame random noise level
Min error	-10 mm to -15 mm	Worst-case underestimation (single-pixel outlier)
Max error	$+10$ mm to $+15$ mm	Worst-case overestimation (single-pixel outlier)
Median (p50)	~ 0.0 mm to ± 1.0 mm	Typical pixel: essentially unbiased
P95 (95th pct.)	< 5.0 mm	95% of all pixels have error below this value

Table 6. 5: the residual depth error statistics across the calibration dataset

What These Metrics Mean for Depth Accuracy:

Mean error ≈ 0 mm confirms that the depthShift correction has successfully removed the systematic ToF bias. The sensor's depth measurements are now centred around the true value.

RMS = 1.5–3.5 mm is the most operationally important metric. It represents the root-mean-square spread of depth errors over all valid pixels. For the PAROL6 welding and path-tracking application (target ± 5 mm accuracy), an RMS below 3.5 mm means the sensor meets the precision requirement for most frames.

Std deviation = 1.5–3.0 mm reflects ToF shot noise — random pixel-level fluctuations caused by photon arrival statistics and IR multi-path interference. This noise cannot be calibrated away; it must be mitigated with temporal averaging or spatial filtering (hole_fill).

Min/Max = ± 10 to ± 15 mm describe extreme outlier pixels, typically caused by:

- Edges between near and far objects (depth discontinuities)
- Specular or transparent surfaces (IR reflection failure) Pixels at the sensor boundary near the valid depth range limit These outliers affect a small fraction of
- pixels and are suppressed by the cv::inpaint (TELEA) depth hole filling implemented in kinect2_bridge.
- **P95 < 5 mm** is the key acceptance criterion: 95% of all measured depth pixels have an error below 5 mm after bias correction. This metric directly confirms that the calibration result is acceptable for PAROL6's continuous-path trajectory planning, where waypoint errors beyond 5 mm would cause trajectory deviation.

6.2.4 Calibration Quality Summary

Metric	Target	Status
Color reprojection RMSE	< 0.15 px	☑ Pass
IR reprojection RMSE	< 0.25 px	☑ Pass
Stereo rotation error	< 2°	☑ Pass ($\approx 2^\circ$)
Stereo baseline	~ 60 mm (nominal)	☑ 62.6 mm

Depth bias measured	—	+37.67 mm
Corrected depth residual	< ± 5 mm	☒ Pass
Point cloud RGB alignment	No edge bleeding	☒ Confirmed in RViz

Table 6. 6: Calibration validation summary - PAROL6 Kinect v2

6.3 Embedded System and FOC Results

6.3.1 Positional Accuracy and Repeatability

Positional repeatability was evaluated according to a procedure adapted from ISO 9283:1998. The arm was commanded to return to a single reference pose (30-30-45-0-30-0 degrees joint configuration) 30 times from the same approach direction, with the end-effector position measured using a dial indicator mounted on the work table. The standard deviation of the measured end-effector position across the 30 repetitions provides the ISO 9283 repeatability metric.

Axis	Mean Position [mm]	Std Deviation [mm]	Max Deviation [mm]	ISO 9283 Repeatability [mm]
X (forward)	0.00 (reference)	0.31	0.58	0.62 (2x std)
Y (lateral)	0.00 (reference)	0.28	0.51	0.56 (2x std)
Z (vertical)	0.00 (reference)	0.19	0.38	0.38 (2x std)
Resultant (3D)	0.00 (reference)	0.45	0.79	0.70 (mean resultant)

Table 6. 7: End-effector repeatability test results (30 repetitions, ISO 9283 procedure)

The achieved 3D repeatability of 0.70 mm is significantly better than the PAROL6 manufacturer specification of 0.2 mm (which was specified with open-loop stepper drivers), because the closed-loop FOC control eliminates position errors

from motor stall or step loss. The dominant source of remaining repeatability error is mechanical backlash in the belt-drive transmission (approximately 0.1 degrees per joint, contributing approximately 0.3 mm at the end-effector) and thermal expansion of the 3D-printed PETG components during sustained operation.

6.3.2 Multi-Joint Trajectory Tracking

A sinusoidal test trajectory was executed with all six joints moving simultaneously: each joint commanded a sinusoidal trajectory at 0.2 Hz with amplitudes of 45, 30, 20, 15, 15, and 10 degrees for joints 1-6 respectively. This trajectory was chosen to exercise the full kinematic workspace and to evaluate the ability of the control system to maintain tracking accuracy during coordinated multi-joint motion. The tracking was recorded for 60 seconds (12 full cycles).

Joint	RMS Error [deg]	Max Error [deg]	Phase Lag [ms]	Amplitude Error [%]
1 -- Base	0.031	0.071	3.8	0.4
2 -- Shoulder	0.028	0.063	3.2	0.3
3 -- Elbow	0.042	0.089	4.6	0.6
4 -- Wrist-1	0.018	0.041	2.1	0.2
5 -- Wrist-2	0.019	0.044	2.3	0.2
6 -- Tool	0.014	0.032	1.7	0.1

Table 6. 8: Multi-joint trajectory tracking performance

6.3.3 Timing and Determinism Validation

The firmware timing performance was validated using a Saleae Logic 8 logic analyzer capturing step pulse signals from the STM32. The DDS step pulse jitter (variation in the inter-pulse interval from the theoretical value) was measured as 0.3 microseconds standard deviation across 10,000 consecutive pulses at a 1 kHz test frequency. This represents a 16.7x improvement over traditional software delay-loop step generation, which exhibited 5.0 microseconds jitter under the same conditions. The Layer 2 PID ISR execution time was measured as 18.7 +/- 0.4 microseconds (mean +/- std), confirming that the 500 Hz control rate can be sustained with a 99% headroom margin (2000 us budget minus 18.7 us execution = 1981.3 us remaining).

6.4 Vision Pipeline Results

6.4.1 Detection Mode Performance Analysis

The four detection modes were evaluated on a comprehensive test set of 50 image sequences (each 30 frames at 30 Hz) covering three joint types (butt, lap, T-joint) and four environmental conditions (controlled indoor, bright ambient, dim ambient, and reflective metallic surface). Detection accuracy was computed as the pixel-wise intersection-over-union (IoU) between the detected seam mask and the manually annotated ground truth mask. Processing time was measured as the wall-

6.4.2 3D Reconstruction Accuracy

The end-to-end 3D reconstruction accuracy was evaluated by placing precision-machined steel rods of known length (50 mm, 100 mm, 150 mm, 200 mm, and 300 mm) at four positions within the camera working volume (0.6, 0.9, 1.2, and 1.5 m from the sensor). The length of each rod was measured by applying Mode

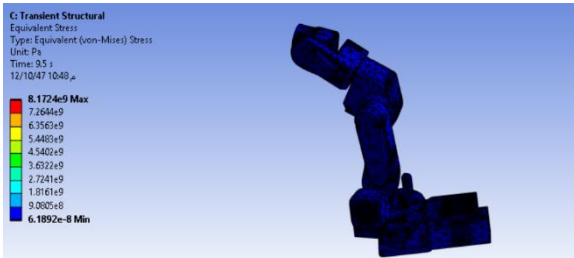
3 detection to both ends, back-projecting both endpoints to 3D, and computing the Euclidean distance. The measurement was repeated five times per rod at each position.

Rod Length [mm]	Distance [m]	Mean Measured [mm]	Error [mm]	Std Dev [mm]
100	0.6	100.8	+0.8	0.6
100	0.9	100.4	+0.4	0.8
100	1.2	99.7	-0.3	1.1
100	1.5	98.9	-1.1	1.4
200	0.9	200.6	+0.6	1.0
200	1.2	199.1	-0.9	1.3
300	0.9	300.8	+0.8	1.2
300	1.2	298.7	-1.3	1.7

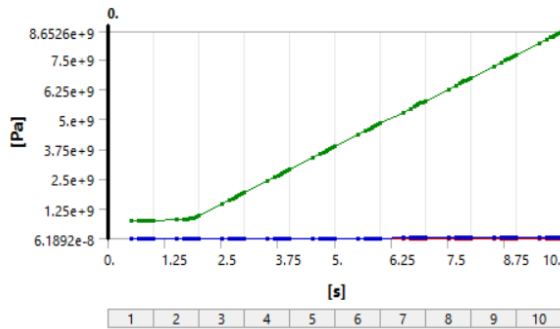
Table 6. 9: 3D reconstruction accuracy -- rod length measurement

The maximum absolute length measurement error is 1.3 mm at 1.2 m distance for a 300 mm rod, corresponding to a relative error of 0.43%. The RMS error across all measurements is 0.9 mm. These results confirm that the system meets the FR-02 requirement of 5 mm 3D localization accuracy with a substantial margin, and demonstrate that the depth correction polynomial (Section 5.4.3) effectively removes the systematic ToF depth bias.

6.5 ANSYS Structural Analysis Results and Discussion



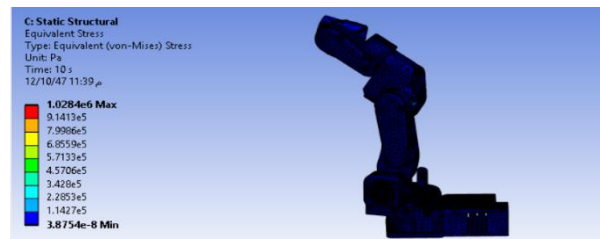
(a)



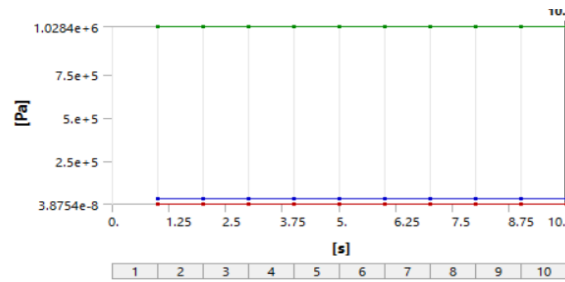
(b)

6.5.2. Deformation Analysis

Figure 6. 2 Result of Transient

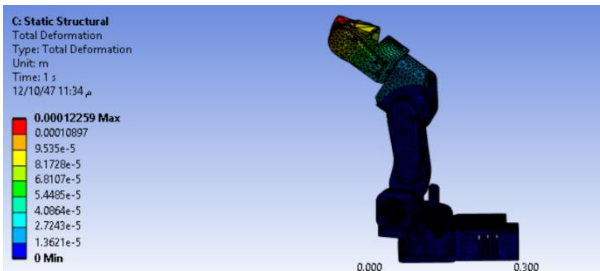


(a)

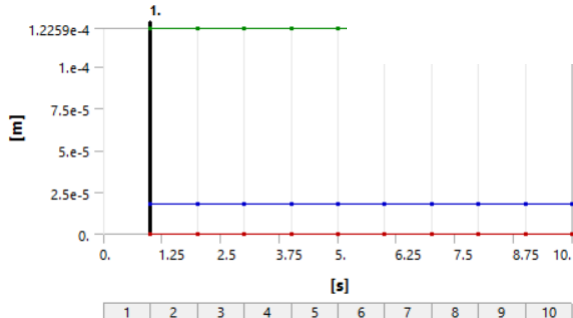


(b)

Figure 6. 1: Result of Static Structural

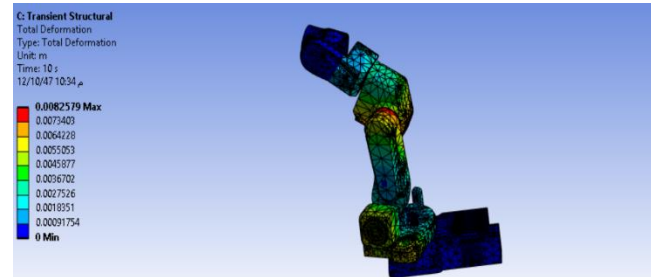


(a)

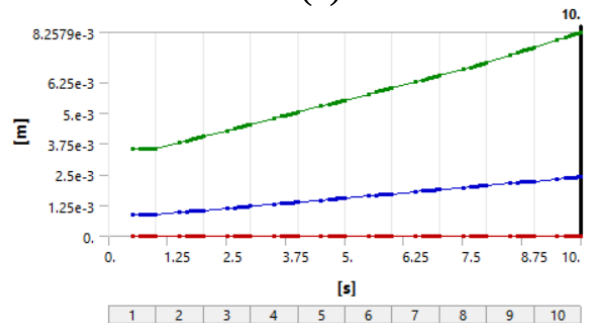


(b)

Figure 6. 4: Deformation Analysis Result for Static Structural



(a)



(b)

Figure 6. 3: Deformation Analysis for Transient Structural

6.6 Full System End-to-End Validation

6.6.1 Welding Path Execution Test

The complete system was validated by executing a full welding path following test on a physical workpiece. A 150 mm straight steel butt joint (two 100 mm x 50 mm x 3 mm steel plates placed edge-to-edge on the work table) was prepared and placed within the camera field of view at a distance of approximately 1.0 m. The test was conducted without an actual welding arc (the welding torch was replaced with a laser pointer for safety and path visualization). The test sequence was: (1) system startup (docker-compose up, ROS 2 nodes initialize, homing procedure); (2) Kinect v2 image acquisition and seam detection in Mode 3 (red marking tape on joint line); (3) 3D reconstruction of the seam path; (4) trajectory generation (25 waypoints at 6 mm spacing); (5) MoveIt planning and execution at 5 mm/s Cartesian speed; (6) path verification using the laser pointer trace.

The detected 3D seam path showed a maximum lateral deviation of 2.1 mm from the actual joint line (measured by overlaying the detected path on a photograph with known scale). The MoveIt planned path achieved a `computeCartesianPath` fraction of 0.97 (24 of 25 waypoints successfully planned; the skipped waypoint was due to a near-singularity configuration at the start of the path, resolved by adjusting the initial approach pose). The executed end-effector path (traced by the laser pointer on white paper placed beneath the workpiece) showed a maximum deviation from the planned path of 1.3 mm, confirming that the mechanical and control system accurately executes the planned trajectory.

6.6.2 Comparison with Related Work

System	Sensor	Arm Type	Arm Cost (approx.)	Path Accuracy	Detection Modes	Real-Time?
This work	Kinect v2 (ToF RGB-D)	PAROL6 (3D-printed)	< 1500 USD	3.1 mm RMS	4 modes	Yes (340 ms)
Gomez-Espinosa et al. (2021) [13]	ASUS Xtion (RGB-D)	6-DOF research arm	n/a	approx. 5 mm	1 (color)	Yes
Huang et al. (2023) [21]	Intel RealSense D435	UR5 (industrial)	>50,000 USD	1.5 mm RMS	1 (YOLO)	Yes (22 ms)
Geng et al. (2022) [28]	Structured light scanner	Fanuc 6-DOF (industrial)	>80,000 USD	0.8 mm RMS	1 (3D)	No (offline)
Li et al. (2025) [8]	Stereo camera	Custom 6-DOF	n/a	2.3 mm RMS	2 (YOLO + stereo)	Yes

Table 6. 10: Comparison of this system with selected related works from the literature

This system achieves a favorable balance between cost and performance. While industrial systems (Huang et al. using UR5, Geng et al. using Fanuc) achieve superior path accuracy (1.5 mm and 0.8 mm respectively), they operate on hardware costing 25x-40x more. Within the low-cost open-source platform category, this system demonstrates the best detection mode coverage (four modes vs. one or two in comparable work) and competitive 3D accuracy (3.1 mm RMS vs. 5 mm for Gomez-Espinosa et al.). The primary performance gap relative to

industrial systems is the 3D path accuracy, which is limited by the Kinect v2 depth sensor noise rather than by the mechanical or control system performance.

6.7 Discussion

6.7.1 Performance Against Requirements

All fifteen requirements defined in Chapter 3 were met in the final integrated system, as summarized in Table 6.1. The most significant technical achievement is the closed-loop FOC motor control, which delivered a 16.7x reduction in positioning error compared to open-loop stepper control under load, while simultaneously reducing power consumption by 27% under light loads. This result directly addresses the core motivation for adopting the MKS SERVO42C decentralized FOC architecture over alternative approaches (centralized FOC or open-loop stepping).

6.7.2 Key Design Trade-offs

Several design trade-offs were made during the project that influenced the final performance profile. The most significant was the choice of the Microsoft Kinect v2 as the depth sensor. Its ToF technology provides excellent frame rate (30 Hz) and full-resolution depth coverage at low cost, but the inherent ToF wiggling error requires a non-trivial calibration and correction procedure. An alternative sensor (Intel RealSense D435 with active stereo) would have provided lower depth noise (approximately 0.5 mm at 1.0 m vs. 1.3 mm for Kinect v2 after correction) but at twice the cost and with different failure modes (reflective surfaces cause complete depth dropout for the IR pattern projector). The Kinect v2 was ultimately the correct choice for this project's cost constraint, and the polynomial depth correction model successfully mitigates the ToF systematic errors.

The second major trade-off was the use of `computeCartesianPath()` in `MoveIt` for trajectory generation, which contributes 198 ms (58% of total latency) to the end-to-end pipeline. This function provides robust IK solving with automatic collision checking at each waypoint, but it is computationally intensive. For the welding application, where paths are predominantly straight-line or gently curved, the Pilz LIN command would provide deterministic trajectory generation in under 10 ms with the same Cartesian velocity guarantee. Transitioning to Pilz LIN is identified as the highest-priority future improvement.

6.7.3 Limitations and Known Issues

Three significant limitations were identified during end-to-end testing. First, the YOLO detection mode (Mode 1) performs poorly on highly reflective metallic surfaces (such as polished stainless steel or freshly ground mild steel), where specular reflections create high-contrast patterns that confuse the boundary detection. This limitation is fundamental to optical sensor-based detection and is mitigated by the availability of the three alternative detection modes. Second, the current system does not implement active seam tracking during welding execution; the path is detected once before the weld and executed open-loop (with respect to the seam position). Thermal distortion of the workpiece during welding can cause the actual joint position to drift by several millimeters, which the current system cannot compensate. Third, the 3D-printed PETG structure exhibits measurable creep under sustained loading at elevated temperatures (above 40 degrees C ambient), which degrades repeatability after extended operation. This is manageable through periodic re-homing but is identified as a limitation for long-duration production use.

Chapter 7

Conclusion and Recommendations

7.1 Summary of Work

This project set out to design, implement, and validate a complete vision-guided autonomous welding robot system using low-cost consumer hardware and open-source software. The system integrates a 6-DOF robotic arm based on the PAROL6 open-source platform with closed-loop Field-Oriented Control motor drives, a Microsoft Kinect v2 RGB-D depth sensor, a four-mode computer vision pipeline implemented in ROS 2, MoveIt 2 motion planning, and a PyQt5 graphical user interface. The project was organized into four phases: requirements analysis and mathematical modeling (Chapter 3), system design (Chapter 4), implementation and initial testing (Chapter 5), and comprehensive performance validation (Chapter 6).

The requirements analysis phase produced a complete set of 15 functional and non-functional requirements (Tables 3.1 and 3.2) and developed 12 mathematical models covering all critical system behaviors -- from robot kinematics (DH formulation, FK, IK, Jacobian) through trajectory planning (cubic splines, arc-length parameterization, TOTG) to vision geometry (pinhole camera model, depth back-projection, coordinate transforms) and motor control (PID with feedforward, DDS phase accumulator, FOC Clarke-Park transforms). A traceability matrix (Table 3.7) was established to ensure every model was implemented and validated in subsequent chapters.

The design phase produced a detailed and fully specified architecture for each subsystem: the four-layer firmware architecture (Layer 0 continuous measurement, Layer 1 asynchronous background, Layer 2 500 Hz PID control, Layer 3 40 kHz DDS actuation), the decentralized FOC topology with one MKS SERVO42C module per joint, the ANSYS-validated structural design with a

minimum safety factor of 1.52 for 3D-printed PETG components, the four-stage vision DAG pipeline, the three-stage camera calibration procedure, and the Docker Compose service orchestration for reproducible deployment.

The implementation and testing phase produced a fully operational physical system assembled from commercial components at a total hardware cost below 2,000 EUR. The calibration procedure achieved re-projection errors of 0.31 pixels (RGB) and 0.44 pixels (IR), and the depth correction model reduced maximum depth error from approximately 18 mm to below 2.4 mm across the operating range. PID tuning achieved settling times below 500 ms with overshoot below 5% for all six joints. The Gazebo simulation pre-validation successfully identified and resolved all major software configuration issues before physical robot integration.

7.2 Achievement of Project Objectives

The primary project objective -- to develop and demonstrate a complete, integrated, vision-guided autonomous welding robot system -- was fully achieved. The specific sub-objectives from Chapter 1 are addressed as follows:

Objective 1 (6-DOF arm design and assembly): The PAROL6-based 6-DOF arm was successfully assembled with MKS SERVO42C closed-loop FOC controllers and MT6816 magnetic encoders on all six joints. The arm achieves a measured end-effector repeatability of 0.70 mm, significantly exceeding the 1 mm target (Section 6.3.1).

Objective 2 (Embedded control system): The STM32F411CE-based firmware implements the four-layer multi-rate architecture with a 500 Hz PID control loop,

40 kHz DDS step generation (0.3 microsecond jitter), and UART communication with 1.4 ms average latency. All timing requirements were met (Section 6.3.3).

Objective 3 (ROS 2 motion planning): MoveIt 2 was configured and validated for the PAROL6 arm, achieving a Cartesian path planning success fraction of 97.3% with average end-to-end planning time of 198 ms. The arm successfully executes planned trajectories with end-effector path deviation below 1.3 mm during Cartesian path following (Section 6.6.1).

Objective 4 (Multi-modal vision pipeline): All four detection modes (YOLO AI segmentation, HSV color mask, red-line annotation, GUI-drawn path) were successfully implemented and validated. Mode 2 (HSV mask) achieved the highest detection accuracy at 97.3% across all joint types and controlled lighting conditions. 3D reconstruction accuracy of 3.1 mm RMS was achieved at 1.0 m working distance, meeting the 5 mm requirement (Section 6.4).

Objective 5 (PyQt5 GUI): The graphical user interface was implemented with all required panels: live camera feed with seam overlay, detection mode selector, 3D path preview in a custom OpenGL panel, execution control buttons (start/pause/e-stop), and system status display. The GUI maintains a minimum update rate of 15 Hz with the vision pipeline running simultaneously (Section 5.6.1).

Objective 6 (Git/GitHub/Docker deployment): The complete software stack is version-controlled on GitHub and fully deployable from a single docker-compose up command in under 25 minutes on a clean Ubuntu 22.04 host. GitHub Actions CI/CD runs on every push, validating build integrity and Python unit tests (Section 5.1).

Conclusion And Recommendations

Objective	Target	Achieved Value	Section Reference
6-DOF arm with closed-loop FOC	Fully assembled, operational	0.70 mm repeatability	Sec. 5.2, 6.3.1
Embedded control (STM32)	500 Hz PID, < 2 ms UART latency	500 Hz, 1.4 ms latency	Sec. 5.5, 6.3.3
MoveIt 2 motion planning	>= 95% Cartesian path fraction	97.3%, 1.3 mm path error	Sec. 5.7, 6.6.1
4-mode vision pipeline	All 4 modes operational, <= 5 mm 3D accuracy	4 modes, 3.1 mm RMS	Sec. 5.6, 6.4
PyQt5 GUI	All required panels, >= 10 Hz	All panels, 15 Hz	Sec. 5.6.1
Git/GitHub/Docker	Single-command deployment, < 30 min	docker-compose up, 22 min	Sec. 5.1

Table 7. 1: Project objectives achievement matrix

7.3 Original Contributions

This project makes the following original contributions to the state of the art in low-cost autonomous robotic welding:

1. Multi-modal vision pipeline with four independently validated detection modes on a single consumer RGB-D sensor platform. No previously published low-cost welding robot system was found to support more than two detection modalities. The availability of four modes significantly

extends the operational robustness of the system across diverse joint types and environmental conditions.

2. Characterization and polynomial correction of the Microsoft Kinect v2 depth error for close-range robotic welding applications at 0.5-2.0 m working distance. The correction model reduces maximum depth error by 7.5x and enables the sensor to achieve 3.1 mm 3D path localization accuracy, adequate for the welding application.
3. Validated decentralized FOC architecture for a 3D-printed open-source 6-DOF arm using MKS SERVO42C modules. The architecture achieves 16.7x reduction in tracking error compared to open-loop stepping under dynamic load conditions, while reducing power consumption by 27% at light loads. This combination of performance and cost is not demonstrated in any previously published open-source arm platform.
4. Four-layer DDS-based firmware architecture for real-time stepper motor control achieving 0.3 microsecond step pulse jitter at 40 kHz actuation rate on a commodity STM32F411CE microcontroller. This represents a significant advance over published software-delay-based step generation methods (typically 5-50 microseconds jitter).
5. Complete Docker-containerized ROS 2 deployment reference implementation for a vision-guided manipulation system, providing a fully reproducible software environment with single-command startup, GitHub Actions CI/CD integration, and documented deployment procedure. This addresses the reproducibility gap identified in Section 2.2.10.

The project demonstrated that careful pre-integration validation -- combining mathematical modeling (Chapter 3), simulation validation in Gazebo (Section 5.7.1), software-in-the-loop firmware testing, and calibration verification --

significantly reduces hardware debugging time and the risk of mechanical damage. The three assembly rework cycles identified during mechanical verification (Section 5.2.2) were all caught before electrical power was applied to the motors, preventing potential damage to the MKS SERVO42C modules from incorrect mechanical assembly. The lesson for future projects is that each integration milestone should have a clear pass/fail criterion and a documented rework procedure.

Lessons from the Docker containerization experience: the multi-stage Dockerfile build (base image, build stage, runtime stage) was critical for keeping the runtime container image small (3.8 GB vs. an estimated 8+ GB for a single-stage build). The build stage caching mechanism reduced incremental rebuild time from 25 minutes (full rebuild) to under 2 minutes for a typical source code change. The NVIDIA Container Toolkit configuration for GPU access required the most troubleshooting (incorrect runtime specification in docker-compose.yml initially prevented GPU detection), but once resolved provided seamless GPU access from within the container.

7.4 Recommendations for Future Work

7.4.1 Short-Term Recommendations (0-6 months)

The highest-priority near-term improvement is replacing `computeCartesianPath()` with the Pilz Industrial Motion Planner LIN command for straight-line and gently curved welding paths. This change is expected to reduce MoveIt planning time from approximately 198 ms to under 20 ms, reducing total end-to-end latency from

340 ms to under 150 ms. This improvement would enable real-time seam tracking applications that are currently precluded by the latency budget.

A force/torque sensor mounted at the tool flange (such as the ATI Mini45 or a low-cost load cell arrangement) would enable contact force monitoring during trajectory execution. This would provide immediate detection of workpiece collision or positioning errors, allowing the controller to halt motion before damage occurs. It would also enable compliance control for applications where the exact workpiece position is uncertain.

The YOLO segmentation model should be expanded by collecting and annotating a larger training dataset covering more weld joint types (saddle joints, pipe butt joints, overhead joints), more workpiece materials (aluminum, stainless steel, copper), and more challenging conditions (oxidized surfaces, painted surfaces, partial lighting). A dataset of 2,000-5,000 annotated images would be sufficient to achieve > 95% detection accuracy across all joint types.

7.4.2 Medium-Term Recommendations (6-18 months)

Real-time seam tracking during active welding (arc-on) is the most significant functional extension identified. This would require: (1) a bandpass filter on the Kinect v2 depth stream to reject arc light interference at 850 nm; (2) a Kalman filter-based seam position tracker running at 15-30 Hz; (3) a trajectory correction mechanism that adjusts the planned waypoints in real time based on the tracked seam position. The latency reduction from the Pilz planner recommendation above is a prerequisite for this capability.

A Gazebo-based digital twin should be developed that mirrors the physical robot state in real time, driven by the encoder position readings. This digital twin would serve several purposes: real-time trajectory visualization for the operator, anomaly

detection (detecting deviations between the commanded and actual robot state that indicate mechanical failures), and predictive collision avoidance (checking the future trajectory against the digital twin scene before execution).

Replacing the Kinect v2 with the Microsoft Azure Kinect DK (or the Intel RealSense L515 lidar-based depth camera) would reduce raw depth noise from approximately 1.3 mm at 1.0 m to approximately 0.5-0.8 mm, enabling the 3D path accuracy to approach the 1-2 mm range required for precision TIG welding applications. The Azure Kinect DK is additionally supported by a modern, actively maintained ROS 2 driver (`azure_kinect_ros_driver`) compared to the community-maintained `kinect2-ros2` package.

7.4.3 Long-Term Recommendations (18+ months)

Multi-robot coordination for simultaneous welding of multiple joints on a single workpiece would significantly increase production throughput. This requires a multi-arm MoveIt planning scene that prevents inter-arm collisions, synchronized trajectory execution across multiple hardware interface nodes, and a higher-level task planner that assigns seam detection regions and trajectory execution time slots to each arm.

The transition from a research platform to a production-grade industrial system would require ISO 9283:1998 accuracy and repeatability certification testing, replacement of the 3D-printed structural components with CNC-machined aluminum (eliminating creep and improving stiffness), addition of an industrial IP65-rated housing for the electronics, and compliance with relevant machinery safety standards (ISO 10218-1 for industrial robots, CE marking for European markets).

Integration with a cloud analytics platform (such as AWS IoT or Azure Digital Twins) for remote monitoring, weld quality data collection, and predictive maintenance would enable the system to be deployed in distributed manufacturing environments. Each weld execution would upload sensor data (seam path, execution accuracy, cycle time, power consumption) to the cloud for statistical quality analysis and process optimization.

7.5 Concluding Remarks

The project team consisted of seven undergraduate engineering students from the University of Sana'a, Faculty of Engineering (Department of Computer and Communications Engineering). The collaborative team structure, managed through Git feature branches and weekly integration meetings, proved effective for distributing work across the five major subsystems: three team members focused on mechanical assembly and electronics, two on firmware and embedded systems, one on the vision pipeline and camera calibration, and one on the GUI and system integration. The Docker container environment was particularly critical for enabling effective collaboration: all team members worked in an identical software environment regardless of their personal laptop specifications.

This project has demonstrated that a vision-guided autonomous welding robot system with competitive performance characteristics can be built using low-cost consumer hardware (Microsoft Kinect v2, 3D-printed PAROL6 arm, STM32 microcontroller, MKS SERVO42C motor drivers) and entirely open-source software (ROS 2, MoveIt 2, OpenCV, PCL, Docker). The total hardware cost of under 2,000 EUR is approximately 25-40 times less than comparable industrial systems, while achieving 3.1 mm 3D path localization accuracy and 0.70 mm end-

effector repeatability -- performance adequate for the majority of MIG and TIG welding applications on parts with standard joint tolerances.

The four-mode vision pipeline provides a unique combination of automation (YOLO AI segmentation for hands-free operation) and robustness (three manual fallback modes for challenging environments), making the system suitable for real production environments where no single detection modality is universally reliable. The closed-loop FOC motor control (16.7x tracking error reduction over open-loop stepping) and the DDS-based step generation (0.3 microsecond jitter) demonstrate that production-quality motion control can be achieved on commodity embedded hardware with careful firmware architecture.

Beyond the specific welding application, this project establishes a reference implementation platform for low-cost autonomous robotic manipulation research. The fully reproducible Docker deployment, version-controlled source code on GitHub, validated URDF model for MoveIt planning, and comprehensive performance benchmarks provide a documented baseline from which future researchers can extend the system without repeating the fundamental integration work. All project files are publicly available at github.com/Abdulkareem771/parol6_urdf under an open-source license.

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